
TECHNICAL UNIVERSITY OF CLUJ-NAPOCA
FACULTY OF INDUSTRIAL ENGINEERING, ROBOTICS AND MANAGEMENT OF PRODUCTION
DEPARTMENT OF DESIGN ENGINEERING AND ROBOTICS

ROBOTICS

BACHELOR THESIS

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TECHNICAL UNIVERSITY OF CLUJ-NAPOCA
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ROBOTICS

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Hybrid Agentic AI Robot for Medical Environments

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Student's statement of originality

The undersigned

Graduate of the bachelor's degree in Robotics:

I declare that my bachelor thesis with the title:

Represents my original contribution and it was not plagiarized.

The bachelor thesis was developed by myself under the coordination _____ and I
received the assistance of the consultants listed below.

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BACHELOR THESIS ABSTRACT

The purpose of the proposed theme

This theme was chosen due to the need to streamline the healthcare system, where staff are often overloaded with administrative and logistical tasks. The main reason is to exploit how a hybrid robotic platform can take over repetitive workloads, allowing nurses to focus on direct clinical care. This theme attempts to converge the handling capabilities of the Moxi robot, the social interface of Pepper and the empathic monitoring functions of Grace, to provide an integrated solution to the problem of medical burnout.

General objective

The overall objective is to develop a theoretical and functional concept of a hybrid robot, capable of both autonomous and manual navigation in the hospital environment and of intuitive interaction with patients. It also aims to demonstrate the viability of a robotic assistant that can perform both logistical tasks, monitor vital parameters, and emotionally engage in conversations. It acts as a bridge between technology and human needs.

Specific objectives

The specific objectives, through which the general objective was achieved, are:

1. Synthesis of hybrid features: Structural analysis of the robot to integrate an articulated arm on a mobile platform with anthropomorphic features, ensuring both logistical functionality and social integration.
2. Logistics flow optimization: Designing navigation and manipulation algorithms that allow the robot to perform tasks such as drug supply and sample collection, thus reducing unnecessary travel by nurses.
3. Interaction mode development: Implementation of an interface based on facial recognition and natural language processing to provide informational and emotional support to patients.
4. Validation of safety and ethical criteria: Assessing the impact of the presence of the robot in proximity to patients and establishing strict protocols for handling sensitive data.

Methodology/Approach/Design

The development follows a New Product Development process structured around seven sequential stages, from problem identification to validated design. Design for Six Sigma (DFSS-DMADV) was applied to translate clinical stakeholder needs into verifiable technical requirements, while the V-Model links each design decision to a corresponding validation activity. Mechanical design was performed in Fusion 360 and CATIA V5, with structural calculations conducted under ISO 13482:2014 safety requirements. The software implementation relies on a Python/Flask pipeline integrating YOLOv8, DeepFace, Vosk, and Ollama/Llama 3.2, with low-level motor control handled via Arduino. A .NET MAUI mobile application provides manual Bluetooth control, while an Android tablet running Termux handles voice and camera streaming to the server.

Main results

The principal result is a fully documented conceptual design of a hybrid healthcare robotic assistant, validated at the theoretical level through structural calculations, CAD modelling, and a complete hardware architecture specification. In parallel, a functional software prototype was developed and tested, demonstrating real-time facial emotion recognition, offline speech recognition, object and person detection, and LLM-driven natural language conversation within a single Python/Flask inference pipeline. Low-level motor control was implemented and validated on Arduino hardware, while two ESP32 modules drive the robot's animated face display, providing a functional emotional expression interface. The complete software implementation is publicly available on GitHub, where the modular architecture allows the inference pipeline to be adapted independently of the physical platform – functioning as a standalone agentic AI system configurable for different robotic or assistive applications.

Results limitations

The primary limitation is that no full-scale physical prototype was built: validation was carried out at reduced scale, on a functional demonstrator in which each subsystem of the full-scale design is represented by an accessible analogue component rather than by the industrial-grade actuators, sensors and structure specified in Chapter 3. Consequently, the mechanical design, the selected commercial components and the load cases are validated by calculation and CAD rather than by measurement on the final hardware, and the perception and interaction functions are demonstrated on substitute sensors and commodity compute. Future development directions include construction of the full-scale mechatronic assembly and integration testing, ROS 2-based navigation stack deployment [62], extension of the vital-signs monitoring suite, and regulatory evaluation under ISO 13482:2014 for clinical deployment certification

Practical implications

The use of this hybrid in medical spaces would significantly reduce the time lost by nurses with logistical tasks, allowing them to focus more on critical care. From a technological point of view, there is the possibility of integrating precise manipulation systems on social interaction platforms, resulting in multifunctional medical robots. Economically, although the initial investment is high, the amortization is achieved by reducing medication distribution errors, decreasing absenteeism rates caused by overload, and a more efficient logistical organization.

Other implications

From a social point of view, the use of such robots can improve the psychological state of patients hospitalized for long periods, providing an interactive presence that reduces the feeling of isolation. Scientifically, the work contributes to the study of ethics in medical robotics, analyzing the limits of acceptability and integration of a humanoid by patients. The quality of life of nurses increases directly by eliminating repetitive physical effort, and reduces psychological stress, leading to a more efficient medical act.

Originality/value

The originality of the work lies in the concept of functional hybridization. It is not limited to a robot that only carries objects or only communicates. A model is proposed that can do both actions

simultaneously. Also, the robot will be able to decide for itself whether social interaction with a patient is more urgent than a logistical task.

Chapters abstract

Chapter 1 – Introduction establishes the conceptual and technological context of the project. It traces the origins of robotics and the ethical framing derived from Asimov’s laws, situates the work within the ISO 8373 classification of medical robotics, and reviews the state of art through a benchmark of eleven deployed clinical platforms (Moxi, Grace, Stevie, TUG, Dinsow and others). The Social vs. Physical capabilities matrix synthesises this analysis and identifies the structural gap the project targets: no deployed system simultaneously covers logistics, monitoring and social interaction on a single platform.

Chapter 2 – Project Planning defines the problem and the methodological foundation. The clinical problem (staff overload, lack of continuous monitoring, patient isolation) is established through a 5-Whys root-cause analysis and quantified with literature data. The development is then structured around a New Product Development process supported by DFSS-DMADV, the V-Model and IEEE/ISO/IEC 15288, and the stakeholder needs are translated into fourteen measurable Voice-of-the-Customer requirements (R1-R14). The chapter closes with the architectural selection that retains the hybrid autonomous configuration and the set of technologies and validation procedures adopted for each functional layer.

Chapter 3 – Design and Implementation. It contains the technical contribution. It opens with the design constraints imposed by the applicable standards and regulatory framework (ISO 13482, IEC 60601-1, NP 015-2022) and the user analysis. The mechanical design dimensions every load-bearing assembly – head, neck, torso, mobile base and chassis – verifying each against the yield strength of the selected aluminium alloy, and resolves the manipulator through an explicit make-vs-buy evaluation. It establishes the mass budget, centre of gravity and tipping stability across five load cases, sizes the drive system and power/autonomy budget. It also presents the technological process and the geometric tolerancing of a representative component. The software sub-chapter describes the layered ROS 2 architecture [61], the offline AI pipeline (Vosk, YOLOv8, DeepFace, Llama 3.2) and the deterministic safety layer, followed by the sensor suite, the innovation aspects of the platform and a make-vs-buy cost analysis for the manipulator.

Chapter 4 – Prototype. This chapter presents the design, construction and programming of a functional scale demonstrator that validates the architectural decisions taken for the full-scale SKY robot in Chapter 3. Rather than a miniature mechanical copy, the demonstrator is a functional analogue in which every major subsystem is represented by an accessible component fulfilling the same engineering role. It covers the layered electrical architecture, the actuation and multi-sensor perception layers, the human–machine interface, and a single non-blocking firmware on an Arduino Mega 2560 supported by a companion software layer for the AI and communication functions. The integrated system executes the complete target scenario — autonomous and tele-operated navigation, continuous physiological monitoring, and an access-controlled arm sequence — as a functional validation of the full-scale concept.

The three major challenges condense the most demanding problems solved during implementation: the wrist-roll bearing interface for the in-house manipulator, the real-time integration of four AI models on commodity hardware, and the multi-device communication architecture of the physical demonstrator.

The Conclusions summarise the results obtained at both the conceptual and functional level, state the limitations, and define the further-development path toward a clinically deployable system.

GENERAL OBJECTIVE AND SPECIFIC OBJECTIVES

The general objective is to design and develop a robotic healthcare system, designed to support repetitive activities performed in medical environments. The aim is to obtain a functional prototype, capable of monitoring patients' physiological parameters (temperature, oxygen saturation, pulse), detecting and avoiding objects, recognizing people and their expressions, and also having the ability to interact verbally. Thus, monitoring and communication with both patients and medical staff is facilitated.

In general terms, it aims to contribute to the digitalization and automation of medical processes, demonstrating the real potential of robots in improving both the quality of medical care and life, by implementing an efficient and safe technological solution.

Specific objects:

- 1) Analysis of the application domain - identification of current requirements in the medical environment regarding automation and patient monitoring, as well as the study of existing technological solutions.
- 2) Defining the general architecture of the system - establishing the hardware and software components necessary to create the mechatronic system (user interface, control systems, mechanical structure, sensors, actuators, power supply).
- 3) Development and implementation of all four systems (mechanical, actuation, sensor, command and control) - design and development of command algorithms and their implementation in a dedicated environment, to ensure good functionality of the robot.
- 4) Simulation and functional testing - evaluating performance in simulated environments, tracking parameters such as: reaction speed, movement accuracy, communication and interaction capacity with both people and ancillary systems, decision making.
- 5) Analysing results and proposing improvement solutions - interpreting data obtained from tests, formulating conclusions and identifying possible development steps.

Part I



General aspects in the context of the theme

INTRODUCTION

1.1 Basic elements in context of the project theme

The conceptual origins of robotics have their roots in the literary domain. The term “robot” was introduced by the Czech writer Karel Čapek in his 1920 play R.U.R (Rossum’s Universal Robots), derived from the Czech word “robota”, meaning forced labor [1]. The term “robotics” was formalized by Isaac Asimov in his 1941 short story “Liar!”, alongside the “Three Laws of Robotics” [2] – which provide an ethical basis that is relevant even today.

- 1) First Law: “A robot may not injure a human being or, through inaction, allow a human being to come to harm.”
 - The “inaction” part is very significant in a medical context. A robot equipped with a vision sensor that detects a patient falling must alert staff, as inaction is a violation of this principle. This law establishes the integration of redundant safety sensors, including 360° LiDAR scanners and ultrasonic sensors, for real-time collision.
- 2) Second Law: “A robot must obey orders given by human beings, except where they conflict with the First Law.”
 - In clinical environments, the question of authority is a practical challenge: if a patient refuses a scheduled procedure while a doctor has mandated it, the robot encounters a logical conflict. To solve this problem, a “Safety Mode” is included that prioritizes patient consent, records the refusal for legal protection, and initiates a withdrawal.
- 3) Third Law: “A robot must protect its own existence, as long as this does not conflict with the First or Second Law.”
 - An inoperative robot indirectly compromises patient safety. Self-preservation is therefore implemented through intelligent battery management, automatic docking prior to critical discharge, and mechanical arm locking as a form to prevent aggression from patients.
- 4) Zero Law: “A robot may not harm humanity, or, by inaction, allow humanity to come to harm.”
 - The implementation of this law is realized at a macro-systematic level, where the collective safety of the hospital takes priority over individual orders. This ethical hierarchy is directly applied in two scenarios: biological protection, by detecting and isolating epidemiological risks (such as a patient unauthorizedly leaving a quarantine zone), and cybersecurity protection, through the adoption of an offline-first architecture that stores sensitive biometric and medical data locally. In both cases, the robot acts to prevent harm to the entire hospital community by refusing commands that could compromise either the quarantine or the integrity of the data network.

In current technical practice, these concepts are formally defined by ISO 8373:2021, which establishes that a robot is an “actuated mechanism programmable in two or more axes with a degree of autonomy, moving within its environment, to perform intended tasks”. This standard defines also the domain of robotics as the “science and practice of designing, manufacturing, and applying robots” (ISO 8373:2021). Based on these definitions, the field is classified into two major categories.

Industrial Robotics addresses structured, controlled environments where precision and repeatability are the primary design objectives. Industrial robots are classified according to their kinematic architecture as follows:

- a) Articulated Robots: having at least three rotary joints and a complex six degree of freedom (6-DOF) workspace as illustrated in the figure 1.1:

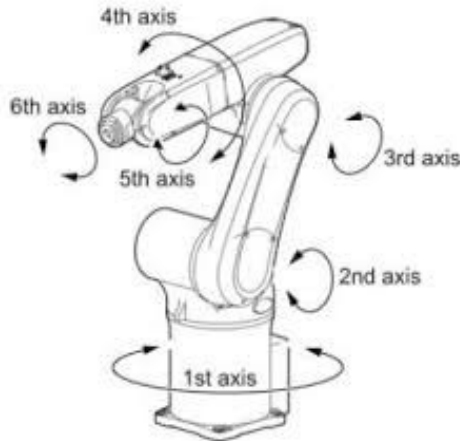


Figure 1.1 – Articulated robot

- b) SCARA (Selective Compliance Articulated Robot Arm) Robots: providing rigidity on the vertical axis (Z) and compliance in the horizontal plane (X-Y), as presented in the figure 1.2:

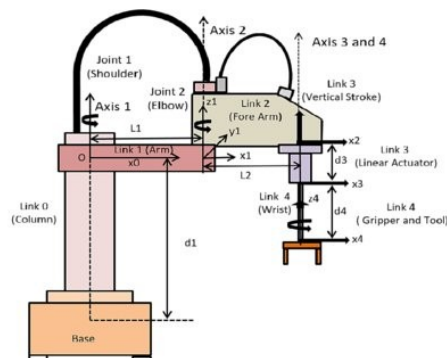


Figure 1.2 - SCARA robot

- c) Delta Robots: having parallel kinematic chains for extremely high-speed pick-and-place operations, as shown in the figure 1.3:

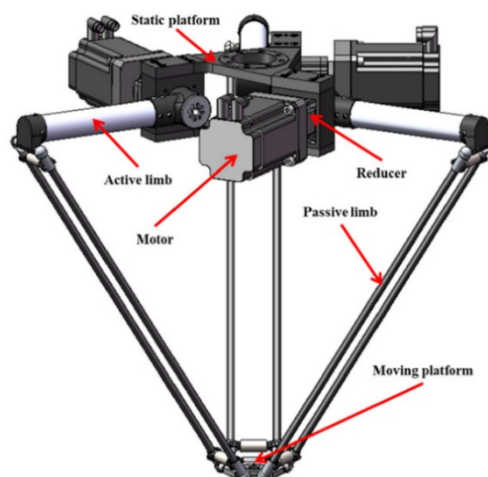


Figure 1.3 - Delta robot

- d) Cobots (Collaborative Robots): as presented in the figure 1.4, these are equipped with integrated force-torque sensors for safe direct interaction with human operators.

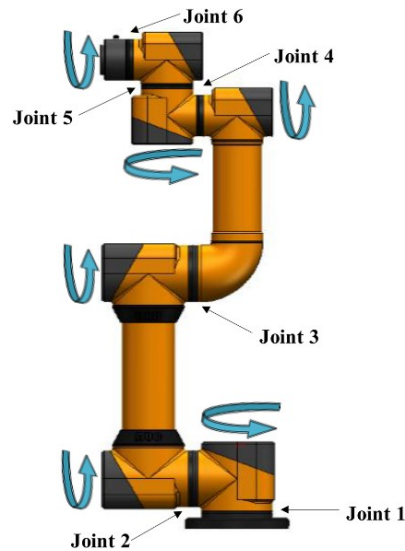


Figure 1.4 - Cobot

Service Robotics, as defined by ISO 8373:2021, includes robots that “perform useful tasks for humans or equipment, excluding industrial automation applications.”. These systems operate in unstructured and dynamic environments, where engineering focuses on safety, adaptability, and autonomous navigation, rather than precision.

Service robots are classified into two operational categories:

- 1) Personal Service Robots: intended for non-commercial tasks performed by untrained users (e.g., household robots).
- 2) Professional Service Robots: operated by trained personnel for commercial or institutional purposes (e.g., surgical systems, logistics platforms).

Medical Robotics constitutes a specialized sub-domain of Professional Service Robotics, further divided into three distinct categories:

- a) Interventional Robots: such as tele-operated surgical systems, operate in a master-slave configuration in which the surgeon controls a remote console while the robot executes movements with millimetric precision, eliminating physiological hand tremor. The critical technical requirements are minimum data transmission latency and haptic feedback.
- b) Rehabilitation Robots: such as exoskeletons, are designed to assist patients in recovering motor functions following trauma or neurological conditions. These systems have integrated force-torque sensors to detect the user’s effort and provide synchronized mechanical assistance. The most important technical requirement to prevent any injuries is backdrivability. In other words, it refers to the ability of an actuator or mechanism to allow the transmission of motion or force from the output back to the input [4].
- c) Assistance and Care Robots: are developed to support both medical employees and patients in repetitive clinical tasks. Unlike the other categories, these systems require a high degree of autonomy to navigate, perform monitoring, and sustain social interaction.

A comparative overview of these three sub-domains is presented in table 1.1, based on autonomy level, primary purpose, workspace, interaction type, key sensors and representative examples:

Table 1.1 - Comparative analysis of medical robotics sub-domains

Criterion	Interventional Robots	Rehabilitation Robots	Assistance and Care Robots
Primary purpose	Surgery and invasive diagnostics	Motor recovery and physical support	Logistic support, monitoring, social interaction
Autonomy level	Low – direct tele-operation	Medium – user-synchronized	High – autonomous navigation and decision-making
Workspace	Structured, operating rooms	Controlled, clinics or home	Unstructured, dynamic environments
Interaction type	Indirect, via instrumentation	Direct, via permanent contact	Social and logistical
Key sensors	Endoscopic cameras, haptic sensors	Force-torque, tactile sensors	LiDAR, vision, IR thermal sensors
Examples	Da Vinci Surgical System	Exoskeletons, ROBEAR	Moxi, Pepper, Grace, Dinsow

The development of medical robotics has progressed through some important technological milestones, each of which has significantly expanded the role of autonomous systems in clinical practice:

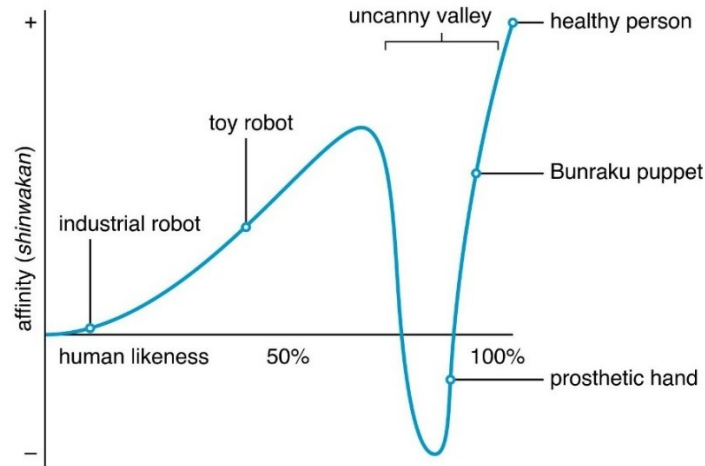
- **1920** – Karel Čapek introduces the concept of the robot as a mechanical labor substitute (Čapek,1920).
- **1941** – Asimov validates the term “robotics” and proposes “The Three Laws”, establishing the foundation of ethics – for human-machine interaction [2].
- **1985** – The PUMA 560 industrial arm was used to guide a needle during a neurosurgical biopsy, marking the first documented application of robotics in an operating room [5].
- **1992** – ROBODOC receives FDA (Food and Drug Administration) approval as the first dedicated surgical robotic system, applied to femoral cavity preparation in hip replacement procedures [6].
- **2000** – Intuitive Surgical introduces the Da Vinci Surgical System, establishing the standard for minimally invasive tele-operated surgery through master-slave configuration and three-dimensional high-definition visualization [7].
- **2014-2018** – The field transitions toward service robotics. Pepper [8] for social interaction and Moxi [9] for autonomous hospital logistics marks the start of the humanoid assistant as a clinically viable platform.
- **2019-2020** – The COVID-19 pandemic accelerates the adoption of autonomous robots in hospitals, with systems such as UVD (Ultraviolet Disinfection) Robots being deployed for autonomous UV disinfection, and telepresence robots reducing direct staff-patient contact [10].
- **2021** – Hanson Robotics introduces Grace, a humanoid robot specifically designed for eldercare and clinical interaction, featuring over 48 facial expressions and integrated thermal monitoring sensors [11].
- **2022-2023** – Advanced large language models (LLMs) such as GPT-4 begin to be integrated into robotic interaction, enabling natural, context-aware medical conversations and preliminary triage support [12].
- **2024-present** – The field moves towards fully integrated hybrid platforms, combining autonomous navigation, manipulation, monitoring, and AI-driven social interaction within a single clinical system (Henschel et al., 2021; Diligent Robotics, 2023; Hanson Robotics, 2021).

Assistance robots in hospitals are becoming more necessary due to two main reasons: technological progress and real staffing problems. According to the World Health Organization, the global shortage of healthcare workers is expected to reach 10 million by 2030, with nurses making up most of this gap [15]. A study carried out in 36 hospitals also found that nurses spend around 35% of their time on tasks like delivering supplies or moving equipment – work that does not require medical training and could be handled by a robot [16]. A robotic assistant that can take over these tasks would allow nurses to spend more time on direct patient care, reducing both physical and mental fatigue [14]. For patients, having a robot present also means better safety: features like fall detection, temperature checks, and emergency alerts provide continuous monitoring that a small team of nurses simply cannot maintain around the clock. In long-term care, robots designed for social interaction have also been shown to reduce feelings of isolation [17].

A hospital assistant robot needs three main technologies to work properly. The first is autonomous navigation, which uses SLAM (Simultaneous Localization and Mapping) to build a map of the environment and track the robot's position at the same time. The second is Human-Robot Interaction (HRI), which includes natural language processing (NLP) for conversation and computer vision for reading facial expressions. The third is medical sensing, such as non-contact temperature measurement and fall detection through body pose analysis. Each of these technologies works well on its own, but combining all three into one reliable system that operates in real time is still a significant engineering challenge [13]. For example, recognizing emotions via computer vision introduces processing latency that can conflict with the response time of the speech synthesis module – if the system detects fear in a patient's expression while simultaneously generating a vocal response, prioritizing one process over the other requires careful management of real-time resources.

Even with all this progress, assistance robots still have important limitations. Moving safely in busy hospital corridors is difficult because people move in unpredictable ways. Not all patients are comfortable around robots either – older patients in particular tend to be more hesitant about robotic interaction [18]. There are also open questions about data privacy, legal responsibility if something goes wrong, and how these systems should be regulated under standards like ISO 13482:2014. It defines the safety requirements specifically for personal care robots, covering aspects such as physical interaction with humans, autonomous navigation in shared spaces, and emergency stop mechanisms. These challenges show that while the technology exists, deploying it effectively in a real clinical setting requires careful design decisions. Most existing systems are addressed either for logistic tasks or for social interaction, but rarely within a single integrated platform. This is a gap that this project directly addresses.

One of the most important design considerations concerns the robot's morphology and its effect on patient acceptance. The Uncanny Valley hypothesis, proposed by Mori in 1970, establishes that as a robot's appearance approaches human likeness, minor imperfections in movement or form produce an increase in observer discomfort. This phenomenon defines a functional design boundary: the robot must be sufficiently anthropomorphic to be perceived as approachable and empathetic, while remaining clearly identifiable as a machine [20]. This concept is explained in figure 1.5:



Source: Masahiro Mori, "The Uncanny Valley," *IEEE Robotics & Automation Magazine*, 19(2):98–100 (June 6, 2012).

Figure 1.5 – The “Uncanny Valley” graph (Masahiro Mori, 1970)

The following sub-chapter presents a comparative analysis of the most representative humanoid and hybrid robotic systems currently deployed in clinical environments, to identify the design benchmarks that will inform the architecture of the proposed solution.

1.2. Highlights of technological developments in the project theme area

This sub-chapter presents a comparative analysis of robotic systems currently representative of the state of the art in clinical and assistive robotics. All the robots within table 1.2 were selected based on four criteria: they are commercially developed and have been deployed in real hospitals; they are internationally recognized and documented in technical literature; cover all major functional categories relevant to hospital assistance; and each one addresses at least one of the core requirements that the present project aims to integrate.

Table 1.2 - Overview of representative robotic systems

No.	Robot	Type	Key feature	Autonomy level	Deployment location
1	Moxi	Hybrid	Autonomous logistics and arm manipulator	High	USA (100+ hospitals)
2	ROBEAR	Specialist	Patient lifting	Low	Japan (research)
3	Pepper	Social Humanoid	Emotion recognition and omnidirectional interaction	Medium	Japan, Belgium, UK
4	Grace	Advanced Humanoid	Nursing AI with 48+ facial expressions and thermal monitoring	High	Hong Kong, USA
5	TUG	Logistic AMR	Heavy logistics	High	USA (500+ hospitals)
6	Stevie	Social Hybrid	Fall detection and therapeutic social engagement	Medium	Ireland, USA
7	Relay	Delivery AMR	Last-mile delivery in tight clinical spaces	High	USA, Asia
8	Aeolus	Hybrid (multi-purpose)	7-DOF arm for object recognition and manipulation	High	USA, Taiwan

9	Care-O-bot 4	Modular Hybrid	360 safety sensing and modular arm configuration	High	Germany (research)
10	HOSPI	Secure AMR	Encrypted transport of sensitive medical materials	High	Japan
11	Dinsow	Medical Assistance	Integrated vitals monitoring and telepresence	Medium	Thailand, Asia

- 1) Moxi [9] – shown in figure 1.6 – is one of the most currently representative hybrid assistant robots. It stands around 140-150 cm tall, which was a design choice to avoid intimidating patients and staff [14]. Its most distinctive feature is a single robotic arm with an adaptive gripper, which allows it to open doors, press elevator buttons, and retrieve items from shelves or storage units. For secure storage, it has three locked drawers that can only be opened with a badge or PIN code. Navigation is made by LiDAR-based SLAM, and a set of 3D depth cameras allows it to recognize and manipulate objects. The robot operates 24/7 with automatic docking for recharging and is currently deployed in over 100 hospitals across USA. Here it handles Personal Protective Equipment (PPE) logistics, supply restocking, and medication delivery [14].



Figure 1.2 – Moxi robot

- 2) ROBEAR [21] – shown in figure 1.7 – is a specialist robot developed in Japan, designed for lifting and transferring patients. Its capacity is up to 80 kg. It uses Smart Rubber capacitive tactile sensors to detect pressure and adjust its gripping in real time, making physical contact safer. Unlike the other systems, ROBEAR requires a human operator for each task, which limits its independent deployment in a clinical setting [21].



Figure 1.3 - ROBEAR robot

- 3) Pepper [8] – shown in figure 1.8 – is a social humanoid robot designed for human interaction. It moves on an omnidirectional base and uses a combination of cameras, microphones, and a 3D depth sensor to analyse facial expressions and voice tone. This allows it to adapt its response emotionally. It has been deployed in hospitals in Japan, Belgium, and UK, mainly for reception and patient guidance roles [8].

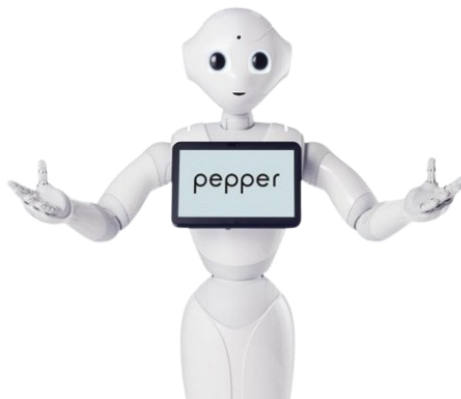


Figure 1.4 - Pepper robot

- 4) Grace [11] – shown in figure 1.9 – is a humanoid robot built for healthcare. It can produce over 48 facial expressions using a bio-inspired skin material, and monitors patient vitals through a chest-mounted thermal camera and optical sensors. Its NLP system is optimized for clinical terminology, making it capable of conducting patient intake interviews and providing emotional support. It has been tested in eldercare facilities in Hong Kong and USA [11].



Figure 1.9 – Grace robot

- 5) TUG [22] – shown in figure 1.10 – is currently one of the most widely used hospital robots in the world, operating in over 500 facilities in the United States. Some of its main characteristics are its capability to carry payloads of up to 453 kg and autonomously navigates between floors through direct integration with elevator control systems and automatic doors. Sensitive compartments are secured with multi-factor authentication, including PIN, badge and biometric access, making it suitable for transporting drugs and controlled substances [22].



Figure 1.5 - TUG robot

- 6) Stevie [17] – shown in figure 1.11 – is a social hybrid robot developed for long-term care environments. It uses a dual-display system containing an LED matrix on the head for simple facial expressions and a chest-mounted tablet for video calls and detailed information. Its key feature is real-time fall detection through 3D depth cameras and AI algorithms trained for abnormal posture analysis. It also supports telepresence [17].



Figure 1.6 – Stevie robot

- 7) Relay [23] – shown in figure 1.12 – is a delivery robot designed for last-mile transport inside hospitals. Its slim cylindrical profile allows it to navigate narrow corridors and elevators without disrupting staff or patients. It carries a secure top-loading compartment with a payload of around 10 kg, for medication, lab samples, and blood units. The compartment can only be opened through a specific interaction or badge scan [23].



Figure 1.7 – Relay robot

- 8) Aeolus [24] – shown in figure 1.13 – is a multi-purpose hybrid robot that combines a humanoid torso with a heavy-duty mobile base. Its key feature is a 7-DOF robotic arm with an adaptive gripper, capable of handling even delicate objects. It uses AI-driven RGB-D cameras to recognize over 1000 unique objects and can perform complex tasks such as retrieving misplaced medical supplies or cleaning floors. It also includes human pose recognition algorithms for detecting fallen patients in real time [24].



Figure 1.13 – Aeolus robot

- 9) Care-O-bot 4 [25] – shown in figure 1.14 – is a modular research platform developed in Germany. It can be configured with zero, one, or two robotic arms depending on the task. It also features spherical joints in the torso and neck that allow a 360° rotation. The navigation is made by integrated laser scanners. Its interface uses abstract facial expressions on a head-mounted screen to avoid the Uncanny Valley effect. The software runs on an open-source ROS platform, making it easy to extend and adapt for research applications [25].



Figure 1.8 – Care-O-bot 4 robot

- 10) HOSPI [26] – shown in figure 1.15 – is a secure transport robot designed for sensitive medical materials. Its fully enclosed casing prevents contamination, and its compartment is locked electronically and can be opened only by ID authorization. It follows pre-programmed sterile route logic and integrates with elevators and automatic doors [26].



Figure 1.9 – HOSPI robot

- 11) Dinsow [27] – shown in figure 1.16 – is a medical assistant robot designed for patient monitoring and social interaction. It includes built-in non-contact IR thermometers and blood pressure monitors, and a 10-inch touchscreen for telepresence consultations. It also uses facial recognition to detect patient distress and send alerts directly to caregiver smartphones [27].

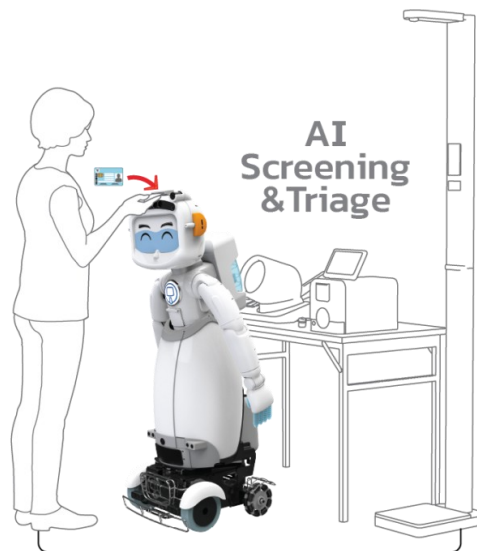


Figure 1.10 – Dinsow robot

The matrix presented in figure 1.17 visualizes each system according to physical capability – such as manipulation, logistics, and autonomous navigation; and social capability – interaction, emotion recognition, and patient communication. The distribution shows us that robots like TUG, Relay, and HOSPI score high on physical capability but have almost no social capability. In the opposite corner, Pepper, Grace, and Stevie have strong social skills but are limited in physical tasks. The hybrid platforms such as Moxi, Care-O-bot 4, and Aeolus, are closer to the cent, but none of them reaches the top-right corner, which represents full integration of both dimensions. This is mainly because these systems were created by improving an existing logistics platform or a social robot, instead of being designed from the beginning as a fully integrated solution. Dinsow is perhaps the most balanced system in terms of combining monitoring with interaction, but its physical manipulation capabilities are limited. The dashed circle marks the target zone that present project aims to occupy.

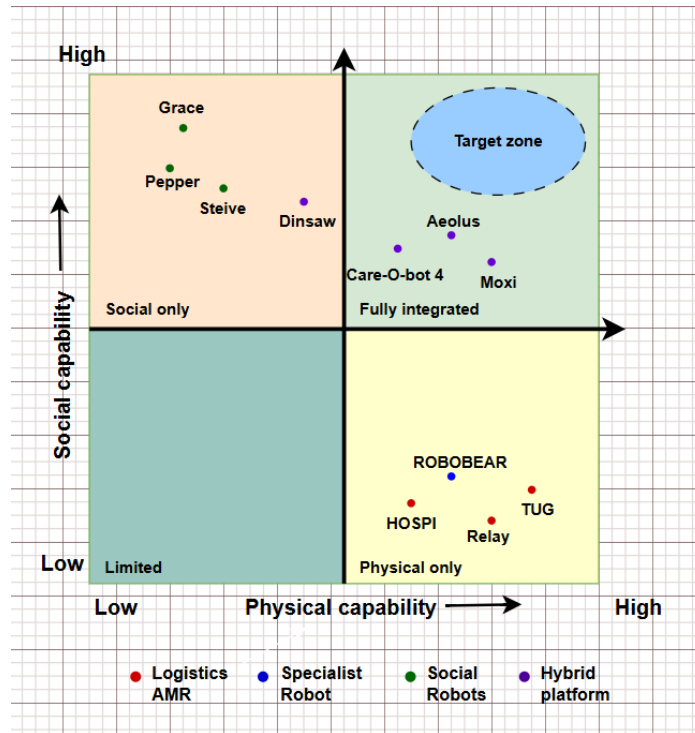


Figure 1.11 – Benchmark analysis: Social vs. Physical capability matrix

What this analysis shows is that no single existing system fully covers all the functional requirements of a hospital assistant robot. Current solutions are either strong in logistics but limited in interaction, or capable of social engagement but unable to perform physical tasks or monitor patients. The closest systems to a complete solution – Grace, Dinsow, and Moxi – each address two or three of these dimensions, but none integrates all of them within a single platform. This gap defines the core motivation of the project: to design a hybrid robotic assistant that combines autonomous navigation, basic manipulation, vital signs monitoring, and social interaction within one system.

PROJECT PLANNING

2.1. Problem to solve

In order to identify the root cause of the problem, the “5 WHY’s” methodology was applied, as illustrated in the figure 2.1:

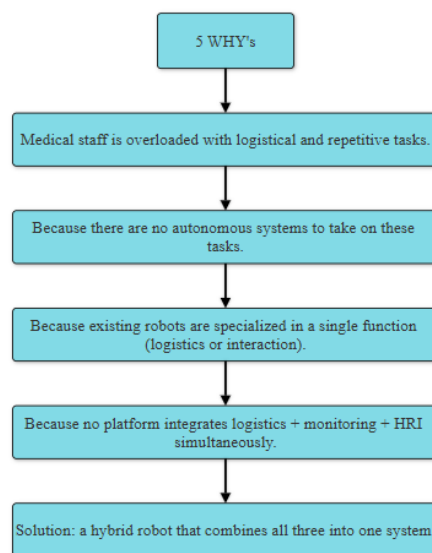


Figure 2.1 - 5 WHY’s methodology

Based on the root cause identified in figure 2.1, the solution tree presented in figure 2.2 maps the three functional pillars of the proposed system – discussed separately in detail below.

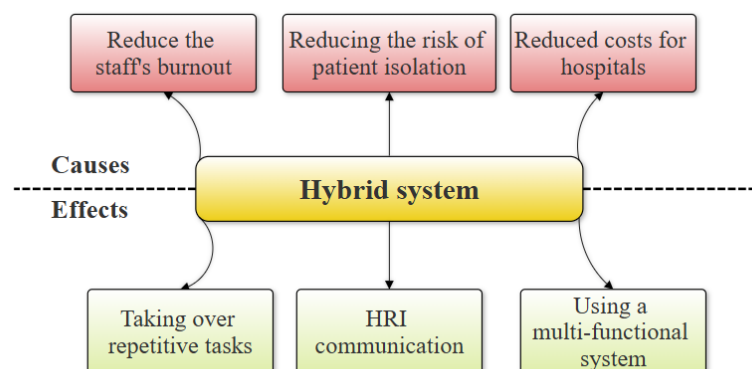


Figure 2.2 – The solution tree

Modern healthcare systems are under increasing pressure. Hospitals worldwide face a growing imbalance between the number of available medical staff and the volume of tasks that need to be

completed daily. As shown in figure 2.3, a large-scale time-and-motion study conducted across 36 hospitals found that nurses spend only 35% of their time on direct patient care, while remaining 65% is split between documentation, supply logistics, care coordination, and other non-clinical activities [16]. Tasks such as delivering supplies and transporting medication between wards do not require clinical expertise and represent a clear opportunity for robotic domain.

This workload imbalance has measurable consequences for nursing staff. As presented in figure 2.4, burnout rates among nurses range from 25% to 43% across European and North American healthcare systems, with workload and time pressure identified as the primary contributing factors [28]. Burnout directly affects the quality of care delivered, increasing the likelihood of medical errors and reducing the level of attention given to individual patients.

Continuous patient monitoring presents a separate but related challenge. A nurse cannot always be present in every room, which means that events such as falls, sudden changes in vital signs, or distress situations can go unnoticed for several minutes. This is particularly problematic in long-term care units, where patients may spend weeks or even more with limited human contact. As shown in figure 2.5, social isolation increases the risk of depression by 29%, slows physical recovery by 22%, and is associated with a 26% increase in mortality risk among hospitalized patients [29].

The problem this project addresses can therefore be summarized as follows: there is no single robotic system currently available that can simultaneously handle hospital logistics, provide continuous monitoring, and maintain meaningful social interaction with patients on one single platform. Existing solutions, as shown in the benchmark analysis of sub-chapter 1.2, solve only one or two of these dimensions at the time. The proposed hybrid robotic assistant is designed to bridge this gap by integrating all three functional areas into one coherent system.

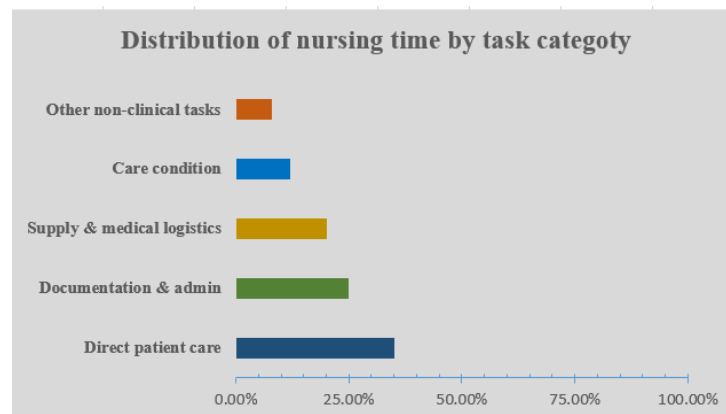


Figure 2.3 – Distribution of nursing time by task category

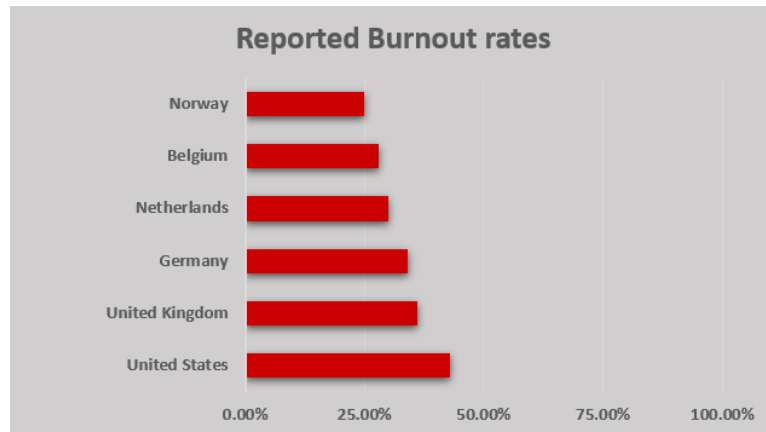


Figure 2.4 – Reported burnout rates among nursing staff by country

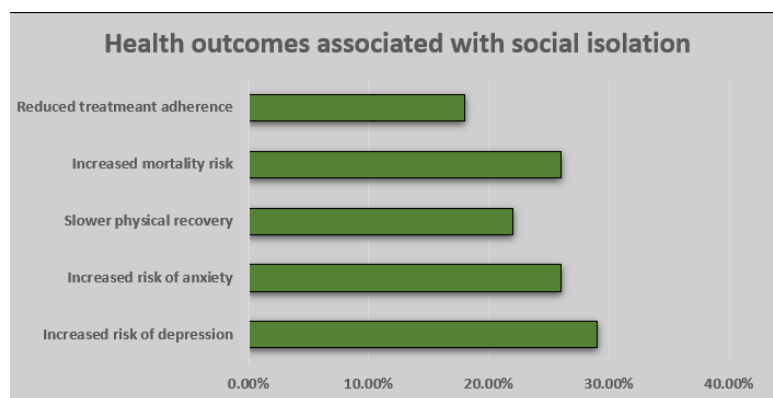


Figure 2.5 – Health outcomes associated with social isolation in hospitalized patients

2.2. Methodology

The development of this robotic assistant follows a New Product Development (NPD) process, adapted to the specific needs of medical robotics engineering. NPD is a structured approach that organizes a project from the first identification of a problem to concept development, detailed design, and final validation [30]. In its original form, NPD is used across many industries to make sure that new products are developed in a logical sequence, with clear goals and documented results. For this project, the commercial phases – such as market testing and product launch, are replaced with equivalents focused on requirements, architecture selection, and design validation.

Within NPD structure, three additional frameworks are used to address specific aspects of the development. Design for Six Sigma (DFSS) provides methodology for turning identified needs into verifiable design requirements. The V-Model organizes the relationship between design decision and their corresponding validation activities. IEEE/ISO/IEC 15288:2023 serves as the normative reference for structuring the overall system life cycle according to international engineering standards.

A Work Breakdown Structure (WBS) has been developed to organize the project into manageable work steps. As shown in figure 2.6, the project is divided into four main pillars: Research, Mechanical, Electrical, and Software. Within this method is ensuring a clear roadmap for the implementation phase.

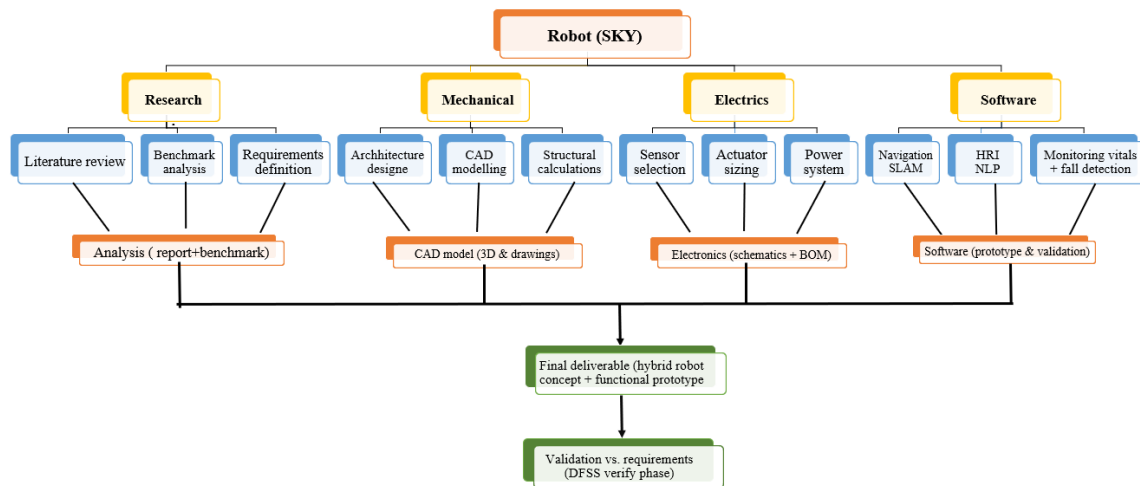


Figure 2.6 – Work Breakdown Structure (WBS)

Each framework works at a different level of the process, and together they form a methodology foundation. The following sub-sections describe each one in detail.

2.2.1. Directions of study

The development of the proposed system follows a sequential path structured around seven key stages, as illustrated in figure 2.6. Each stage produces documented outputs, ensuring full traceability from the initial problem statement to the final validated design.

The process begins with the identification of the clinical problem and stakeholder needs, as presented in sub-chapter 2.1. This stage establishes the functional context of the project by documenting the specific deficiencies in current hospital workflows – staff overload, lack of continuous patient monitoring, and patient isolation – that the proposed system is expected to address. The outputs of this stage are the problem statement and the initial list of functional needs, which serve as the starting point for all subsequent design decisions.

The second stage is the benchmark analysis of existing robotic systems, documented in sub-chapter 1.2. Eleven representative clinical robotic platforms are evaluated against a common set of criteria, covering autonomy level, functional category, deployment context, and key technical features. The results are synthesized in the Social vs. Physical capability matrix presented in figure 1.17, which visualizes the gap between what currently exists and what SKY aims to achieve. This stage represents the starting point in choosing the final system architecture elaborated later.

The third stage is the definition of functional and technical requirements. Based on the gap identified in the benchmark analysis and the stakeholders needs described in section 2.1, a Voice of the Customer (VOC) analysis was performed so structure the main characteristics. The resulting set of requirements is summarized in table 2.1. Each of them is formulated to be measurable, so that it can serve as a verifiable criterion in the final validation stage.

Table 2.1- Voice of the Customer (VOC) set

No	Requirement
1	Non-invasive monitoring of vital functions: temperature, oxygen saturation, pulse.
2	Preliminary triage and history taking assisted by access to medical platforms.
3	Telepresence: if necessary, medical personnel can connect to the robot and speak through it.
4	Detecting patients in need both in the hallways and in the rooms (fall, lack of food, etc.) and alerting them to emergencies via sound alarm or direct call.
5	Companionship and emotional support.
6	Checking the presence or absence of patients' medications.
7	Live video monitoring.
8	Manual remote control via app.
9	Be able to detect and avoid obstacles.
10	To be able to detect and differentiate human emotions and know how to respond to them.
11	To be able to move.
12	To be able to record video and audio in case the patient refuses to cooperate with the robot (to ensure the legal protection of the hospital), without intervening in a brutal way.
13	To inform both the patient and the medical staff of each action they are about to take, awaiting agreement/disagreement.
14	In case the patient does not want to cooperate with the robot, it should enter a withdrawal mode and contact the medical staff.

The fourth stage is the selection of the system architecture. The hybrid platform concept – combining a mobile base with one manipulator arms, a medical sensing suite, and a social interaction module – is selected as the architectural approach that best satisfied the defined requirements. This decision is directly justified by the benchmark matrix and evaluating different possible implementations of the system, which are detailed in table 2.2, by three different types of configurations. The final configuration type being “The hybrid autonomous”.

Table 2.2 - Architectural comparison of robotic configurations

Requirement	High-Tech Autonomous Specialist	Hybrid Autonomous	The Secure Clinical Monitor
R1: Vitals monitoring	FLIR Lepton Thermal camera	FLIR Lepton 3.4	Digital Non-contact Thermometer
R2: Triage Logic	Claude API	Llama 3.2 – Ollama	Infermedica API
R3: Telepresence	MIPI DSI Display	Tablet with VESA Mount	Zoom for Healthcare
R4: Fall Detection	MediaPipe Pose	YOLOv8	MPU-6050 Accelerometer
R5: Emotional Support	Synthesia AI Virtual Avatar	Bark AI (Expressive Voice)	Google Cloud TTS
R6: Medication Management	Vision AI Identification	RFID access	RFID/NFC Access Control
R7: Live Video	OV5647 5MP IR Camera	RealSense + IMX	Hikvision/Dahua IP Camera
R8: Manual Control	Custom Mobile Application	.NET MAUI app	MQTT Protocol (Secure)
R9: Obstacle Avoidance	ToF VL53L1X Sensors	LiDAR + SLAM (SICK TiM571)	HC-SR04 ultrasonic
R10: Emotion AI	Vision AI (Aggressive Motion)	DeepFace	Keyword STT Identification

R11: Mobility System	DC Motors with Encoders	Mecanum wheels	Differential Drive Chassis
R12: Legal Protection	AWS S3 Cloud Storage	TTS Legal Disclaimer	Encrypted Local SSD
R13: Interaction Info	Automated Action Planner	Gesture Detection (Vision AI)	Touchscreen Agree/Disagree
R14: Safety Mode	Withdrawal + Mechanical Lock	Pause & wait + LED Signal	Silent Alert + Emergency Stop

The fifth stage is the mechanical design phase, which includes the structural dimensioning of the mobile base, the manipulator arms configuration, and the sensor mounting geometry. The output of this stage are the CAD model and the drawings, which provide the manufacturing reference for the physical prototype.

The sixth stage is the electronic and sensor integration phase. This includes the LiDAR scanner, depth cameras, infrared thermometer, pulse oximeter interface, and motor drivers. The output is the electronic schematic and the bill of materials, which define the hardware configuration of the system.

The last one is the development of navigation and interaction algorithms, which enable the robot to operate autonomously in a dynamic hospital environment. This includes the SLAM-based navigation stack for localization and path planning, the fall detection module based on body pose analysis, the emotion recognition pipeline using computer vision, and the NLP-based conversational interface powered by a local language model. The outputs are the software modules and the validation results, which are evaluated based on the VOC defined in the stage three.

2.2.2. Used theories, methods and instruments

The methodological foundation combines several engineering theories, design methodologies, and development frameworks. These approaches support the structured design of the system architecture and ensure that the proposed solution addresses the needs identified during the Voice of the Customer (VOC) analysis.

Design for Six Sigma (DFSS – DMADV) – presented in figure 2.7 – is the primary design methodology. Unlike traditional Six Sigma, which improves existing processes, DFSS is built for developing new systems, making sure that quality and reliability are part of the design from the start [32]. It follows the DMADV cycle – Define, Measure, Analyze, Design, Verify. In the Define phase, the clinical problem is stated, and the objectives are set, as in section 2.1. In the Measure phase, the performance of existing models is evaluated through the benchmark analysis, giving a baseline for comparison. The Analyze phase takes the capability gap from the benchmark matrix and turns it into a structured set of design requirements. The Design phase covers the mechanical, electronic, and algorithmic development. Finally, the Verify phase checks whether the final design actually meets the requirements.

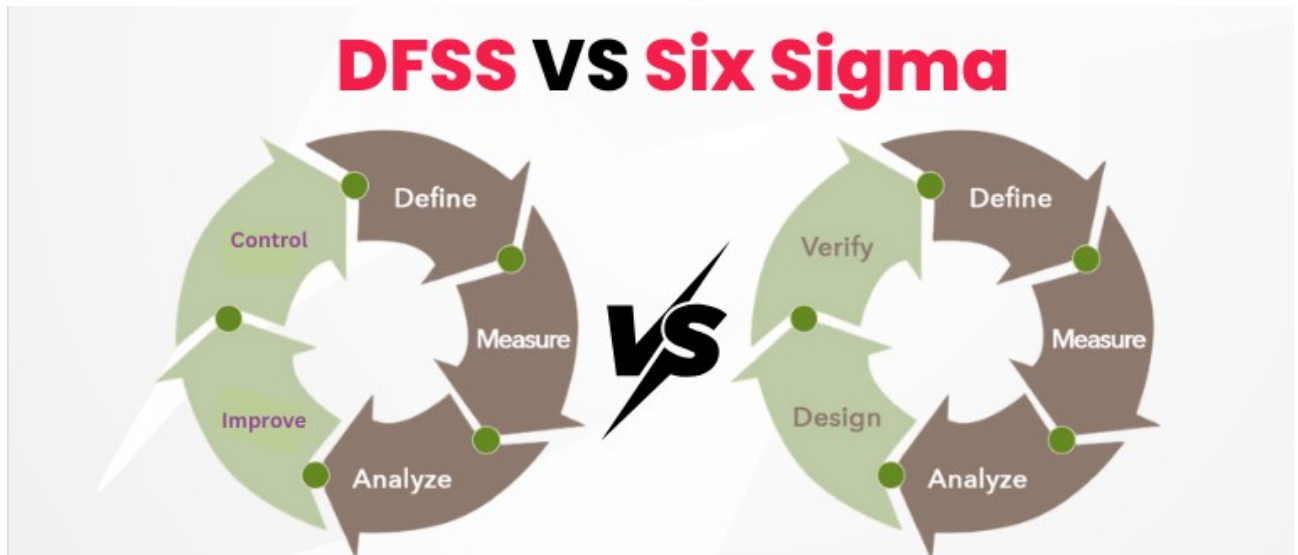


Figure 2.7 – DFSS vs. Six Sigma

The V-Model - presented in figure 2.8 – is used alongside DFSS to organize validation activities. In the V-Model, each design stage on the left side of the V has a matching verification activity on the right side [33]. System requirements are verified through system-level testing. Architectural decisions are validated through integration testing. Individual components are checked through unit-level tests. This structure makes sure that every design decision has a clear plan for how it will be verified.

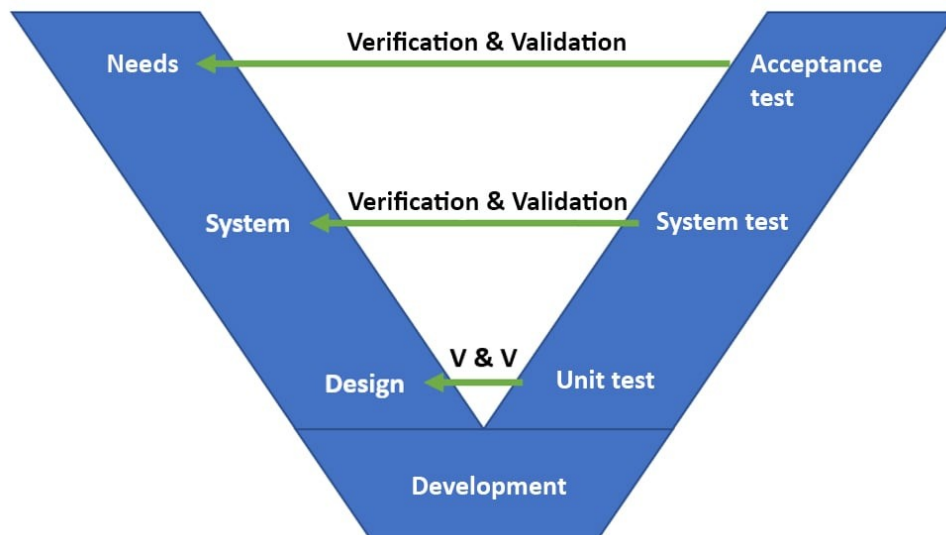


Figure 2.8 – V-model representation

IEEE/ISO/IEC 15288:2023 provides the normative reference for organizing the system life cycle. This international standard defines the processes needed for the conception, development, and verification of complex engineered systems [31]. For this project, it is used to ensure that development activities follow recognized engineering practice. This is particularly relevant for a medical robotic system, where traceability of design decision is a key factor in any future regulatory evaluation.

New Product Development (NPD) structures the project as a sequence of phases, ensuring that each stage produces documented outputs that feed into the next [30]. It is applied in an engineering-oriented adaptation (without the commercial phase), focusing on the systematic progression from problem definition to validated design solution.

System Engineering Decomposition is applied to break down the overall system into manageable subsystems. Following the IEEE/ISO/IEC 15288:2023 framework, the robot is decomposed into four

functional subsystems: the mechanical structure, the actuation and power system, the sensing and processing unit, and the software and interaction layer. This methodology ensures that each subsystem can be design, tested, and validated independently before the final integration, reducing the risk of conflicts and a proper function overall.

Human-Robot Interaction (HRI) design principles guide the development of the social and communicative functions. This ensures that the robot's intentions must be readable by non-expert users and social presence – the robot must convey a sense of engagement without triggering discomfort [13]. In practice, this translates into the use of facial expressions recognition to adapt the robot's responses to the user's emotional state, natural language processing for context-aware verbal communication, and a morphological design constrained by the Uncanny Valley boundary discussed in sub-chapter 1.1. The HRI framework also defines the priority logic between social interaction and logistical tasks, ensuring that safety-critical events always take precedence.

Concept selection is performed using a structured decision matrix, in which each configuration is evaluated against the functional requirements defined in stage three. The evaluation criteria include: coverage of functional requirements, technical feasibility within available hardware constraints, scalability for future development, and compliance with ISO 13482:2014 safety requirements. The hybrid platform concept – mobile base with one manipulator arm, sensing suite, and interaction module – is the optimal configuration, achieving the highest score across all criteria.

Engineering tools and technologies used throughout the development include Fusion 360 for 3D mechanical modelling and structural analysis, Python as the primary programming language for all software modules, and the following open-source libraries: YOLOv8 for object detection, Vosk for offline speech recognition, DeepFace for facial expression and emotion analysis, and Ollama with the Llama 3.2 language model for natural language interaction. The onboard NVIDIA Jetson Orin NX runs all high-level computation and issues motion commands through ROS 2, while low-level motor control is executed by dedicated motor controllers – integrated within the Kinova Gen3 actuators for the arm and Maxon drives for the mobile base. Patient interaction is provided through two Android tablets connected to the onboard system over the local network.

Safety-oriented design principles are integrated throughout the development process rather than applied as a post-design correction. The primary reference is ISO 13482:2014, which defines safety requirements for personal care robots operating in proximity to humans. At the mechanical level, this is applied through collision avoidance via redundant LiDAR and ultrasonic sensing, force-limited actuation in the manipulator arms, and emergency stop button. At the software level, safety is implemented through a dedicated Safety Mode – as described in the context of Asimov's Second Law in sub-chapter 1.1 – which prioritizes patient consent, logs all refusals for legal protection, and initiates a controlled withdrawal in case of conflict.

2.2.3. Technologies, experiments, tests

The proposed system relies on a combination of mechanical design software, programming environments, AI frameworks, and embedded hardware, each addressing a specific functional layer. The selection criteria prioritize offline operation, real-time performance, and suitability for a clinical environment where patient-data confidentiality and independence from internet connectivity are essential. For each technology, a corresponding validation procedure is planned.

Fusion 360 is used for the 3D mechanical modelling, assembly, and execution drawings. The structural verification of the critical components is performed analytically in Mathcad: each member is checked against the yield strength of the selected aluminium alloy under the worst-case load, the resulting safety factors must remain above the adopted threshold, and the tipping-stability calculation must confirm that the centre of mass stays within the support polygon for all defined load cases.

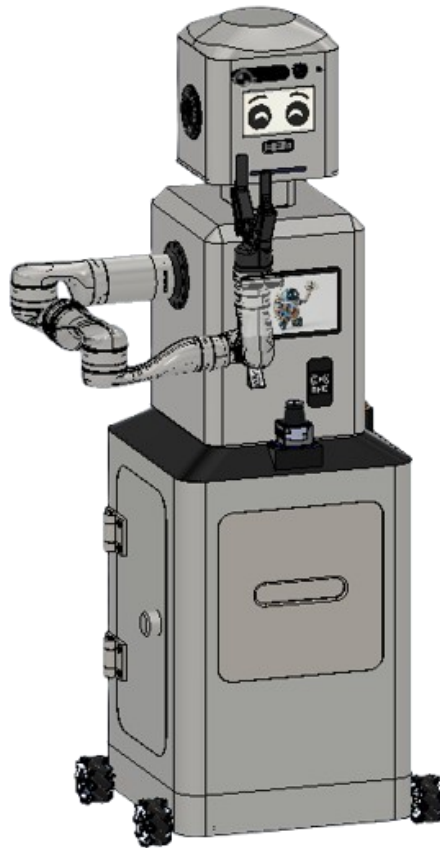
Python integrates the high-level software into a single real-time pipeline running on the onboard computer, combining offline speech recognition, computer vision, emotion analysis, and natural-language response generation, with a WebSocket connection pushing data to the tablet interface and

a serial link to the motion controller. The AI stack comprises four offline frameworks, each with its own validation experiment: Vosk for speech-to-text, validated by word-error rate and by reliable detection of the safety-critical emergency keywords; YOLOv8 for person, object, and fall detection, validated by precision and recall by the true/false-positive rates of the fall logic on staged sequences; DeepFace for facial-expression analysis, validated against labelled samples across the seven emotion categories (happy, sad, angry, neutral, fear, surprise, disgust); and Ollama, running a local Llama model, validated for persona consistency and response latency. Each module is first tested in isolation, then within the full pipeline for end-to-end latency and stability.

Manipulation is provided by a commercial Kinova Gen3 7-DOF arm [50], validated against its specified payload, reach, and repeatability, complemented by a reachability study confirming that the required task position lies within its workspace. Interaction is handled by two Android tablets – one mounted on the head for rendering the robot’s facial expressions, and one mounted on the chest as the patient-facing interface, displaying vital signs and battery status and allowing the patient to interact with the robot by typing or selecting options. Physiological and environmental sensing relies on a FLIR Lepton 3.5 LWIR camera [58], an OB1203 pulse/SpO₂ sensor, and a LiDAR unit; each is validated against a reference instrument, with the error required to stay within the tolerance acceptable for an assistive screening role. The low-level layer handles locomotion, the medication drawer, and the safety subsystem, where the obstacle-detection logic must trigger a stop within the defined threshold and the emergency-stop chain must override all motion.

The validation is planned in two stages. First, each module – structural, perception, dialogue, manipulation, sensing, and safety – is verified in isolation against its own acceptance criterion, using analytical, model-based, and software methods that do not require the complete physical robot. Second, the integration loop is evaluated end to end, from voice or visual input through decision-making to the verbal and physical response, confirming that the subsystems operate correctly together and within acceptable timing.

Part II



Contributions of the project theme

3. Design and Implementation

3.1 Design constraints, applicable standards and regulatory framework

Because the platform is intended to operate in direct proximity to patients and alongside medical staff, its design was constrained from the outset by safety standards, regulatory requirements and ethical principles, rather than treating these as post-design verification step. This section consolidates the rules that governed the development and shows, for each, the concrete design decision it imposed.

ISO 13482:2014 – safety personal care robots. This standard is the primary normative reference for the project, as the robot falls within the category of personal care robots operating in shared human-robot spaces. Its requirements propagate directly into the quantitative design values of table 3.1. The kinematic envelope was bounded below human-risk thresholds: the base velocity was capped at 1.5 m/s so as not to exceed pedestrian walking speed, the base acceleration at 0.5 m/s^2 to remain below the human discomfort threshold, and the arm angular acceleration and velocity were limited to 1.5 rad/s^2 and 1.0 rad/s respectively – values that, in turn, drove the actuator-sizing calculations throughout sub-chapter 3.4. The standard further imposed force-limited actuation in the manipulator, realised through per-actuator torque sensing. Redundant collision-avoidance sensing, satisfied by combining the LiDAR scanner with the depth camera; a hardware emergency-stop chain that overrides all motion; and a structural safety factor of 2.0 on the base, against the 1.5 applied to the actuators, reflecting the elevated human-risk zone. Stability is treated as a safety requirement: the tipping analysis was required to keep the centre of mass within the support polygon for all load cases with a safety factor above 1.5, and the safety-supervisor enforces a mechanical interlock that retracts the arm and locks the drawer before any base motion is permitted.

IEC 60601-1 – medical electrical equipment. As the platform carries electrical equipment that operates in the patient environment, IEC 60601-1 governs its electrical safety. The most visible consequence is the choice of self-locking lead-screw actuation for the gripper, so that a grasped object cannot be released in the event of a power failure. The standard also shaped the treatment of any patient-contacting electronics – in particular the contact pulse/ SpO_2 sensor – with respect to electrical isolation, leakage current and single-fault safety, and reinforced the segregated, fuse power architecture adopted for the platform.

Medical-data confidentiality. Patient-data protection requirements drove one of the defining architectural decisions of the system: an offline-first AI stack in which all speech, vision, emotion and biometric processing run locally on the onboard computer, so that no patient audio, video or biometric data ever leaves the robot. A cloud language model was rejected for this reason in favour of a locally hosted one. The requirement to record audio and video for the hospital's legal protection (requirement R12) was balanced against this principle by keeping such data on encrypted local storage rather than transmitting it externally.

IEEE/ISO/IEC 15288:2023 – system life-cycled processes. This standard provided the framework for organising the development as a traceable sequence of life-cycle processes, decomposing the robot into four verifiable subsystems. Traceability of design decisions is not merely

good practice here: it is a precondition for any future regulatory evaluation of a medical robotic system, and the V-Model verification structure was adopted specifically to preserve it.

Ethical and human-robot-interaction constraints. Beyond formal standards, the design was bound by ethical principles translated into concrete behavioural rules. The ethical framework derived from Asimov's law in sub-chapter 1.1 – in particular the duty not to allow harm through inaction – mandated the redundant fall-detection and alerting logic. Respect for patient autonomy and dignity imposed a consent-first interaction model: the robot announces each intended action and awaits the patient's agreement before proceeding (requirement R13), enters a non-confrontational withdrawal mode if the patient refuses to cooperate (requirement R14), and relies on non-invasive monitoring that requires no action from the patient. Morphologically, the Uncanny Valley constraint discussed in sub-chapter 1.1 dictated a humanoid upper body mounted on a non-humanoid omnidirectional base, so that the robot remains clearly identifiable as a machine and reassuring rather than unsettling.

Operational constraints for medical staff. Acceptance by medical staff imposed its own design rules, derived from the user analysis. The robot had to operate with minimal cognitive overhead and without constant supervision, since a system requiring frequent manual intervention would increase rather than reduce workload. It also had to provide a clear, audible alarm path for emergencies; and access to stored medication had to be secure and traceable, which is the reason the drawer is gated by an RFID-secured lock rather than left open. Together, these constraints ensured that the platform reduces staff burden, the founding objective of this project, rather than adding to it.

Romanian healthcare-construction normative NP 015-2022 – physical-integration constraints. Because the robot must circulate through the same routes as patients, staff and beds, its external dimensions were bounded by the design rules for hospital construction, governed in Romania by the normative NP 015-2022 (*Normativ pentru construcții spitalicești*, approved by Order 2412/2022, published in *Monitorul Oficial* no. 919 bis/2022, replacing NP 015-1997). This normative requires the main hospital circulation routes to be dimensioned according to the means of transport that must pass through them – explicitly a stretcher, a wheelchair and a mobile hospital bed – and that the patient's circulation be continuous and equally accessible on foot, by wheelchair, by stretcher and by bed. Taking the mobile hospital bed (approximately 0.9-1.0 m wide) as the reference gauge against which doors and corridors are sized, the 500x500 mm base was dimensioned well within this envelope, so that the robot moves on the same routes as the staff and beds without imposing any constraint on hospital circulation. Accessibility-related clearances – door clear width and wheelchair maneuvering radius – follow the normative NP 051-2012 on the adaption of civil buildings to the needs of person with disabilities, to which NP 015-2022 refers, and the omnidirectional mecanum drive was selected specifically to achieve a near-zero turning radius within the constrained maneuvering space of ward intersections.

Colour and visual-signalling choices. While no standard prescribes the body colour of a service robot, the visual design was treated as a non-trivial acceptance factor. A neutral, light colour scheme with a cold accent was adopted to convey a calm, hygienic and non-threatening presence consistent with the clinical environment and with the Uncanny Valley considerations of sub-chapter 1.1, deliberately avoiding aggressive or alarm-associated colours on the body. Colour is, however, used functionally and per convention where safety is concerned: the emergency-stop actuator follows the red-actuator-on-yellow-background convention of ISO 13850, and the animated expression display uses a consistent colour code to communicate the robot's internal state to patients and staff.

3.2 Sources of Inspiration and Design References

The design of the SKY did not emerge from a blank slate. It was informed by a systematic analysis of eleven existing clinical robotic systems, presented in sub-chapter 1.2.

Among the analysed systems, several provided direct design references for specific functional modules. Moxi, developed by Diligent Robotics, demonstrated that a hybrid morphology – combining a mobile base with a single manipulator arm – is viable in a real hospital environment and accepted by clinical staff. This, directly informed the design of SKY's storage compartment and navigation architecture. Grace, developed by Hanson Robotics, established a reference for the social interaction layer – specifically the use of bio-inspired facial expression and NLP optimized for clinical terminology. Stevie, developed by Trinity College Dublin, provided the reference for the fall detection module and the dual-display interaction concept. Dinsow contributed the benchmark for integrated vital signs monitoring within a socially capable platform.

Beyond individual systems, the Social vs. Physical capability matrix presented in figure 1.17 revealed a structural gap in the field: no existing commercially deployed system occupies the top-right quadrant, where both capabilities are presented at a high level simultaneously. This gap reflects the fact that existing hybrid platforms were developed by expanding either a logistics robot or a social one, rather than being designed as integrated solutions from the ground up. SKY is designed to fill this gap, with architecture decisions driven by the requirement to cover all four functional dimensions: autonomous navigation, physical manipulation, medical sensing, and human-robot interaction. The morphological concept – a humanoid upper body mounted on a non-humanoid omnidirectional base – inspired from Care-O-bot 4 and Stevie, both of which deliberately avoid full humanoid morphology to remain clearly identifiable as machines, consistent with the Uncanny Valley design constraint discussed in sub-chapter 1.1. The arm configuration references the Robotiq 2F-140 gripper [51] benchmark for payload, while the head assembly references the Intel RealSense D435i and Sony IMX519 sensor suite used in research-grade HRI platforms.

3.3 User Analysis and Clinical Context

There are three primary user categories that interact with the system in a clinical environment: patients, medical staff, and hospital administrators. Each category has distinct needs, interaction patterns, and acceptance thresholds that directly influenced the design decisions documented below.

Patients are the primary interaction partners of the robot. In a hospital setting, patients range from kids to elderly, post-operative, or long-term care patients with limited mobility and potentially reduced tolerance for technological interaction. The key requirements emerging from this user group are: non-invasive monitoring that does not require the patient to perform any action, clear and empathetic verbal communication, the ability to call for help at any moment, and a physical appearance that is reassuring rather than intimidating. The Uncanny Valley constraint is particularly relevant here – a robot that is too human-like risks triggering discomfort, while a robot that is too mechanical risks being perceived as cold and unapproachable.

Medical staff – nurses, doctors, and paramedical personnel – interact with the robot primarily through the .NET MAUI control application and through the RFID-secured storage compartment. Their key requirements are reliability, predictability, and minimal cognitive overhead. A robot that requires constant supervision of frequent manual intervention would increase rather than decrease workload. For this user group, the most critical design requirements are: autonomous navigation that does not require staff guidance, a clear and audible alarm system for emergencies, secure and traceable access to stored medications, and a withdrawal mode that removes the robot from a situation without creating additional problems. The requirement for the robot to announce its actions and await

agreement or disagreement before proceeding – requirement R13 from table 2.1 – emerged directly from staff usability considerations.

Hospital administrators represent the institutional user of the system. Their requirements are centred on operational efficiency, regulatory compliance, and data traceability. The offline-first architecture – with all AI processing running locally on the Jetson Orin NX in the final configuration – addresses data privacy requirements under medical data regulations.

3.4 Design and calculation of the main mechanical components

The mechanical structure of the robot platform is organized into four main assemblies: the head, the neck, the upper-body manipulator interface, and the mobile base. Each assembly was dimensioned based on the mass distribution of its components, the kinematic requirements of its degree of freedom, and the structural constraints imposed by the selected materials and manufacturing methods. Unless otherwise stated, all load-bearing structural components are made from wrought aluminium alloy EN AW-6061 T6 ($E = 69.000 \text{ MPa}$, $\sigma_{\text{yield}} = 241 \text{ MPa}$, $\rho = 2700 \text{ kg/m}^3$), selected for its availability as standard extruded profile. The single exception is the arm-mounting flange (detailed in sub-chapter 3.6), machined from EN AW-7075 T651 because the amplification of any angular error over the arm reach demands the higher specific stiffness of that alloy.

The global design assumptions applied throughout the structural and drive calculations are summarised in table 3.1. The kinematic limits drive from ISO 13482:2014 and Siciliano et al (2009, chapter 8); the rolling-resistance and bearing-friction coefficients from Shigley's Mechanical Engineering Design (table 2-17) and the SKF General Catalogue 6000; and the safety factors from standard mechanical-design practice for service equipment operating in shared human spaces.

Table 3.1 – Global design values

Parameter	Value	Source / justification
Arm angular acceleration a_{max}	1.5 rad/s^2	ISO 13482:2014; Siciliano et al. (2009), chapter 8
Arm angular velocity v_{max}	1.0 rad/s	ISO 13482:2014; Siciliano et al. (2009)
Base velocity v_{max}	1.5 m/s	$\leq$ pedestrian walking speed (ISO 13482; Thrun et al., 2005)
Base acceleration a_{max}	0.5 m/s^2	Below human discomfort threshold
Rolling resistance μ_r	0.02	Hard smooth floor – Shigley's, table 2-17
Bearing friction μ	0.002	Precision ball bearing – SKF General Catalogue 6000
Mechanical efficiency η	0.85	Planetary/harmonic transmission datasheet + margin
Safety factor SF (actuators)	1.5	Dynamic load uncertainty, friction, wear margin
Safety factor SF (base)	2.0	Shock loading, human-risk zone

3.4.1. Head Assembly

The head assembly integrates all primary sensing and interaction components of the robot. Its complete bill of mass is presented in table 3.2:

Table 3.2 – Head Assembly bill of mass

Component	Mass (kg)
Intel RealSense D435i (RGB-D camera)	0.072
Sony IMX519 (emotion camera)	0.055
Arducam IMX477 (wide-angle camera)	0.045
ReSpeaker 4-mic linear array	0.015
Dayton Audio CE Series speakers	0.180
LCD display	0.290
HDMI-to-MIPI driver board	0.025
NVIDIA Jetson Orin NX	0.100
Seeed reComputer J4012 carrier board	0.085
Heatsink + 40mm fan	0.180
DC-DC USB converter	0.045
ABS plastic front casing	0.280
2mm aluminum rear casing	0.320
3mm aluminum mounting bracket	0.150
Cables and connectors	~0.100
M2/M3 screws	~0.050
FLIR Lepton 3.5	0.015
Total	2.000

3.4.1.1. Thermal Analysis of the Head Assembly

The NVIDIA Jetson Orin NX [54] is the dominant heat source inside the sealed head shell. Operating in eco mode at a thermal design power of 10W, it must be cooled by forced convection without exceeding the 85°C maximum junction temperature, assuming a 40°C worst-case ambient (permissible rise $\Delta T = 45^\circ\text{C}$). The required airflow to dissipate the thermal load, the effective intake/exhaust vent areas (five 30x8mm intake slots and one 40x40mm exhaust grille), the resulting through-vent velocity and the static-pressure requirement are derived below.

Power:

$$P_u = 10W \quad (3.1)$$

Where P_u is the eco-power of the Jetson Orin NX.

Require power:

$$P_{u_{req}} = P_u * SF = 10 * 1.5 = 15W \quad (3.2)$$

Where SF is the safety factor.

Ambient temperature:

$$T_{\text{ambient}} = 40^\circ\text{C} \quad (3.3)$$

Maximum temperature:

$$T_{max}=85^{\circ}C \quad (3.4)$$

Permissible temperature:

$$\Delta T=45^{\circ}C \quad (3.5)$$

Density of the air at 40°C:

$$\rho_{air}=1.2 \frac{kg}{m^3} \quad (3.6)$$

Specific air's temperature:

$$Cp_{air}=1005 \frac{J}{kg \cdot K} \quad (3.7)$$

Required airflow:

$$Q = \frac{Pu_{req}}{\rho_{air} * Cp_{air} * \Delta T} = \frac{15}{1.2 * 1005 * 45} = 0.000276 \frac{m^3}{s} \quad (3.8)$$

Vent area calculation:

$$Cd_{slot_{coeff}}=0.6 \quad (3.9)$$

$$Cd_{grille_{coeff}}=0.5 \quad (3.10)$$

Theoretical area of slots:

$$A_{slots_{th}}=5 * L_s * l_s=5 * 30mm * 8mm=1200 mm^2 \quad (3.11)$$

Where L_s is the length of the slot and l_s is the width of the slot.

Theoretical area of grille:

$$A_{grille_{th}}=L_g * l_g=40mm * 40mm=1600mm^2 \quad (3.12)$$

Where L_g is the length of the grille and l_g is the width of the grille.

$$A_{slots_{eff}}=Cd_{slot_{coeff}} * A_{slots_{th}}=0.6 * 1200mm^2=720mm^2 \quad (3.13)$$

Effective area of grille:

$$A_{grille_{eff}}=Cd_{grille_{coeff}} * A_{grille_{th}}=0.5 * 1600mm^2=800mm^2 \quad (3.14)$$

Effective area ratio:

$$R = \frac{A_{slots_{eff}}}{A_{grille_{eff}}} = \frac{720}{800} * 100 = 90\% \quad (3.15)$$

Condition is checked because $R \geq 80\%$.

The maximum airflow velocity through vents:

$$v_{slots} = \frac{Q}{A_{slots_{eff}}} = \frac{0.000276 m^3/s}{0.00072 m^2} = 0.384 \frac{m}{s} \quad (3.16)$$

$$v_{grille} = \frac{Q}{A_{grille_{eff}}} = \frac{0.000276 m^3/s}{0.00080 m^2} = 0.345 \frac{m}{s} \quad (3.17)$$

Condition is checked because $\begin{cases} v_{slots} \leq 2 \frac{m}{s} \\ v_{grille} \leq 2 \frac{m}{s} \end{cases}$

The resulting static pressure requirement:

$$\Delta P = \frac{1}{2} * \rho_{air} * v_{slots}^2 = \frac{1}{2} * 1.2 \frac{kg}{m^3} * 0.384^2 \frac{m^2}{s^2} = 0.088 Pa \quad (3.18)$$

A 40mm fan, rated for 10-50 Pa, satisfies this requirement while operating at only ~30% PWM, leaving substantial thermal headroom.

3.4.2. Neck structure

The neck is a hollow square aluminium profile of 126x126mm cross-section with a 2mm wall thickness and a length of 70mm. It connects the head assembly to the upper torso and must withstand the combined gravitational, inertial and shear forces generated during head pan motion. The design load case combines gravitational force of 2.0kg head, the centripetal and inertial forces produced by the pan acceleration and velocity ($\alpha_{pan} = 3 rad/s^2$, $\omega_{pan} = 1.5 rad/s$), and the resulting bending, axial and shear stresses, which are combined into an equivalent von Mises stress.

Density of Aluminium:

$$\rho_{Al} = 0.0027 \frac{g}{mm^3} \quad (3.19)$$

Gravitational force:

$$F_{grav} = M_{total_{head}} * g = 2.0 kg * 9.81 \frac{m}{s^2} = 19.61 N \quad (3.20)$$

Where g is the gravitational acceleration.

Inertial moment:

$$M_{inert_{pan}} = \alpha_{pan} * I_{pan} = 0.037 N*m \quad (3.21)$$

Where α_{pan} is the acceleration of the pan motor and I_{pan} is the total mass moment of inertia of the head assembly.

Centripetal, inertial and total forces:

$$F_{centrip} = M_{total_{head}} * \omega_{pan}^2 * L_{neck} = 2.0 kg * 1.5^2 \frac{rad^2}{s^2} * 70 mm = 0.315 N \quad (3.22)$$

Where ω_{pan} is the angular velocity and L_{neck} is the height of the neck.

$$F_{inert} = M_{total_{head}} * \alpha_{pan} * L_{gat} = 2.0kg * 3 \frac{rad}{s^2} * 70mm = 0.42N \quad (3.23)$$

$$F_{total} = \sqrt{F_{centrip}^2 + F_{inert}^2} = \sqrt{(0.315^2 N^2 + 0.42^2 N^2)} = 0.525N \quad (3.24)$$

The section modulus and cross-sectional area of the neck profile:

$$W = \left(L * \frac{L^2}{6} \right) - \left[(L - 2 * thickness) * \frac{(L - 2 * thickness)^2}{6} \right] \quad (3.25)$$

$$= \left(126mm * \frac{126^2 mm^2}{6} \right) - \left(122mm * \frac{122^2 mm^2}{6} \right) = 30754.667 mm^3$$

Where L is the length of the neck and thickness represents the neck's wall thickness.

$$A_{section} = L^2 - (L - 2 * thickness)^2 = 126^2 mm^2 - 122^2 mm^2 = 992 mm^2 \quad (3.26)$$

The bending, axial and shear stresses:

$$\sigma_{bend} = \frac{M_{inert_{pan}}}{W} = \frac{0.037 N*m}{30754.667 mm^3} = 0.001 MPa \quad (3.27)$$

$$\sigma_{axial} = \frac{F_{grav}}{A_{section}} = \frac{19.61 N}{992 mm^2} = 0.02 MPa \quad (3.28)$$

$$\tau_{shear} = \frac{F_{total}}{A_{section}} = \frac{0.525 N}{992 mm^2} = 0.0005 MPa \quad (3.29)$$

The von Mises equivalent stress:

$$\sigma_{VM} = \sqrt{\sigma_{bend}^2 + \sigma_{axial}^2 + 3 * \tau_{shear}^2} = \sqrt{0.001^2 MPa + 0.02^2 MPa + 3 * 0.0005^2 MPa} = 0.02 MPa \quad (3.30)$$

Mass and volume:

$$V_{neck} = A_{section} * L_{neck} = 992 mm^2 * 70mm = 69440 mm^3 \quad (3.31)$$

$$m_{neck} = V_{neck} * \rho_{Al} = 69440 mm^3 * 0.0027 \frac{g}{mm^3} = 0.187kg \quad (3.32)$$

Structure profile checking:

$$SF_{neck} = \frac{\sigma_{yield}}{\sigma_{VM}} = \frac{241}{0.02} = 12050 (times) \quad (3.33)$$

The resulting von Mises stress is only 0.02 MPa, giving a structural safety factor of a several thousand relative to the 241 MPa yield strength. The neck is therefore not a strength-critical element; its cross-section is governed by the need to house the pan bearing and the cable pass-through rather than by stress.

3.4.3. Manipulator selection – Make-vs-buy concept evaluation

The manipulator is the most kinematically demanding subsystem of SKY, this is the reason why two concepts were developed and compared.

Concept A – in-house manipulator. A fully custom 6-DOF arm was dimensioned from the end-effector inward, using three aluminium links (arm = 300mm, forearm = 250mm, wrist = 100mm) and a mixed actuator set (Dynamixel PH54-200-S500-R for shoulder pitch and elbow, Harmonic Drive FHA-32C for shoulder roll, Dynamixel MX-28 for wrist roll) carrying a Robotiq 2F-140 gripper. The complete sizing – link bending, deflection and buckling checks, joint-torque, inertia and bearing-friction calculations, and actuator selection for every DOF – is documented in appendix A. This concept demonstrates that the required performance is achievable with an in-house design and is retained as the future-development path.

Concept B – commercial manipulator (selected). For the present model, a Kinova Gen3 7-DOF manipulator was selected. Gen3 is the modern successor of the arm class used on the reference robot Moxi and is superior on the criteria that matter the most: it integrates torque sensors in every actuator with 1 kHz closed-loop control, offers kinematic redundancy (7 DOF avoids singularities), provides a native ROS 2/ Kortex driver that connects directly to the AI stack, couples natively to the Robotiq 2F-140 gripper, and consumes only ~36W. The key data of the manipulator is presented below, in table 3.3.

Table 3.3 – Key data of Kinova Gen3

Parameter	Value
Mass	8.2 kg
Continuous payload	4.0 kg
Maximum reach	0.902 m
Nominal power	≈ 36 W (24 VDC)
End-effector	Robotiq 2F-140 (≈ 1.0 kg, 140 mm, 125 N)
Control	Integrated torque sensing, 1 kHz loop, ROS 2 / Kortex

The make-vs-buy outcome propagates into the downstream structural calculations as follows: the upper-body load used for the torso and base verifications is the realised arm – 8.2 kg (arm) and 1 kg gripper, plus 4 kg payload in the worst case – acting at the Gen3 maximum reach of 0.902 m. This is lighter than the custom arm (~10.2 kg), so the structural and stability margins for the below sub-chapters are conservative with respect to it.

3.4.3.1. Make vs. buy cost analysis of the manipulator

The manipulator is the single most expensive subsystem of the platform, which is why the make vs. buy decision is examined here also on economic grounds. The comparison is restricted to the arm subsystem alone: the Robotiq 2F-140 gripper, the onboard computer and the sensor suite are common on both concepts and therefore cancel out of the differential.

The bill of materials of the in-house arm (Concept A) is dominated by its actuator set, as shown in table 3.4.

Table 3.4 – Indicative bill of materials for Concept A (parts only)

Item	Quantity	Unit cost	Subtotal
Harmonic Drive SDH-32-100-2UH (shoulder pitch and abduction)	2	~ €1800-2500	~ €3600-5000
Harmonic Drive SDH-17-100-2UH (wrist pitch)	1	~ €1200-1800	~ €1200-1800
Dynamixel PH54-200-S500-R (elbow)	1	~ €1300	~ €1 300
Dynamixel MX-28 (wrist roll)	1	~ €250	~ €250
Servo drives / controllers for the Harmonic Drive units	3	~ €200–500	~ €600–1 500
Machined EN AW-6061 T6 links + brackets (CNC)	-	-	~ €400–800
Bearings (SKF 6008 / 6004 / 6002 / 61802) + custom bushing	-	-	~ €100–200
Fasteners, cabling, connectors	-	-	~ €100
TOTAL			~ €7500–10000

The commercial alternative (Concept B), a Kinova Gen3 7-DOF arm supplied with an integrated controller, pre-actuator torque sensing and a native ROS 2 / Kortex driver, is estimated at ~ €25000–30000.

On parts cost alone, the in-house arm is therefore roughly three to four times cheaper. However, that comparison is misleading on its own, because the two options differ fundamentally in what the price includes. This comparison must weigh the recurring (per-unit) cost against the non-recurring engineering cost and the qualitative factors summarised in table 3.5:

Table 3.5 – Make-vs-buy comparison of the manipulator

Criterion	In-house arm	Kinova Gen3
Recurring (per-unit) cost	Lower (~ €7.5-10K)	Higher (~ €25-30K)
Non-recurring engineering	High – design, control software, calibration, validation	None – delivered validated
Control software	Must be developed	Native ROS 2 / Kortex driver
Integration time	Weeks-months	Immediate
Performance guarantee	None	Manufacturer-specified
Intellectual-property ownership	Full	None
Risk	Higher (unproven assembly)	Low (field-proven)

The decisive factor, however, is not cost but integration. Building the in-house arm would have required developing and validating an entire control stack from scratch – low-level joint drivers, torque-loop tuning, kinematic calibration and independent safety validation – none of which exists for a custom assembly and all of which would have had to be produced and verified before the arm could perform even a single supervised task. The Kinova Gen3, by contrast, ships with a native ROS 2 / Kortex driver that connects directly to the layered software architecture developed in sub-chapter 3.7, exposing per-actuator torque sensing and manufacturer-validated, warranted motion immediately, without any low-level development effort. For a platform whose contribution lies in the integration of perception, decision-making and manipulation rather than in the manipulator hardware

itself, this immediate and validated integration is precisely what made the commercial arm the correct choice for the realised system, and it is the reason Concept B was adopted. The fully dimensioned in-house arm of Appendix A is therefore retained as a documented design alternative, ready to be manufactured and integrated once a dedicated control and safety stack has been developed for it.

3.4.4. Torso Structure

The torso is the primary structural element of the robot, connecting the mobile base to the arm shoulder interface and to the head/neck assembly. It is a hollow rectangular aluminium box of 340x380 mm cross-section, 5 mm wall thickness and 450 mm length. It must withstand the combined gravitational and inertial loads of all mounted components and the eccentric bending moment generated when the arm is extended. The component masses supported by the torso are listed in the table 3.6.

Table 3.6 – Masses supported by the torso

Component	Mass (kg)
Arm assembly	9.2
Arm payload (maximum)	4.0
Head and neck	2.3
Drawer and payload	7.0
TOTAL	22.2

Exterior area:

$$A_{torso_{ext}} = L * l = 380mm * 340mm = 129200 \text{ mm}^2 \quad (3.34)$$

Where L is the length of the torso cross-section, and l is the width of the cross-section.

Interior area:

$$A_{torso_{int}} = (L - 2 * th) * (l - 2 * th) = (380mm - 2 * 5mm) * (340mm - 2 * 5mm) = 122100 \text{ mm}^2 \quad (3.35)$$

Where th represents the torso's wall thickness.

Section area:

$$A_{torso_{sec}} = A_{torso_{ext}} - A_{torso_{int}} = 129200mm^2 - 122100mm^2 = 7100mm^2 \quad (3.36)$$

Volume and mass:

$$V_{torso} = A_{torso_{sec}} * H_{torso} = 7100mm^2 * 450mm = 3195000mm^3 \quad (3.37)$$

Where H_{torso} represents the height of the torso.

$$m_{torso} = V_{torso} * \rho_{Al} = 3195000mm^3 * 0.0027 \frac{g}{mm^3} = 8.627 \text{ kg} \quad (3.38)$$

Total arm force:

$$F_{arm_{total}} = m_{arm_{total}} * g = 13.2 \text{ kg} * 9.81 \frac{m}{s^2} = 129.45 \text{ N} \quad (3.39)$$

Where $m_{arm_{total}}$ represents the sum of the manipulator, gripper, and maximum payload.

Total head and neck force:

$$F_{head_{neck}} = (m_{head} + m_{neck}) * g = 2.3 \text{ kg} * 9.81 \frac{m}{s^2} = 22.57 \text{ N} \quad (3.40)$$

Torso force:

$$F_{torso} = m_{torso} * g = 8.627 \text{ kg} * 9.81 \frac{\text{m}}{\text{s}^2} = 84.63 \text{ N} \quad (3.41)$$

Total axial force:

$$F_{axial_{tot}} = F_{arm_{total}} + F_{head_{neck}} + F_{torso} = 129.45 \text{ N} + 22.57 \text{ N} + 84.63 \text{ N} = 236.65 \text{ N} \quad (3.42)$$

The worst-case bending load on the torso occurs when the arm is fully extended in the horizontal plane while carrying its rated payload.

Front-back bending moment:

$$M_{front_{back}} = F_{arm_{total}} * reach = 129.45 \text{ N} * 0.902 \text{ m} = 116.76 \text{ N*m} \quad (3.43)$$

Lateral bending moment:

$$M_{lateral} = F_{arm_{total}} * dist_{shoulder} = 129.45 \text{ N} * 0.16 \text{ m} = 20.712 \text{ N*m} \quad (3.44)$$

Resultant bending moment:

$$M_{resultant} = \sqrt{M_{front_{back}}^2 + M_{lateral}^2} = \sqrt{116.76^2 \text{ N*m} + 20.712^2 \text{ N*m}} = 118.58 \text{ N*m} \quad (3.45)$$

Second moment areas:

$$I_{lateral} = \left[L * \frac{l^3}{12} \right] - \left[(L-2*th) * \frac{(L-2*th)^3}{12} \right] = \left(380 \text{ mm} * \frac{340^3 \text{ mm}^3}{12} \right) - \left(370 \text{ mm} * \frac{330^3 \text{ mm}^3}{12} \right) = 1.366 * 10^8 \text{ mm}^4 \quad (3.46)$$

$$I_{front_{back}} = \left[l * \frac{L^3}{12} \right] - \left[(l-2*th) * \frac{(l-2*th)^3}{12} \right] = \left(340 \text{ mm} * \frac{380^3 \text{ mm}^3}{12} \right) - \left(330 \text{ mm} * \frac{370^3 \text{ mm}^3}{12} \right) = 1.617 * 10^8 \text{ mm}^4 \quad (3.47)$$

Section modulus:

$$W_{lateral} = \frac{\frac{I_{lateral}}{l}}{2} = \frac{1.366 * 10^8 \text{ mm}^4}{340 \text{ mm} / 2} = 8.033 * 10^5 \text{ mm}^3 \quad (3.48)$$

$$W_{front_{back}} = \frac{\frac{I_{front_{back}}}{L}}{2} = \frac{1.617 * 10^8 \text{ mm}^4}{380 \text{ mm} / 2} = 8.513 * 10^5 \text{ mm}^3 \quad (3.49)$$

Stress verification:

$$\sigma_{front_{back}} = \frac{M_{front_{back}}}{W_{front_{back}}} = \frac{116.76 \text{ N}}{8.513 * 10^5 \text{ mm}^3} = 0.137 \text{ MPa} \quad (3.50)$$

Condition is checked because $\sigma_{front_{back}} \ll 241 \text{ MPa}$

$$\sigma_{lateral} = \frac{M_{lateral}}{W_{lateral}} = \frac{20.712 \text{ N}}{8.033 * 10^5 \text{ mm}^3} = 0.026 \text{ MPa} \quad (3.51)$$

Condition is checked because $\sigma_{lateral} \ll 241 \text{ MPa}$

$$\sigma_{axial} = \frac{F_{axial_{tot}}}{A_{torso_{sec}}} = \frac{236.65N}{7100mm^2} = 0.034 \text{ MPa} \quad (3.52)$$

Condition is checked because $\sigma_{axial} \ll 241 \text{ MPa}$

3.4.5. Mobile Base

The mobile base is a 500x500 mm hollow aluminium box with a 3mm wall thickness, 675 mm tall (excluding wheels and chassis), carrying the battery, the driver electronics and the drawer. The maximum navigation velocity $v_{max} = 1.5 \text{ m/s}$ and maximum acceleration $a_{max} = 0.5 \text{ m/s}^2$ were selected in accordance with ISO 13482:2014 safety requirements for personal care robots operating in shared human-robot spaces, ensuring that the platform does not exceed pedestrian walking speed and maintains acceleration levels below the human discomfort threshold [41]. Two verifications are performed: the structural stress in the base walls and the tipping stability of the whole platform.

Exterior, interior and section area:

$$A_{base_{ext}} = L * l = 500mm * 500mm = 250000mm^2 \quad (3.53)$$

Where L is the length of the base and l is the width of it.

$$A_{base_{int}} = (L - 2 * thickness) * (l - 2 * thickness) = 494mm * 494mm = 244036mm^2 \quad (3.54)$$

Where thickness represents the base's wall thickness.

$$A_{base_{sec}} = A_{base_{ext}} - A_{base_{int}} = 250000mm^2 - 244036mm^2 = 5964mm^2 \quad (3.55)$$

Volume and mass:

$$V_{base} = A_{base_{sec}} * h_{base} = 5964mm^2 * 675mm = 4025700mm^3 \quad (3.56)$$

$$m_{base} = V_{base} * \rho_{Al} = 4025700mm^3 * 0.0027 \frac{g}{mm^3} = 10.9kg \quad (3.57)$$

Total mass:

$$m_{base_{total}} = 41kg \quad (3.58)$$

Axial compressive load:

$$F_{axial_{base}} = m_{base_{total}} * g = 41kg * 9.81 \frac{m}{s} = 402.21N \quad (3.59)$$

Axial compressive stress:

$$\sigma_{axial_{base}} = \frac{F_{axial_{base}}}{A_{base_{sec}}} = \frac{402.21N}{5964mm^2} = 0.067MPa \quad (3.60)$$

The worst case for the drawer bending is the fully open position on full-extension slides, where the drawer and payload centre of gravity (a mass of 7kg) reaches about 0.4 m forward of the base axis:

Drawer's force:

$$F_{drawer} = (m_{drawer} + m_{payload}) * g = 7 \text{ kg} * 9.81 \frac{\text{m}}{\text{s}^2} = 68.67 \text{ N} \quad (3.61)$$

Drawer's moment:

$$M_{drawer} = F_{drawer} * d_{drawer} = 68.67 \text{ N} * 0.4 \text{ m} = 27.5 \text{ N*m} \quad (3.62)$$

Section modulus:

$$W_{base} = \frac{I_{base}}{y_{max}} = \frac{2.455 * 10^8 \text{ mm}^4}{250 \text{ mm}} = 9.82 * 10^5 \text{ mm}^3 \quad (3.63)$$

Second moment of area:

$$I_{base} = \frac{L * l^3}{12} - \frac{[(L - 2 * thickness) * (l - 2 * thickness)^3]}{12} = \frac{500 \text{ mm} * 500^3 \text{ mm}^3}{12} - \frac{494 \text{ mm} * 494^3 \text{ mm}^3}{12} = 2.455 * 10^8 \text{ mm}^4 \quad (3.64)$$

Bending stress:

$$\sigma_{basebend} = \frac{M_{drawer}}{W_{base}} = \frac{27.5 \text{ N*m}}{9.82 * 10^5 \text{ mm}^3} = 0.028 \text{ MPa} \quad (3.65)$$

Total bending stress:

$$\sigma_{base_{total}} = \sigma_{axial_{base}} + \sigma_{basebend} = 0.067 \text{ MPa} + 0.028 \text{ MPa} = 0.095 \text{ MPa} \quad (3.66)$$

Structural safety:

$$SF_{base} = \frac{241}{0.095} = 2537 \text{ (times)} \quad (3.67)$$

3.4.6. Chassis

The chassis is built from 40x40 mm aluminium I-profile and includes a central traverse beam that connects the two longitudinal frame rails and provides the primary support surface for the battery pack and the drive electronics. The traverse carries a distributed load of 68.7 N. Modelled as a simply supported beam under uniform load; the maximum bending stress and mid-span deflection are derived below; a lateral load case from the wheel-reaction forces is also verified.

Section area:

$$A_{chassis_{sec}} = 2 * (b_{profile} * t_2) + (h_{profile} - 2 * t_2) * t_1 = 2 * (40 \text{ mm} * 4 \text{ mm}) + (40 \text{ mm} - 2 * 4 \text{ mm}) * 3 \text{ mm} = 416 \text{ mm}^2 \quad (3.68)$$

Where:

- $h_{profile}$ – the height of the profile (40mm);
- $b_{profile}$ – the base of the profile (40mm);
- t_1 – web thickness (3mm);
- t_2 – flange thickness (4mm).

Second moment of area (vertical bending):

$$I_{vertical} = \left(b_{profile} * \frac{h_{profile}^3}{12} \right) - 2 * \left[\frac{b_{profile} - t_1}{12} * \frac{(h_{profile} - 2 * t_2)^3}{12} \right] = \frac{(40mm * 40^3 mm^3)}{12} - 2 * \left[\frac{37mm}{12} * \frac{(40mm - 2 * 4mm)^3}{12} \right] = 112298.667 mm^4 \quad (3.69)$$

Section modulus (vertical):

$$W_{vertical} = \left(\frac{I_{vertical}}{\frac{h_{profile}}{2}} \right) = \frac{112298.667 mm^4}{\frac{40mm}{2}} = 5614.933 mm^3 \quad (3.70)$$

Second moment of area (lateral bending):

$$I_{lateral} = 2 * \left(t_2 * \frac{b_{profile}^3}{12} \right) + (h_{profile} - 2 * t_2) * \frac{t_1^3}{12} = 2 * \left(4mm * \frac{40^3 mm^3}{12} \right) + (40mm - 2 * 4mm) * \frac{3^3 mm^3}{12} = 42738.667 mm^4 \quad (3.71)$$

Section modulus (lateral):

$$W_{lateral} = \left(\frac{I_{lateral}}{\frac{b_{profile}}{2}} \right) = \frac{42738.667 mm^4}{\frac{40mm}{2}} = 2136.933 mm^3 \quad (3.72)$$

Volume and mass:

$$V_{profile} = A_{chassis_{sec}} * L_{total} = 416 mm^2 * 2000 mm = 832000 mm^3 \quad (3.73)$$

Where:

$L_{profile} = 4 * 400 mm = 1600 mm$ (length of the profile);

$L_{beam} = 400 mm$ (length of the beam);

$L_{total} = L_{profile} + L_{beam} = 2000 mm$ (total length).

$$m_{profile} = V_{profile} * \rho_{Al} = 832000 mm^3 * 0.0027 \frac{g}{mm} = 2.246 kg \quad (3.74)$$

Load on the traverse:

$$m_{load} = \frac{m_{battery} + m_{electronics}}{2} = 5.5 kg \quad (3.75)$$

$$F_{load} = m_{load} * g = 5.5 kg * 9.81 \frac{m}{s} = 53.937 N \quad (3.76)$$

Distributed load:

$$q = \frac{F_{load}}{L_{beam}} = \frac{53.937 N}{400 mm} = 0.13 \frac{N}{mm} \quad (3.77)$$

Maximum bending moment (simply supported beam, uniform load):

$$M_{bend_{max}} = q * \frac{L_{beam}^2}{8} = 0.13 \frac{N}{mm} * \frac{400^2 mm^2}{8} = 2600 N*mm \quad (3.78)$$

Bending stress:

$$\sigma_{beam} = \frac{M_{bend_{max}}}{W_{vertical}} = \frac{2600 N*mm}{5614.933 mm^3} = 0.46 MPa \quad (3.79)$$

The condition is checked because $\sigma_{beam} \ll 241 MPa$

Safety verification:

$$SF_{beam} = \frac{241}{\sigma_{beam}} = \frac{241}{0.46} = 524 (times) \quad (3.80)$$

Maximum deflection:

$$\delta_{max} = 5 * q * \frac{L_{beam}^4}{384 * E * I_{vertical}} = \quad (3.81)$$

$$5 * 0.13 \frac{N}{mm} * \frac{400^4 mm^4}{384 * 69000 MPa * 112298.667 mm^4} = 0.005 mm$$

The result of the deflection can be negligible.

3.4.7. Mass budget, centre of gravity and tipping stability

The complete mass budget of the platform can be established in the table 3.7. The structural masses of the head/neck, arm and torso are computed in the preceding sub-chapters; the remaining items are component datasheet values or engineering estimates. The fully loaded mass is 56.5 kg and the dry mass is 47.5 kg.

Table 3.7 – Mass budget

Component	Mass (kg)
Head and neck assembly	2.3
Manipulator (Kinova Gen3)	8.2
Gripper (Robotiq 2F-140)	1.0
Torso structure	8.6
Mobile base structure	10.9
Battery	8.0
Drive electronics and controllers	2.0
Drawer	2.0
Wiring, fasteners	1.5
Drive motors	2.0
Mecanum wheels	1.0
Subtotal – dry mass	47.5
Arm payload (rated)	4.0
Drawer payload (rated)	5.0
TOTAL – fully loaded	56.5

Tipping stability. Unlike the wall-stress check, the tipping analysis uses the total platform mass (56.5 kg), because every component contributes to the gravitational moment that resist overturning. The overturning axis is the line of the wheel contacts, at half the track (200 mm) from the centre, and the centre of the gravity height is 0.50 m. The stability factor is the ratio of the gravitational stabilising moment to the overturning moment, evaluated for the operating configurations enforced by the controller, which retracts the arm and locks the drawer before any base motion.

There are 5 cases:

Case 1 – driving + emergency braking (arm retracted and the drawer is closed):

$$M_{\text{overturn}_1} = M_{\text{total}} * a_{\text{max}} * h_{\text{CG}} = 56.5 \text{ kg} * 0.5 \frac{\text{m}}{\text{s}^2} * 0.50 \text{ m} = 14.125 \text{ N*m} \quad (3.82)$$

Where M_{total} is the total mass of the robot, including both max. payloads, a_{max} is the maximum acceleration of the base, and h_{CG} is the height of the centre mass of the robot.

$$M_{\text{stab}_1} = M_{\text{total}} * g * d_{\text{stab}} = 56.5 \text{ kg} * 9.81 \frac{\text{m}}{\text{s}} * 0.2 \text{ m} = 110.85 \text{ N*m} \quad (3.83)$$

Where d_{stab} is the distance within the chassis where the robot is stable, calculated as the half of the distance between two wheels.

$$SF_1 = \frac{M_{\text{stab}_1}}{M_{\text{overturn}_1}} = \frac{110.85 \text{ N*m}}{14.125 \text{ N*m}} = 7.85 \text{ (times)} \quad (3.84)$$

The condition is checked $SF_1 > 1.5$

Case 2 – arm fully extended forward, stationary, drawer closed (worst permitted):

$$M_{\text{overturn}_2} = m_{\text{arm}} * g * (x_{\text{arm}} - d_{\text{stab}}) = 13.2 \text{ kg} * 9.81 \frac{\text{m}}{\text{s}} * (0.517 \text{ m} - 0.2 \text{ m}) = 41.1 \text{ N*m} \quad (3.85)$$

Where x_{arm} is forward of the centre.

$$M_{\text{stab}_2} = (m_{\text{total}} - m_{\text{arm}}) * g * d_{\text{stab}} = (56.5 \text{ kg} - 13.2 \text{ kg}) * 9.81 \frac{\text{m}}{\text{s}} * 0.2 \text{ m} = 84.95 \text{ N*m} \quad (3.86)$$

$$SF_2 = \frac{M_{\text{stab}_2}}{M_{\text{overturn}_2}} = \frac{84.95 \text{ N*m}}{41.1 \text{ N*m}} = 2.06 \text{ (times)} \quad (3.87)$$

The condition is checked $SF_2 > 1.5$

Case 3 – drawer fully open, stationary, arm stowed:

$$M_{\text{overturn}_3} = m_{\text{drawer}} * g * (d_{\text{open}} - d_{\text{stab}}) = 7 \text{ kg} * 9.81 \frac{\text{m}}{\text{s}} * (0.4 \text{ m} - 0.2 \text{ m}) = 13.7 \text{ N*m} \quad (3.88)$$

$$M_{\text{stab}_3} = (m_{\text{total}} - m_{\text{drawer}}) * g * d_{\text{stab}} = (56.5 \text{ kg} - 7 \text{ kg}) * 9.81 \frac{\text{m}}{\text{s}} * 0.2 \text{ m} = 97.119 \text{ N*m} \quad (3.89)$$

$$SF_3 = \frac{M_{\text{stab}_3}}{M_{\text{overturn}_3}} = \frac{97.119 \text{ N*m}}{13.7 \text{ N*m}} = 7.08 \text{ (times)} \quad (3.90)$$

The condition is checked $SF_3 > 1.5$

Case 4 – arm extended + emergency braking (excluded by interlock):

$$M_{\text{overturn}_4} = M_{\text{overturn}_1} + M_{\text{overturn}_2} = 14.125 \text{ N*m} + 41.1 \text{ N*m} = 55.225 \text{ N*m} \quad (3.91)$$

$$M_{\text{stab}_4} = M_{\text{stab}_2} = 84.95 \text{ N*m} \quad (3.92)$$

$$SF_4 = \frac{M_{\text{stab}_4}}{M_{\text{overturn}_4}} = \frac{84.95 \text{ N*m}}{55.225 \text{ N*m}} = 1.53 \text{ (times)} \quad (3.93)$$

The condition is checked $SF_4 > 1.5$

Case 5 – arm extended + drawer open, stationary (excluded by operation):

$$M_{\text{overturn}_5} = M_{\text{overturn}_2} + M_{\text{overturn}_3} = 41.1 \text{ N*m} + 13.7 \text{ N*m} = 54.8 \text{ N*m} \quad (3.94)$$

$$M_{\text{stab}_5} = (m_{\text{total}} - m_{\text{arm}} - m_{\text{drawer}}) * g * d_{\text{stab}} = 36.3 \text{ kg} * 9.81 \frac{\text{m}}{\text{s}^2} * 0.2 \text{ m} = 71.22 \text{ N*m} \quad (3.95)$$

$$S_5 = \frac{M_{\text{stab}_5}}{M_{\text{overturn}_5}} = \frac{71.22 \text{ N*m}}{54.8 \text{ N*m}} = 1.29 \text{ (times)} \quad (3.96)$$

The condition is checked $SF_5 < 1.5$

All operating configurations satisfy the ISO 13482:2014 requirement ($SF > 1.5$); the worst permitted case – arm fully extended while stationary – gives $SF = 2.07$. The two configurations are never reached in normal operation. Case 4 is prevented by the motion interlock, which retracts the arm and locks the drawer before any base motion; braking with the arm deployed therefore cannot physically occur, and even if it did the platform would remain marginally above the limit ($SF = 1.53$). Case 5 is the only configuration that drops below the threshold ($SF = 1.3$); it is excluded operationally, since the arm reaches into the drawer at short range rather than at full extension, so the two forward overturning moments never combine.

3.5. Calculation of the driving system

The actuation of SKY is split between custom-driven subsystems – the head and the mobile base, whose actuators are sized – and the commercial manipulator, whose actuation is supplied integrally by the Kinova Gen3 unit and therefore requires no in-house torque sizing. A safety factor is applied to every actuator sizing: $SF = 1.5$ for the head and arm drives, and $SF = 2$ for the mobile-base, the higher value reflecting the shock loading and the human-risk zone in which the base operated. A mechanical efficiency ($\eta = 0.85$) is used throughout. The maximum angular acceleration and velocity are those table 3.1.

3.5.1. Head – Pan Motor

The head has a single degree of freedom – pan (rotation about the vertical axis). The pan motor must overcome the inertial torque of the head shell and its distributed electronics, plus the friction of the pan bearing and the cable-routing resistance. The required output torque and the corresponding motor-shaft torque (with safety factor and mechanical efficiency) are derived below.

Polar moment of inertia of the head shell about the vertical axis:

$$I_{zz_{\text{head}}} = 0.004713 \text{ kg*m}^2 \quad (3.97)$$

This value is taken from the assembly.

Effective radial distance of the distributed electronics mass from the pan rotation axis:

$$r_{pan} = 0.072m \quad (3.98)$$

This value is an assumption.

Mass moment of inertia of the internal electronic components:

$$I_{electronics} = m_{electronics} * r_{pan}^2 = 1.55 \text{ kg} * 0.072^2 \text{ m}^2 = 0.008 \text{ kg} * \text{m}^2 \quad (3.99)$$

Total mass moment of inertia of the head assembly:

$$I_{total} = I_{zz_{head}} + I_{electronics} = 0.004713 \text{ kg} * \text{m}^2 + 0.008 \text{ kg} * \text{m}^2 = 0.012 \text{ kg} * \text{m}^2 \quad (3.100)$$

Dynamic torque:

$$\tau_{pan_{dyn}} = I_{total} * \alpha_{pan} = 0.012 \text{ kg} * \text{m}^2 * 3 \frac{\text{rad}}{\text{s}^2} = 0.037 \text{ N} * \text{m} \quad (3.101)$$

Axial bearing load:

$$F_{axial_{rul}} = m_{head} * g = 2 \text{ kg} * 9.81 \frac{\text{m}}{\text{s}} = 19.613 \text{ N} \quad (3.102)$$

We will use precision ball bearing, interior diameter of 20 mm, friction coefficient of 0.002.

Bearing friction moment:

$$M_{friction} = 0.5 * \mu * F_{axial_{rul}} * d_{rul} = 0.5 * 0.002 * 19.613 \text{ N} * 0.02 \text{ m} = 3.92 * 10^{-4} \text{ N} * \text{m} \quad (3.103)$$

Total friction torque:

$$\tau_{total} = \tau_{cables} + M_{friction} = 0.030 \text{ N} * \text{m} + 3.92 * 10^{-4} \text{ N} * \text{m} = 0.03 \text{ N} * \text{m} \quad (3.104)$$

Total required output torque:

$$\tau_{req} = \tau_{pan_{dyn}} + \tau_{total} = 0.037 \text{ N} * \text{m} + 0.03 \text{ N} * \text{m} = 0.067 \text{ N} * \text{m} \quad (3.105)$$

Required motor shaft torque:

$$\tau_{motor} = \frac{\tau_{req} * SF}{\eta} = \frac{0.067 \text{ N} * \text{m} * 1.5}{0.85} = 0.118 \text{ N} * \text{m} \quad (3.106)$$

The selected actuator is as in the figure 3.1, the Dynamixel AX-12A, with a rated stall torque of 1.5 N*m – a torque safety margin of about 12.5x over the calculated requirement, chosen for its integrated controller, position feedback and direct compatibility with the Dynamixel control bus.



Figure 3.1 – Servomotor Dynamixel AX-12A

3.5.2. Manipulator actuation

In the realised configuration, all seven degrees of freedom and the gripper are actuated by then Kinova Gen3 unit and the Robotiq 2F-140 – illustrated in figure 3.2 and figure 3.3 – both supplied as integrated, factory-rated assemblies with their own controllers and torque sensing. No in-house actuator sizing is therefore required for the arm. The full torque, inertia, power and actuator selection calculations for the alternative in-house arm (Concept A) – gripper lead-screw actuation, wrist pitch and roll, elbow pitch, and shoulder pitch and abduction – are retained in Appendix A as the engineering basis for the future-development path.



Figure 3.2 – Kinova Gen3 manipulator



Figure 3.3 – Robotiq 2F-140 gripper

3.5.3. Mobile base drive

The mobile base uses four mecanum wheels, each driven by an individual Maxon EC-I 40 brushless motor with a GP32 planetary gearbox. From the total robot mass, the rolling-resistance coefficient ($\mu_r = 0.02$) and the target velocity profile ($v_{\max} = 1.5 \text{ m/s}$, $a_{\max} = 0.5 \text{ m/s}^2$), the rolling-resistance and acceleration forces give a required per-wheel torque of $0.49 \text{ N}\cdot\text{m}$ and mechanical power of 14.75 W per wheel.

Rolling resistance force:

$$F_{rr} = \mu_r * m_{total} * g = 0.02 * 56.5 \text{ kg} * 9.81 \frac{\text{m}}{\text{s}} = 11.08 \text{ N} \quad (3.107)$$

Acceleration force:

$$F_{acc} = m_{total} * a_{\max} = 56.5 \text{ kg} * 0.5 \frac{\text{m}}{\text{s}^2} = 28.25 \text{ N} \quad (3.108)$$

Total drive force:

$$F_{drive_{total}} = F_{rr} + F_{acc} = 11.08 \text{ N} + 28.25 \text{ N} = 39.33 \text{ N} \quad (3.109)$$

Force per wheel:

$$F_{wheel} = \frac{F_{drive_{total}}}{4} = \frac{39.33 \text{ N}}{4} = 9.83 \text{ N} \quad (3.110)$$

Torque at wheel axis:

$$\tau_{wheel} = F_{wheel} * r_{wheel} = 9.83 \text{ N} * 0.05 \text{ m} = 0.49 \text{ N}\cdot\text{m} \quad (3.111)$$

Where r_{wheel} is the radius of the wheel.

Required mechanical power:

$$P_{drive} = F_{drive_{total}} * v_{\max} = 39.33 \text{ N} * 1.5 \frac{\text{m}}{\text{s}} = 59 \text{ W} \quad (3.112)$$

Power per wheel with safety factor and efficiency:

$$P_{wheel} = \frac{P_{drive}}{4} = \frac{59W}{4} = 14.75 W \quad (3.113)$$

$$P_{req_{wheel}} = P_{wheel} * \frac{SF_{base}}{\eta} = 14.75W * \frac{2}{0.85} = 34.7 W \quad (3.114)$$

Angular velocity at wheel:

$$\omega_{wheel} = \frac{v_{max}}{r_{wheel}} = \frac{1.5 \frac{m}{s}}{0.05m} = 30 \text{ rad/s} \quad (3.115)$$

$$n_{wheel} = \omega_{wheel} * \frac{60}{2 * \pi} = 30 \frac{rad}{s} * \frac{60}{2 * \pi} = 286 \text{ rpm} \quad (3.116)$$

A direct drive is rejected. The wheel turns at only 286 rpm, far below the efficient range of a small brushless motor. A reduction stage is therefore required to lower the motor speed into its normal band while multiplying torque. For the motor, the Maxon EC-I 40 is selected – a brushless unit chosen for maintenance-free continuous duty and its integrated ENX 16 encoder (ROS 2 odometry). For the reduction, the Maxon GP32 planetary gearbox (19:1) [53] is selected for its compact coaxial layout and high efficiency, bringing the drive speed to ≈ 5440 rpm.

In figure 3.4, 3.5, and 3.6, the wheels, motor and planetary gearbox will be presented, as selected above.



Figure 3.4 – Mecanum wheel (100 mm diameter)



Figure 3.5 – Motor Maxon EC-I 40 [52]



Figure 3.6 - Maxon GP32 planetary gearbox (19:1)

Nominal torque verification:

$$\tau_{motor} = \tau_{wheel} * \frac{SF_{base}}{N_{drive} * \eta} = 0.45 \text{ N*m} * \frac{2}{19 * 0.85} = 0.055 \text{ N*m} \quad (3.117)$$

Where: N_{drive} is the gearbox ratio.

Its normal torque is verified applying the base safety factor and the efficiency coefficient, resulting in a 2x margin.

3.5.4. System power and battery autonomy

The complete electrical power budget of the platform is given in table 3.8. Each subsystem is listed with its peak and duty-cycle-average draw. A 15% allowance covers DC-DC conversion losses, cabling and the 48 to 24 VDC converter required by the arm.

Table 3.8 – Electrical power budget

Subsystem	Peak (W)	Avg (W)
Jetson Orin NX	25	18
Kinova Gen3	100	36
Mobile base	139	28
SICK TiM571 LiDAR	4	4
Intel RealSense D435i	3.5	3.5
ReSpeaker 4-mic array	1	1
Head tablet	10	7
Chest tablet	10	7
Audio and speakers	15	5
Cooling fan	4	4
FLIR Lepton 3.5	1	0.2
RFID reader, misc sensors, OB1203, MCUs	8	5
Drawer mechanism	10	1
TOTAL	321	120
TOTAL (+15% conversion/cabling losses)	≈ 370	≈ 138

For alimentation we chose a LiFePO4 48V 20Ah battery.

Battery energy:

$$E_{battery} = V_{battery} * C_{battery} = 48V * 20Ah = 960Wh \quad (3.118)$$

Autonomy:

$$t_{\text{autonomy}} = \frac{E_{\text{battery}}}{P_{\text{avg}}} = \frac{960Wh}{138W} \approx 7h \quad (3.119)$$

Maximum current:

$$I_{\text{max}} = \frac{P_{\text{peak}}}{V_{\text{battery}}} = \frac{370W}{48V} = 7.7A \quad (3.120)$$

Battery check:

$$\begin{cases} I_{\text{max}} = 7.7A \\ I_{\text{nominal}} = 70A \end{cases} \rightarrow I_{\text{max}} < I_{\text{nominal}} \quad (3.121)$$

The migration from the in-house actuator set to 36W Kinova arm roughly halves the average system draw relative to the Concept A budget, approximately doubling the autonomy to a full clinical-shift. LiFePO4 chemistry was selected over Li-ion NMC for its superior thermal safety – the absence of thermal-runaway risk is an advantage for a battery operating in close proximity to patients – and for its long cycle life. An automatic docking sequence is initiated at 20% remaining capacity, leaving roughly 1.4h of reserve.

3.6. Technological calculations for a mechanical component

3.6.1. Component description and manufacturing justification

The component selected is the chassis traverse beam of the platform. This structural element is located inside the mobile base chassis, connecting the two longitudinal frame rails and providing the primary support surface for the battery pack and electronics. Based on the structural calculations presented in section 3.3.6, the traverse beam is subjected to a distributed load of 68.7 N, generating a maximum bending stress of 0.46 MPa. The yield strength of the material is 241 MPa and it has a structural safety factor equal to 524.

The selected material is EN AW-6061 (Al 6061-T6), a wrought aluminium alloy widely used in structural aerospace and robotics applications. It has a low density (2700 kg/m³), high specific strength, good machinability and corrosion resistance, which are make it the optimal choice. The nominal hardness of 64 HB place it in the easily machinable range for high-speed cutting operations.

The component geometry is an I-profile with h = b = 40 mm, web thickness = 3 mm, and flange thickness = 4 mm. The final length is 400 mm. Four M6 through-holes are machined at the ends of the traverse for mechanical fastening to the longitudinal chassis rails, replacing a welded joint to avoid the heat-affected zone strength reduction associated with MIG/TIG welding.

Table 3.9. presents the complete component data sheet:

Table 3.9 – Traver beam data sheet

Component name	Chassis traverse beam
Profile type	I-profile 40x40 mm
Wall thickness (web)	3 mm
Wall thickness (flange)	4 mm
Final length	400 mm
Threads	4 x M6 though
Material	EN AW-6061
Hardness	64 HB

Tensile strength	290 MPa
Yield strength	241 MPa
Blank length	402 mm
Surface roughness	Ra 3.2 μm

3.6.2. Starting stock

The starting stock is a commercially extruded EN AW-6061 I-profile, purchased in standard 6000 mm length. The only dimensional machining required is the adjustment of the part length from the stock length to the final 400 mm, through face milling of both ends after sawing.

The blank length is determined by adding a machining allowance of 1 mm per face:

$$L_{blank} = L_{final} + 2 * allowance = 400 \text{ mm} + 2 * 1 \text{ mm} = 402 \text{ mm} \quad (3.122)$$

This allowance is sufficient to remove the saw-cut surface.

3.6.3. Technological process

The complete manufacturing sequence is defined in table 3.10:

Table 0.10 – Technological process

No.	Operation	Tool	Machine	t _b [min]	t _{aux} [min]	t _{total} [min]
1	Sawing: stock to 402 mm	HSS bimetal band saw blade	Band saw	0.500	1.000	1.500
2	Face milling: 402 to 401 mm	CoroMill Dura 1K223- 2500-100-NH H10F	Universal machining center 28 kW/18000 rpm	0.012	0.004	0.016
3	Face milling: 401 to 400 mm	CoroMill Dura 1K223- 2500-100-NH H10F	Universal machining center 28 kW/18000 rpm	0.012	0.004	0.016
4	Drilling 4xφ5 mm through hole	CoroDrill 860 860.1- 0500-040A1- NM H10F	Universal high-perf. Machine 200 kW/30300 rpm	0.024	0.010	0.034
5	Tapping 4xM6x1 through	CoroTap 200 T200- XM101DA- M6 C110	Universal high-perf. Machine 200 kW/1060 rpm	0.325	0.130	0.454
6	Deburring & chamfering 0.5x45° all edges	Hand file + chamfer tool (manual)	Bench / manual operation	2.000	0.500	2.500
TOTAL TIME				2.873	1.648	4.521

3.6.4. Cutting parameters

All cutting parameters were obtained from Sandvik Coromant CoroPlus Tool Guide (coromant.sandvik.com).

Table 3.11 consolidates all Sandvik-recommended cutting parameters per operation.

Table 3.11– Cutting parameters

No.	Tool code	Tool name	vc [m/min]	fz/fn [mm]	vf [mm/min]	ap [mm]	n [rpm]
1	HSS bimetal	Band saw blade	-	-	-	-	-
2/3 (pre)	1K223- 2500-100- NH H10F	CoroMill Dura	1400	0.372	20100	0.5	18000
2/3 (fin)	1K223- 2500-100- NH H10F	CoroMill Dura	1400	0.307	16600	0.5	18000
4	860.1- 0500- 040A1-NM H10F	CoroDrill 860	476	0.237 (fn)	7190	40	30300
5	T200- XM101DA- M6 C110	CoroTap 200	20	1 (fn=pitch)	1060	40	1060

Op.1 – Figure 3.7 and 3.8 are describing the face milling operation, parameters and tool in the platform Sandvik Coromant:

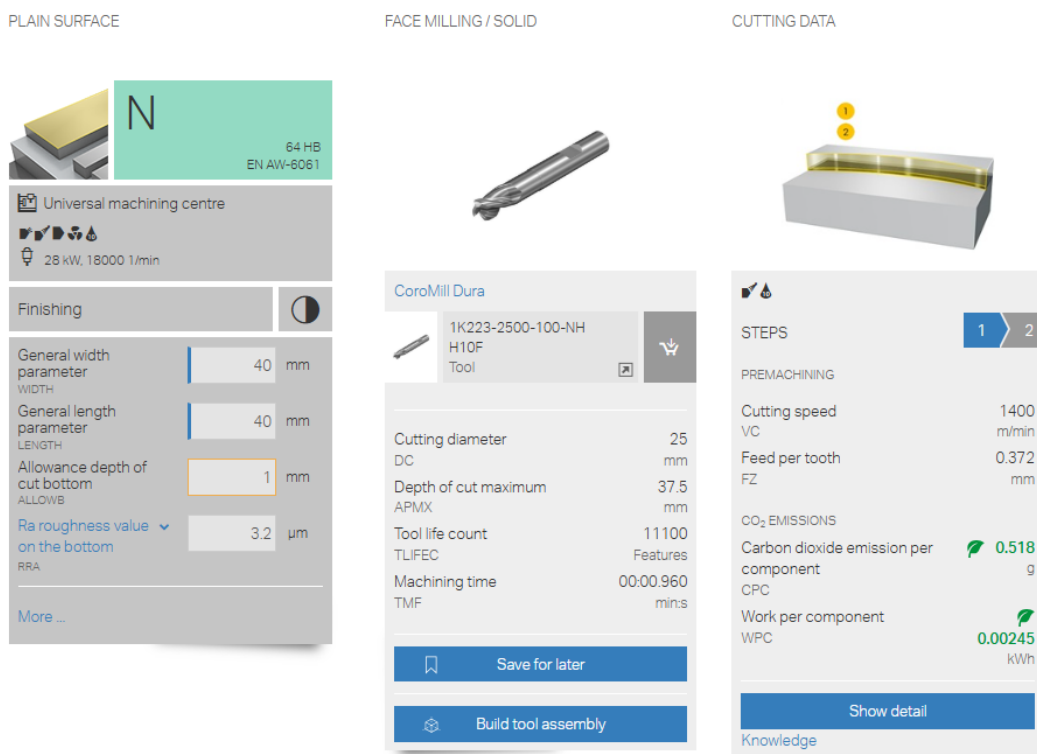


Figure 3.7 - Face milling tool

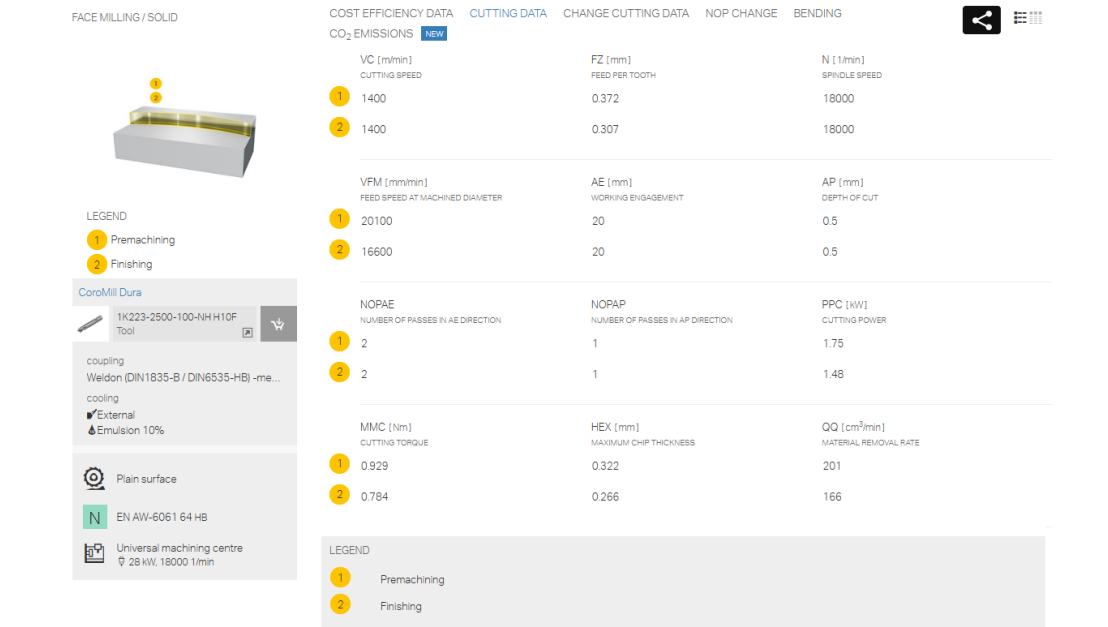


Figure 3.8 - Face milling parameters

Op.2 – Figure 3.9 and 3.10 are describing the operation, parameters and tool in the platform Sandvik Coromant for the M5 through holes:

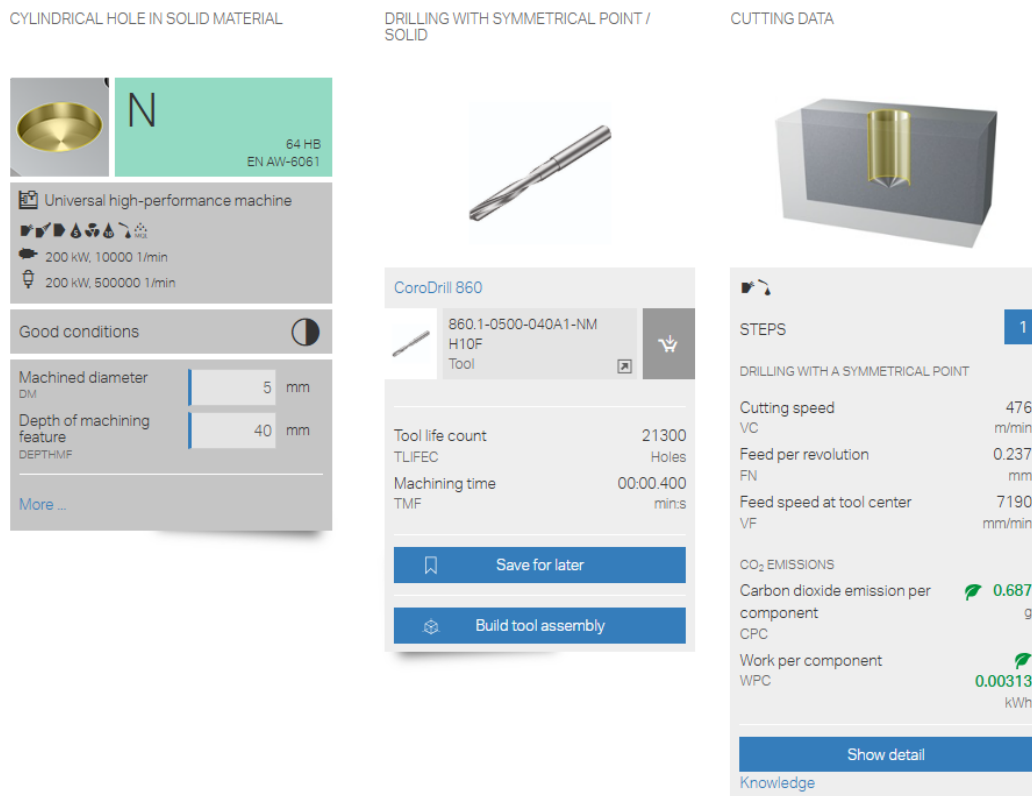


Figure 3.9 - M5 through holes tool

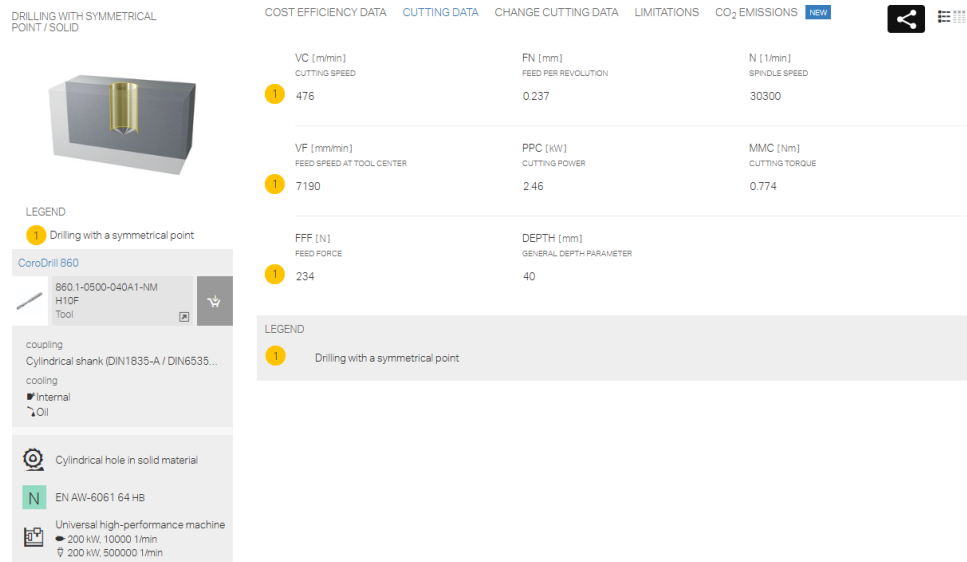


Figure 3.10 - M5 through holes parameters

Op.3 – Figure 3.11 and 3.12 are describing the operation, parameters and tool in the platform Sandvik Coromant for the Tapping M6x1 through:

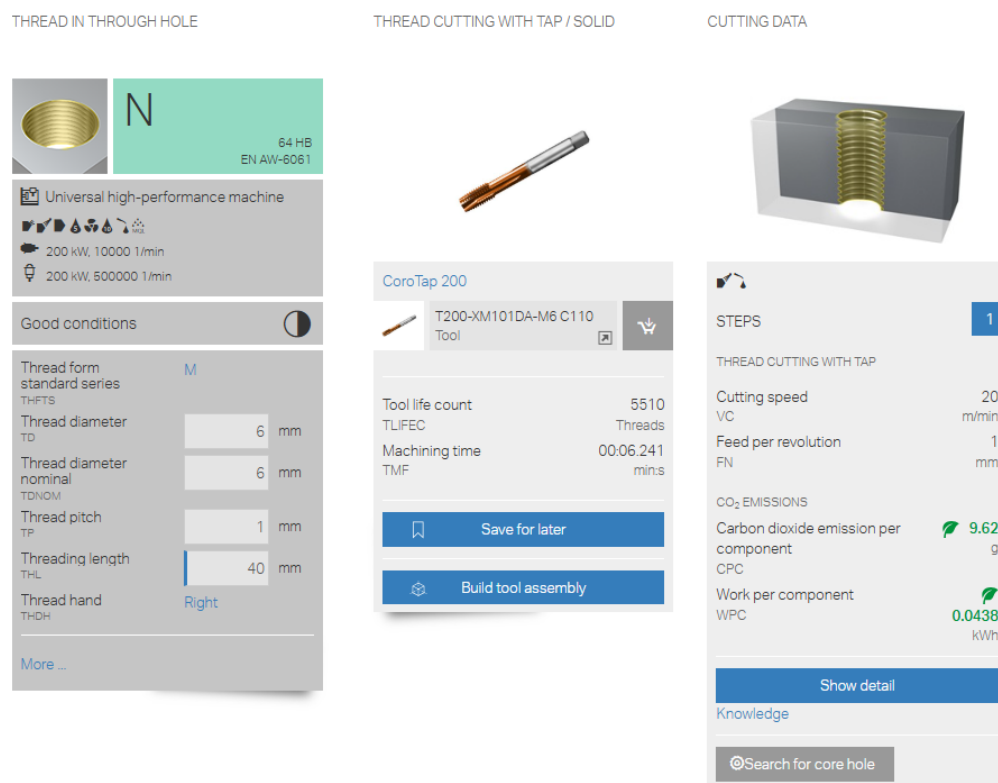


Figure 3.11 - Tapping M6x1 through tool

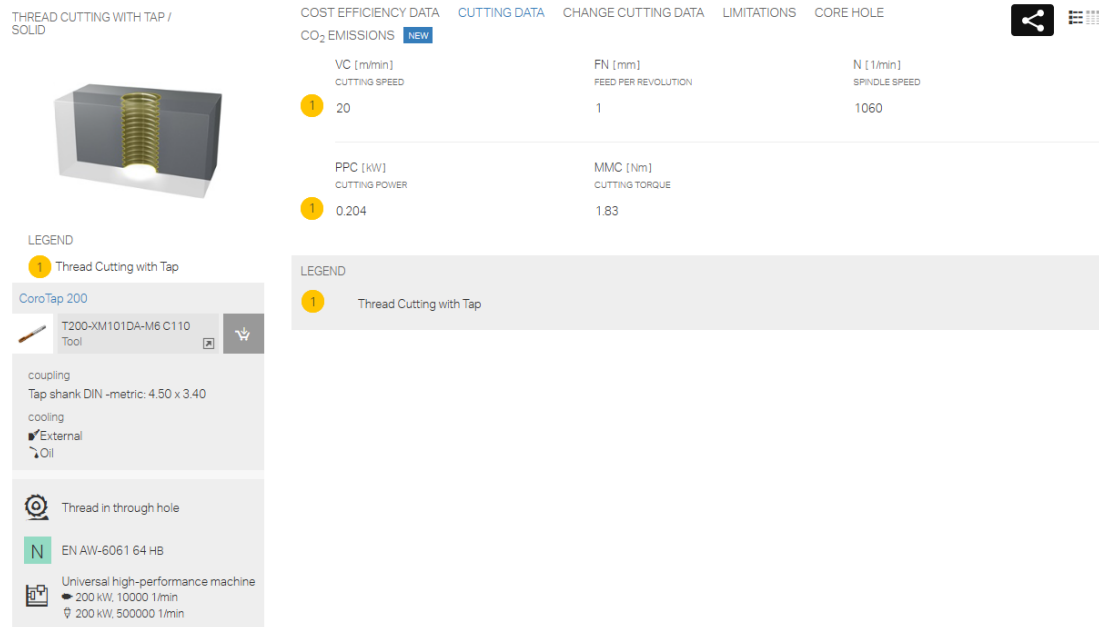


Figure 3.12 - Tapping M6x1 through parameters

3.6.5. Machine time calculations

The basic machining time (t_b) for each operation is calculated using the general formula:

$$t_b = \frac{L_{total}}{vf} [min] \quad (3.123)$$

Where

$$L_{total} = L_{workpiece} + L_{approach} + L_{overturn} [mm]$$

Table 312 presents all calculations:

Table 3.12 – Machining time calculations per operation

OP.	Calculation result	t _b [min]	t _{aux} [min]	t _{total} [min]	Note
1	Standard data	0.500	1.000	1.500	t _{aux} includes claming
2	Pre: 0.0056 Fin: 0.0068	0.012	0.004	0.016	L _{total} = 56.2 mm
3	Pre: 0.0056 Fin: 0.0068	0.012	0.004	0.016	Part repositioned
4	0.024	0.024	0.010	0.034	L _{total} = 43.5 mm
5	0.325	0.325	0.130	0.454	In + out stroke
6	Standard value	2.000	0.500	2.500	Standard value
TOTAL		2.873	1.648	4.521	~4.5 min/part

$$t_b = \text{standard value} \quad (3.124)$$

For operation 1.

$$\begin{cases} t_b = \frac{L_{total} * NOPAE}{vf} = \frac{(56.2 * 2)}{20100} = 0.0056 \text{ (pre)} \\ t_b = \frac{L_{total} * NOAPE}{vf} = \frac{56.2 * 2}{16600} = 0.0068 \text{ (fin)} \end{cases} \quad (3.125)$$

For operation 2.

$$\begin{cases} t_b = \frac{L_{total} * NOPAE}{vf} = \frac{(56.2 * 2)}{20100} = 0.0056 \text{ (pre)} \\ t_b = \frac{L_{total} * NOAPE}{vf} = \frac{56.2 * 2}{16600} = 0.0068 \text{ (fin)} \end{cases} \quad (3.126)$$

For operation 3.

$$t_b = \left(\frac{L_{total}}{vf} \right) * n_{holes} = \left(\frac{43.5}{7190} \right) * 4 = 0.024 \quad (3.127)$$

For operation 4.

$$t_b = \left(\frac{2 * L_{total}}{vf} \right) * n_{holes} = \left(\frac{2 * 43}{1060} \right) * 4 = 0.325 \quad (3.128)$$

For operation 5.

$$t_b = \text{standard value} \quad (3.129)$$

For operation 6.

NOPAE = number of passes in AE direction = 2

For drilling, L_{total} is 43.5 because:

$$L_{approach} = 0.3 * DC = 0.3 * 5 \quad (3.130)$$

$$L_{overrun} = 2 \text{ mm} \quad (3.131)$$

3.7. Definition of geometric parameters and tolerances for a mechanical component

The selected component is the arm-mounting flange, represented in figure 3.13, which mechanically couples the base of the manipulator to the upper plate of the robot torso. It was chosen because it is a single, fully machined part that interfaces a standardized commercial component (the arm base) with a custom structure (the torso): parts of its tolerances are imposed by the supplier interface and part by functional accuracy requirements.

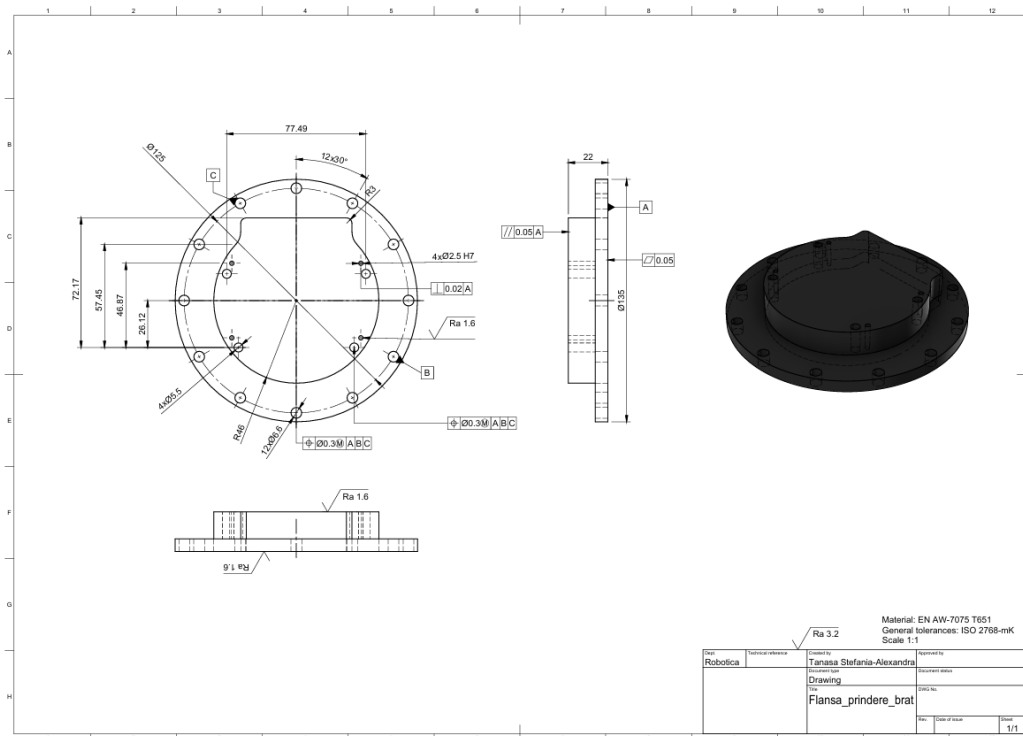


Figure 3.13 - Geometrical parameters and tolerances for the arm-mounting flange

3.7.1. Functional role and loading

The flange must:

- Carry the static weight of the arm, gripper and payload;
- React the bending moment produced by the arm;
- Guarantee a repeatable, accurate position of the arm coordinate frame relative to the robot;
- Provide the mounting interfaces on both of its faces.

Loading data – Kinova Gen3 (7 DOF):

- Mass: 8.2kg
- Continuous payload: 4kg
- Maximum reach: 0.902m
- Robotiq 2F-140 gripper mass: 1kg

The worst-case scenario is when the arm is fully extended in the horizontal plane:

Static moment is:

$$M_{\text{static}} = \text{Mass} * g * \text{center}_{\text{mass}} + (\text{continuous}_{\text{payload}} + \text{gripper}_{\text{mass}}) * g * \text{maximum}_{\text{reach}} \quad (3.132)$$

$$M_{\text{static}} = 72.4 \text{ N*m}$$

With a dynamic factor $k_d = 2$, the dynamic moment is:

$$M_{\text{dyn}} = 145 \text{ N} * m$$

For the EN AW-7075 T651 flange ($\phi 135 \times 22 \text{ mm}$) the resulting bending stress is far below the yield limit, so any angular deflection at the base is amplified into a tool-point error over the 0.902 m

reach. The reaction moment is taken by the twelve M6 torso screws, giving a peak screw load of ≈ 0.4 kN – well within the capacity of class-8.8 M6 fasteners.

3.7.2. Geometry, interfaces and material

The flange is a $\phi 135$ mm disc, 22 mm thick, made of EN AW-7075 T651 aluminium alloy, selected for its high specific stiffness and good machinability. On the arm side it carries a raised cylindrical boss ($\phi 92$, R46 mm) that reproduces the Kinova base footprint. The interface features, taken from the arm, are:

- 4 x M5 arm-mounting holes in an asymmetric pattern that matches the arm base, opened to $\phi 5.5$ clearance;
- 4 x $\phi 2.5$ mm reamed locating-pin holes, used as the centre datums;
- 12 x M6 torso-fixing holes equally spaced ($12 \times 30^\circ$) on a $\phi 125$ mm bolt circle, opened to $\phi 6.6$ mm clearance.

3.7.3. Dimensional tolerances and fits (ISO 286-1)

The only fits with a functional role are the four locating-pin holes, which are reamed to H7 to obtain a precise, repeatable location of the dowel pins (pin tolerance m6, giving an H7/m6 location fit). The screw holes have no fit function and are simply opened to the standard clearance diameters for their thread size. In table 3.13, the dimensional tolerances and fits are detailed:

Table 3.13 – Dimensional tolerances and fits

Feature	Specification	Standard
Locating dowel holes (x4)	$\Phi 2.5$ H7, reamed; mating pin $\phi 2.5$ m6, resulting H7/m6 location fit	ISO 286-1, ISO 2338
Arm fixing holes (x4)	$\Phi 5.5$ clearance for M5	ISO 273
Torso fixing holes (x12)	$\Phi 6.6$ clearance for M6	ISO 273
General linear / angular	ISO 2768-mK (medium, geometrical class K)	ISO 2768-1/-2

3.7.4. Geometric tolerances (ISO 1101)

The geometric tolerances are referenced to the A-B-C datum system and detailed in table 3.14 as following:

Table 3.14 – Geometric tolerances

Feature	Characteristic	Tolerance (mm)	Datum	Rationale
Seating face (datum A)	Flatness	0.05	-	Ensures full-contact seating on the torso, uniform bolt preload and no rocking of the joint.
Arm mounting face	Parallelism	0.05	A	Keeps the arm base parallel to the torso.
Dowel-hole axes (x4)	Perpendicularity	0.02	A	The locating pins must sit square to the mounting plane for correct, repeatable location.
Arm fixing holes (x4)	Position	$\Phi 0.3$ (MMC)	A, B, C.	Clearance holes for the arm screws; the MMC modifier grants a bonus tolerance the eases manufacture and assembly.

Torso fixing holes (x12)	Position	$\Phi 0.3$ (MMC)	A, B, C.	Provide clamping preload to the torso only; not a location function, MMC- modified zone.
--------------------------------	----------	------------------	----------	--

3.7.5. Surface texture, general tolerances and manufacturing conditions

A surface roughness of $Ra\ 1.6\ \mu m$ is specified on the functional contact and location surfaces (datum A, the arm mounting face and the reamed dowel bores) and $Ra\ 3.2\ \mu m$ on the remaining surfaces, in accordance with ISO 1302 (now ISO 21920). General linear and angular dimensions follow ISO 2768-mK. The part is machined from EN AW-7075 T651 plate: turning of the boss and outer diameter, drilling and milling of the hole patterns, and reaming of the locating holes to H7; all edges are deburred and free of sharp edges per ISO 13715.

Reference standards:

- ISO 286-1 (limits and fits);
- ISO 1101 (geometrical tolerancing);
- ISO 2768-1/-2 (general tolerances);
- ISO 1302 / ISO 21920 (surface texture);
- ISO 273 (clearance holes for bolts);
- ISO 2338 (parallel dowel pins);
- ISO 13715 (edges).

3.8. Programming

The software is organized as a layered architecture built on ROS 2 (Robot Operating System 2) as the middleware backbone. ROS 2 was selected over a monolithic application for three reasons: its publish / subscribe and action / service model decouples the concurrent subsystems – navigation, manipulation, perception, interaction – into independent modes that can be developed and tested in isolation; its DDS (Data Distribution Service) transport provides the real-time, fault-tolerant communication required for a safety-relevant mobile platform; and it gives direct access to mature, maintained stacks (Nav2 for navigation, MoveIt 2 [63] for manipulation) together with native Kinova Kortex ROS 2 driver. All computation runs on the onboard NVIDIA Jetson Orin NX, so that no patient audio, video or biometric data leaves the robot – a privacy requirement under medical-data regulations and the reason a cloud language model was rejected in favour of a locally hosted one. The architecture is organised into five layers – perception, cognition / decision, actuation, human-interaction – supervised by a transversal safety layer as presented in figure 3.14. For more details, please check Appendix D.

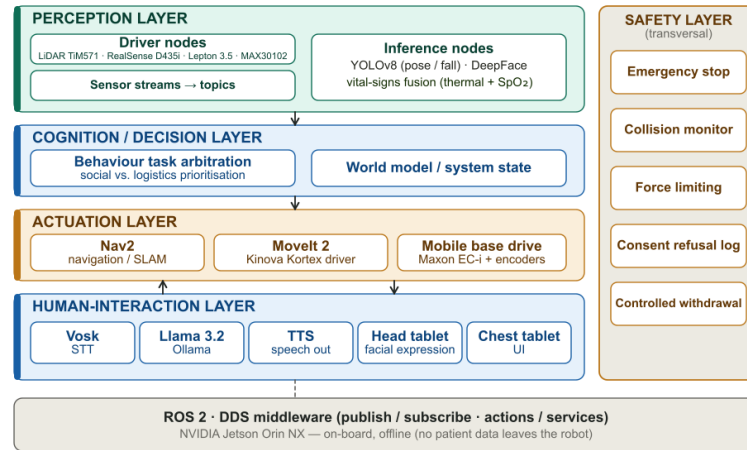


Figure 3.14 – ROS 2 layered architecture

3.8.1. Perception layer

Driver nodes publish the raw sensor streams: the SICK TiM571 LiDAR scan [60], the Intel RealSense D435i colour-and-depth stream, the FLIR Lepton 3.5 thermal frame and the OB1203 vital-signs readings. Three inference nodes consume these streams – a YOLOv8 node [64] for person/object detection and body-pose estimation (the basis of fall detection), a DeepFace node [65] for facial-emotion recognition, and a vital-signs node that fuses the thermal map (febrile screening) with the pulse/SpO₂ readings. Each publishes structured results on dedicated topics for the layers above.

The perception pipeline combines object detection and facial emotion recognition on each camera frame, as presenting in listing 3.1. YOLOv8 detections are filtered to a curated subset of COCO classes relevant to a healthcare environment, while DeepFace estimates the patient’s dominant emotion. The most recent emotion is written to a shared state variable so that it can be polled independently by the animated face display.

Fall detection is implemented as a lightweight heuristic over the temporal history of each person’s bounding box, rather than full pose estimation, in order to keep the method real-time on commodity hardware. As shown in listing 3.2, a fall is flagged only when several conditions hold simultaneously: the bounding box becomes horizontal, its centre drops rapidly, its height shrinks, and the person remain down beyond a time threshold. Requiring all conditions together substantially reduces false positives from ordinary movements such as bending or sitting.

Listing 3.1 – Vision pipeline: YOLOv8 detection & DeepFace emotion analysis

```
def analyze_image(image_bytes):
    """
    Run YOLOv8 object detection and DeepFace emotion analysis on a JPEG/PNG
    frame.
    Accepts raw image bytes (as received from the tablet camera stream).
    Returns a tuple: (list of detected object labels, dominant emotion
    string).
    Updates current_emotion_global so the /emotion polling endpoint stays
    current.
    """
    nparr = np.frombuffer(image_bytes, np.uint8)
    frame = cv2.imdecode(nparr, cv2.IMREAD_COLOR)
    if frame is None:
        return [], "unknown"

    # Object detection – filter to healthcare-relevant COCO classes only
    detected_objects = []
    try:
        results = yolo_model(frame, verbose=False, conf=0.4)
        for result in results:
            for box in result.bboxes:
                cls = int(box.cls[0])
                if cls in RELEVANT_OBJECTS:
                    detected_objects.append(RELEVANT_OBJECTS[cls])
        detected_objects = list(set(detected_objects)) # Remove duplicates
    except Exception as e:
        print(f"YOLO error: {e}")

    # Emotion analysis – enforce_detection=False avoids crashes on unclear
    # faces
    emotion = "unknown"
    try:
        analysis = DeepFace.analyze(frame, actions=["emotion"],
        enforce_detection=False, silent=True)
        if analysis:
            emotion = analysis[0]["dominant_emotion"]
            current_emotion_global["emotion"] = emotion # Update shared
            state for ESP32
    except Exception as e:
        print(f"DeepFace error: {e}")

    return detected_objects, emotion
```

Listing 3.2 – Heuristic fall detection from YOLOv8 bounding-box history

```
def check_fall(self, box, frame_height, person_id):
    x1, y1, x2, y2 = box
    width = x2 - x1
    height = y2 - y1
    center_y = (y1 + y2) / 2

    # Update position (thread-safe, with automatic trimming)
    self.update_person_position(person_id, {
        "time": time.time(),
        "center_y": center_y,
        "height": height,
        "width": width
    })

    positions = self.get_person_positions_snapshot().get(person_id, [])

    if len(positions) < FALL_MIN_HISTORY:
        return False

    is_horizontal = width > height * 2
    old_center = positions[0]["center_y"]
    new_center = positions[-1]["center_y"]

    dropped_fast = (new_center - old_center) > frame_height *
FALL_DROP_THRESHOLD

    old_height = positions[0]["height"]
    new_height = positions[-1]["height"]

    height_reduced = new_height < old_height * FALL_HEIGHT_REDUCTION

    time_on_ground = time.time() - positions[0]["time"]
    staying_down = is_horizontal and time_on_ground > FALL_TIME_ON_GROUND

    if staying_down and (dropped_fast or height_reduced):
        return True

    return False
```

3.8.2. Navigation and localisation

The Nav2 stack performs autonomous navigation: a SLAM module (slam_toolbox) builds and maintains the ward map, AMCL (Adaptive Monte Carlo Localization) provides localisation, and global and local planners generate collision-free paths from costmaps fused from the LiDAR and depth data, with recovery behaviours for blocked situations. An omnidirectional controller maps the planned velocity commands to the four mecanum-wheel speeds, exploiting the holonomic mobility needed in confined clinical spaces.

3.8.3. Manipulation

The Kinova Gen3 is driven through its native Kortex ROS 2 driver under MoveIt 2, which performs collision-aware motion planning (OMPL) and executes grasps with the Robotiq 2F-140 gripper. Manipulation requests are issued as ROS 2 actions, allowing the behaviour layer to monitor and pre-empt them.

3.8.4. Conversational AI and emotion-adaptive interaction

The dialogue pipeline runs entirely on-board: a Vosk node [66] performs offline speech-to-text, a locally hosted Llama 3.2 model [68] (via Ollama [67]) generates the response, and a text-to-speech node renders it. The emotion topic published by the perception layer conditions the generation, so the reply adapts to a patient's detected state (for example, a reassuring tone when fear or distress is recognised). The head tablet subscribes to an expression topic and renders the corresponding animated facial expression, while the microphone (ReSpeaker 4 Linear Array), the loudspeaker (Dayton Audion CE Series) and the chest tablet handle the audio input/output and the touch user interface.

Speech recognition is performed entirely offline using the Vosk engine, which keeps all patient audio on the local machine. As shown in listing 3.3, the audio file is processed in fixed 4000-frame chunks, allowing recordings of arbitrary length to be transcribed without loading the entire file into memory. The partial results are accumulated and flushed at the end to produce the final transcription.

Natural language responses are generated by a local hosted Llama 3.2 model served through Ollama, avoiding any dependency on cloud services. Listing 3.4 shows how the detected emotion and visible objects are injected into the prompt as additional context, enabling the model to produce situationally aware replies. A rolling conversation history is maintained and bounded to a fixed length to preserve recent context while preventing unbounded growth.

Listing 3.3 – Offline speech-to-text with Vosk (chunked recognition)

```
def transcribe_audio(audio_path):
    """
    Transcribe a WAV audio file to text using Vosk (offline STT).
    Reads the file in 4000-frame chunks to handle large recordings.
    Returns the full transcribed string, or an empty string on failure.
    """
    wf = wave.open(audio_path, "rb")
    recognizer = vosk.KaldiRecognizer(vosk_model, wf.getframerate())
    result_text = ""
    while True:
        data = wf.readframes(4000)
        if len(data) == 0:
            break
        if recognizer.AcceptWaveform(data):
            result = json.loads(recognizer.Result())
            result_text += result.get("text", "")
    final = json.loads(recognizer.FinalResult())
    result_text += final.get("text", "")
    return result_text.strip()
```

Listing 3.4 – Emotion-aware response generation via local Ollama / Llama 3.2

```
def get_response(user_input, emotion=None, objects=None):
    """
    Send the patient's message to the local Ollama LLM and return the reply.
    Optionally prepends detected emotion and visible objects as context
    so the LLM can give more relevant responses (e.g. "I see you look sad").
    Maintains a rolling conversation history capped at
    MAX_CONVERSATION_HISTORY
    to avoid context overflow while preserving recent context.
    Falls back to a safe static message if Ollama is unreachable.
    """
    context = ""
    if emotion and emotion != "unknown":
        context += f"Patient seems {emotion}. "
    if objects:
        context += f"Visible objects: {objects}. "

    conversation_history.append({"role": "user", "content": context +
user_input})

    try:
        response = req.post("http://localhost:11434/api/chat", json={
            "model": "llama3.2:1b",
            "messages": conversation_history,
            "stream": False,
            "options": {"num_predict": 50, "temperature": 0.5, "num_ctx":
512}
        }, timeout=60)
        answer = response.json()["message"]["content"]
        conversation_history.append({"role": "assistant", "content": answer})

        # Keep history bounded: remove oldest user/assistant pairs after
system prompt
        while len(conversation_history) > 8:
            conversation_history.pop(1)

        return answer

    except Exception as e:
        print("Ollama error:", e)
        return "I am having a technical issue. Please call a nurse."
```

3.8.5. Behaviour and task arbitration

A behaviour-tree decision layer – detailed in figure 3.15 – arbitrates between the robot's competing objectives – logical tasks (medication and supply delivery) and social/clinical tasks (responding to a detected fall, distress or call). This is the agentic core that realises the originality claim of the project: a patient in distress detected by the perception layer pre-empts a queued logistics task, and the robot announces its intended action and awaits agreement (requirement R13) before proceeding. The behaviour tree issues navigation and manipulation goals as actions and reacts to their outcomes.

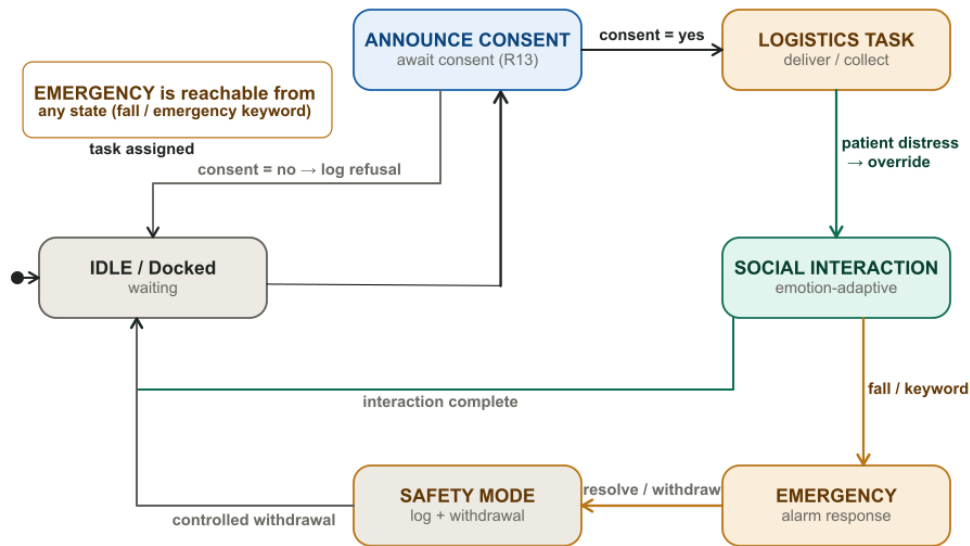


Figure 3.15 – Behavior-tree decision layered

3.8.6. Safety layer

An independent safety-supervisor node, separate from the task logic, enforces the ISO 13482:2014 constraints in real time: it caps velocity and acceleration, enforces the tipping interlocks (arm retracted and drawer locked before any base motion), runs a heartbeat watchdog over the critical nodes, and routes the hardware emergency-stop. Because it sits outside the behaviour layer, a fault or a hostile command in the task logic cannot override it.

Before any input is passed to the language model, the transcribed text is screened against a fixed set of emergency keywords, as shown in listing 3.5. If any keyword is matched, the system bypasses the LLM entirely and returns an immediate alert response, prioritising reaction time over conversational nuance in safety-critical situations. This design ensures that a deterministic safety path always takes precedence over the probabilistic behaviour of the language model.

Listing 3.5 – Safety-critical emergency keyword filter (pre-LLM)

```

# Words that immediately trigger an emergency alert response, bypassing the
# LLM entirely for faster reaction time.
EMERGENCY_KEYWORDS = ["help", "emergency", "pain", "falling", "bleeding",
"chest", "unconscious"]
def check_emergency(text):
    """
    Check whether the transcribed text contains any emergency keyword.
    Returns True if an emergency keyword is found, False otherwise.
    """
    return any(kw in text.lower() for kw in EMERGENCY_KEYWORDS)

```

3.8.7. Manual control

For setup, teleoperation and override, the .NET MAUI [70] smartphone application connects through a bridge to the ROS 2 teleoperation interface, exposing a directional pad, speed sliders, a battery/telemetry readout and an emergency-stop, with the manual command path kept independent of the autonomous stack. The application follows the MVVM (Model-View-View-Model) pattern using the CommunityToolkit.Mvvm library, with UI actions bound to relay commands that translate slider and button input into a compact ASCII command protocol (e.g. MOVE: F, SPD:150,150, ARM:90,90,90, GRIP:50, HEAD:90) transmitted over Bluetooth SPP, as shown in listing 3.7. The interface, presented in figure 3.16, groups base movement, arm joint control, gripper, head orientation and task queue into a single screen, with line connection status and battery telemetry parsed from the data stream returned by the Arduino.

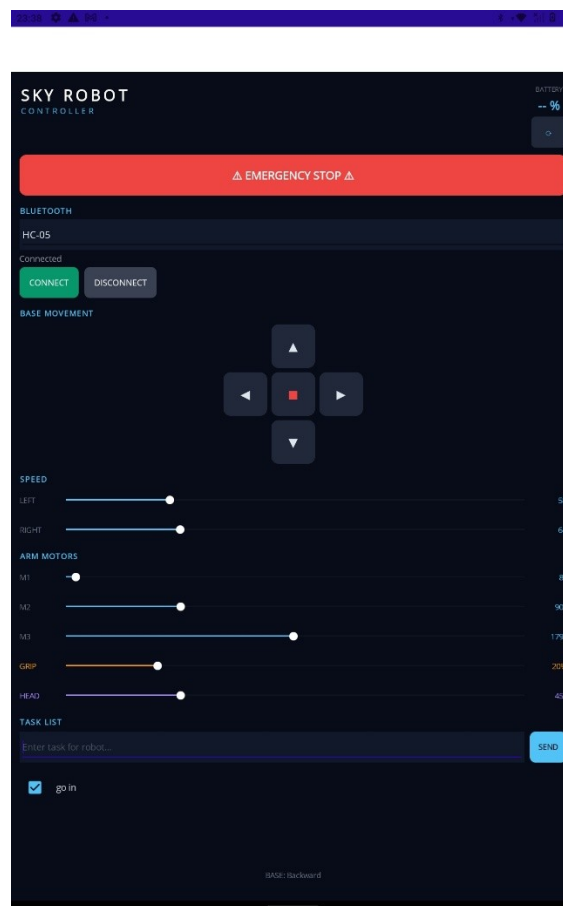


Figure 3.16 - NET MAUI control application interface

Listing 3.6 - Base-movement and speed command dispatch

```
[RelayCommand]
private async Task Move(string direction)
{
    string cmd = $"MOVE:{direction}";
    await _bluetooth.SendCommandAsync(cmd);
    string label = direction switch
    {
        "F" => "Forward",
        "B" => "Backward",
        "L" => "Left",
        "R" => "Right",
        "S" => "Stop",
        _ => direction
    };
    LastCommand = $"BASE: {label}";
}

public async Task SendSpeedCommand()
{
    int l = (int)LeftSpeed;
    int r = (int)RightSpeed;
    string cmd = $"SPD:{l},{r}";
    await _bluetooth.SendCommandAsync(cmd);
    LastCommand = $"SPEED: L={l} R={r}";
}
```

3.9. Sensorics

The platform integrates a heterogeneous sensor suite selected to cover the four functional dimensions established in the requirements analysis: autonomous navigation, three-dimensional scene perception, human-robot interaction, and non-contact physiological monitoring. Each selection was driven by the corresponding functional requirement (R1-R10) and by suitability for a clinical environment, favouring non-contact measurement, hygienic operation, and robotics-grade reliability over the lower-cost alternatives listed in the requirements matrix.

3.9.1. Navigation and spatial detection

Autonomous navigation and obstacle avoidance (R9) are handled by a 2D LiDAR scanner. The SICK TiM571 – shown in figure 3.17 – uses an 850 nm class-1 infrared laser with a 270° aperture angle, a 15 Hz scanning frequency, a 0.33° horizontal angular resolution, a 0.05-25 m working range, and an 8 m scanning range at 10% remission. It carries an IP67 enclosure rating and draws typically 4 W from a 9-28 V DC supply over an Ethernet connection. It was chosen over the ultrasonic alternative because a single unit provides near-panoramic coverage suitable for SLAM-based localisation and path planning. Furthermore, its IP67 housing tolerates the cleaning and incidental spills. For more details, please check Appendix D-J.



Figure 3.17 - SICK TiM571 LiDAR scanner

Intel RealSense D435i [55] – as shown in figure 3.18 – is used for the three-dimensional perception – person and object detection, the spatial basis for fall analysis, and navigation assistance – relies on a stereo depth camera. The D435i provides an $87^\circ \times 58^\circ$ depth field of view, an ideal operating range of 0.3 m to 3 m, a global-shutter sensor, and an integrated IMU; the built-in BMI055 IMU delivers 6-DOF motion data that supports SLAM and point-cloud alignment, and it can stream depth up to 1280×720 at 30 fps over USB 3. The global shutter avoids motion distortion when the head moves, the wide field of view reduces blind spots, and the IMU makes the unit well suited to a moving robot platform – the reason this RGB-D module is a reference choice in research-grade HRI platforms.

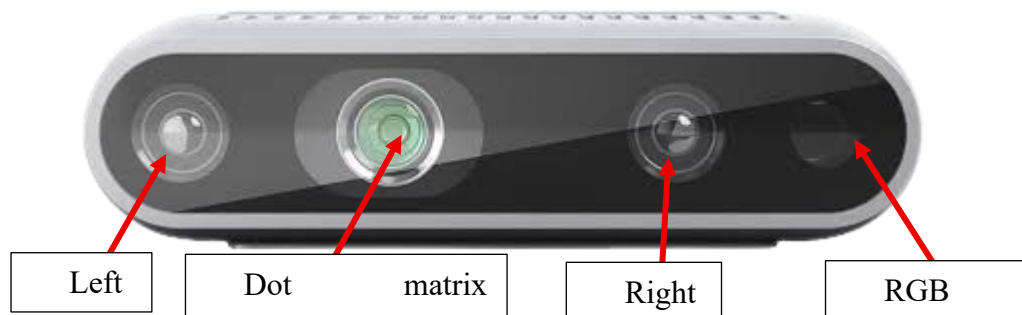


Figure 3.18 – 3D perception Intel RealSense D435i camera

3.9.2. Vision for human-robot interaction

Sony IMX519 [56] – shown in figure 3.19 – is used for facial-expression and emotion recognition (R10). This camera is a high-resolution, with a 16-megapixel autofocus CMOS sensor. The high pixel count and autofocus were prioritised to capture fine facial detail reliably at conversational distance, which is what DeepFace emotion-analysis pipeline requires to classify the seven emotion categories with acceptable confidence.



Figure 3.19 – Sony IMX519 camera

In figure 3.20, Arducam IMX477 [57] is shown. This board provides situational awareness. It has a 12.3-megapixel CMOS sensor paired with a wide-field lens. Whereas the IMX519 is optimised for the patient's face, the IMX477 captures the wider scene, supporting object detection and contextual awareness across the room without the narrow framing of the interaction camera.



Figure 3.20 – Arducam IMX477

3.9.3. Physiological sensing

Requirement 1 (non-invasive monitoring) is performed by FLRI Lepton 3.5 – shown in figure 3.21 – which has a long-wave infrared thermal camera. It is a 160 x 120, 12-micron uncooled microbolometer that provides calibrated radiometric output for the temperature of every pixel, with a thermal sensitivity below 50 mK (0.05 °C), a 56° lens, SPI video and two-wire control interfaces. It has a typical operating power of 140 mW (650 mW during a shutter event) and is sensitive across the 8-14 μm long-wave infrared band. The deciding factor for choosing this camera was the radiometric per-pixel temperature, because it enables febrile screening at a distance without touching the patient, which is both more hygienic and less intrusive than a contact thermometer, while the module's small size and low power allow it to be integrated directly into the head assembly.



Figure 3.21 – FLIR Lepton 3.5 IR thermal camera

As part of R1, Renesas OB1203 [59] – shown in figure 3.22 – is used for pulse rate and peripheral oxygen saturation (SpO_2) measurement. The sensor integrates red and infrared LEDs and communicates with the robot controller via an I²C interface while operating from a 1.7-3.6 V supply. Using reflective photoplethysmography (PPG), the module enables the patient to simply place a fingertip on a dedicated contact pad integrated on the robot's torso, eliminating the need for a conventional finger clip. The sensor also incorporates automatic ambient light cancellation and an integrated proximity detector for reliable finger presence detection, improving measurement stability in varying lighting conditions. Its compact size, low power consumption, and contact-pad configuration make it well suited for integration into an assistive healthcare robot, providing an intuitive and hygienic method for acquiring vital signs. The thesis explicitly positions these readings as indicative monitoring rather than clinical-grade diagnosis, with each measurement validated against a reference instrument.



Figure 3.22 – Renesas OB1203 sensor for pulse rate and SpO_2

3.10. Innovation aspects

The novelty of SKY does not reside in any single component – most of which are commercially available – but in their integration into a configuration that does not currently exist in deployed clinical robotics. Five aspects distinguish the proposed system from the state of the art benchmarked in sub-chapter 1.2.

The first and central innovation is functional hybridisation. As the Social vs. Physical capability matrix in figure 1.17 shows, every existing platform occupies either the physical-logistics axis or the social-interaction axis, but none occupies the high-capability region of both simultaneously. SKY is

designed from the ground up to fill that gap, covering all four functional dimensions – autonomous navigation, physical manipulation, medical sensing and human-robot interaction – on a single platform, rather than extending a logistics robot toward sociability or vice versa.

The second innovation is the agentic decision core. A behaviour-tree arbitration layer (sub-chapter 3.7.5) does not merely sequence pre-programmed tasks; it weighs competing objectives in real time and pre-empts them by priority. A patient distress event detected by the perception layer interrupts a queued logistics task, and the robot announces its intended action and awaits the patient's agreement (requirement R13) before proceeding. This capacity to autonomously decide that a social or clinical need outranks a logistical one is the concrete realisation of the project's originality claim.

The third innovation is a privacy-by-design AI stack that runs entirely on-board. Four offline models – Vosk for speech recognition, YOLOv8 for detection, DeepFace for emotion analysis and a locally hosted Llama 3.2 via Ollama for dialogue – operate without any cloud dependency, so that no patient audio, video or biometric data ever leaves the robot. This is both a practical advantage in network-restricted hospital environments and a direct response to medical-data confidentiality requirements, and it is the reason a cloud language model was rejected.

The fourth innovation is the deterministic safety architecture layered over the probabilistic AI. An independent safety-supervisor node sits outside the task logic and enforces the ISO 13482:2014 limits regardless of what the behaviour layer requires, while a pre-LLM emergency-keyword filter (sub-chapter 3.7.6) guarantees that a safety-critical utterance bypasses the language model entirely and triggers an immediate alert. A deterministic safety path therefore always takes precedence over the statistical behaviour of the neural models.

The fifth innovation is the hardware-decoupled, reusable software architecture. Because the interface pipeline communicates with the physical platform only through well-defined interfaces, it functions as a standalone agentic AI system that can be reconfigured for other robotic or assistive applications independently of this specific body – and is published as such. The system itself carries a continuous upgrade path rather than a single fixed configuration.

3.11. Electrical architecture

The preceding sections defined the mechanical structure of the robot and selected actuators, sensors, and power source individually. This section brings those choices together into a coherent electrical architecture, showing how every component is powered and how the subsystems communicate. Following the system decomposition adopted in subchapter 2.2, the architecture is presented as four component-level schematics, each corresponding to one functional subsystem: power distribution and safety, actuation, sensing and processing, and interaction and communications. The schematics are drawn at component level – with supply rails, buses, decoupling, protection and connector interfaces. Each schematic is described in the subsections below.

3.11.1. Power distribution and safety

The power distribution and safety are presented in figure 3.23, which is built around a LiFePO₄ 48V / 20Ah pack (BT1) protected by a 16-series battery-management system (U1), which performs cell balancing and over-current / over-voltage protection before the energy reaches the rest of the platform. The protected output passes through the main fuse F1 and disconnect switch SW1 to the +48V bus. From this bus, four step-down (buck) converters generate the working rails required by the platform – 24V for the manipulator (U3), 19V for the onboard computer (U4), 12V for the LiDAR, servo and auxiliary loads (U5), and 5V for logic and USB (U6) – while a low-dropout regulator (U7) derives the 3.3V sensor rail. Each converter is shown with its complete application circuit (input/output capacitors, inductor, catch diode and a feedback divider R1/R2 that sets the output voltage). Safety is enforced by the normally-closed emergency-stop switch SW2, which opens the

+48V_MOT branch and removes power to log the event and command a controlled stop. The pack provides roughly 960Wh, giving about seven hours of autonomy at the average draw, with automatic docking triggered at 20% state of charge.

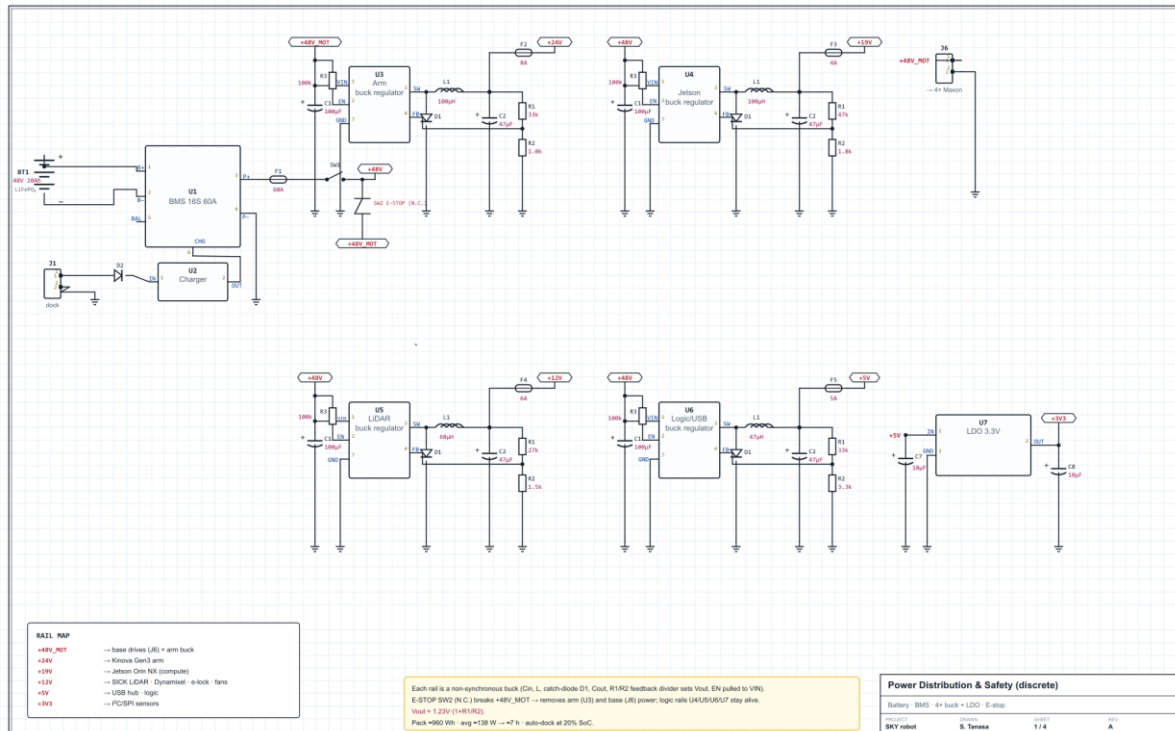


Figure 3.23 – Power distribution and safety scheme

3.11.2. Actuation

The actuation subsystem – detailed in figure 3.24 – covers locomotion, manipulation and head motion. Motion commands are issued by the Jetson (U1) and distributed to the four mecanum-wheel drives over a CAN bus, buffered by a transceiver (U2) and correctly terminated with a split termination (R1/R2 and C2) at the controller end and a 120 Ω resistor (R3) at the far end. Each Maxon EPOS4 servo controller (U3–U6) closes the current and velocity loop locally and drives one EC-i 40 brushless motor (M1–M4) through its three phases, using the Hall sensors for commutation feedback; the 48 V motor rail (+48V_MOT) is the one interrupted by the emergency stop. The manipulator (Kinova Gen3, A1) is powered at 24 V and commanded over Ethernet through connector J7, while the Robotiq 2F-140 gripper (A2) is controlled through the arm's wrist bus over Modbus RTU. Head panning uses a Dynamixel AX-12A servo (M5) on a TTL half-duplex link, generated from the Jetson UART by a tri-state buffer (U8) that handles the bus direction.

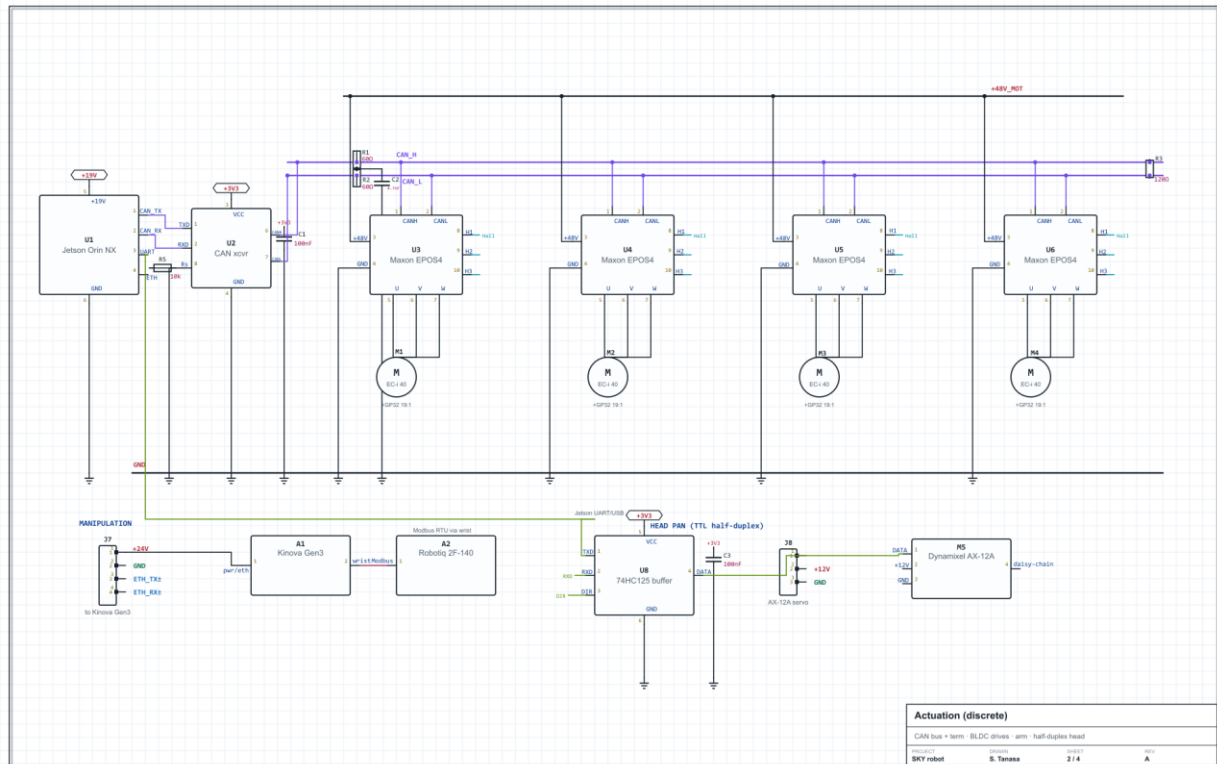


Figure 3.24 - Actuation scheme

3.11.3. Sensing and processing

The sensing subsystem – detailed in figure 3.25 – is organised around the Jetson (U1) as the central processing hub, with each sensor connected on the interface best suited to its bandwidth. The I²C bus (SDA/SCL), pulled up to 3.3 V by R1 and R2, carries the low-rate devices: the OB1203 pulse/SpO₂ sensor (U2) and the control interface of the FLIR Lepton 3.5 thermal camera (U3); the Lepton's thermal video is read separately over SPI to keep the slow control bus uncongested. The two interaction cameras, Sony IMX519 (J2) and Arducam IMX477 (J3), stream over dedicated MIPI CSI-2 lanes, the Intel RealSense D435i over USB 3.0 (J4), and the SICK TiM571 LiDAR over Gigabit Ethernet (J5) with a separate 12 V supply. Every active device carries a local 100 nF decoupling capacitor at its supply pin, and power is taken from the 12 V / 5 V / 3.3 V rails defined in the power schematic.

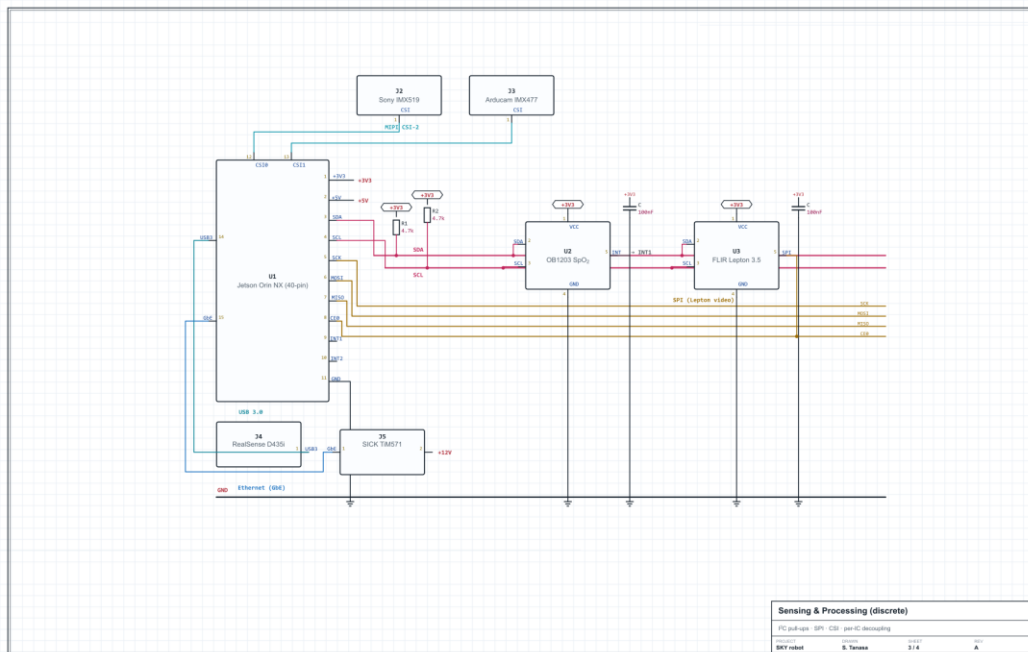


Figure 3.25 - Sensing subsystem schematic

3.11.4. Interaction and communications

The interaction subsystem groups the human-facing functions, the onboard network and the medication-drawer security. The electronic drawer lock is driven by a discrete low-side circuit: a Jetson GPIO drives the gate of an N-channel MOSFET (Q1) through a series resistor (R1), with a pull-down (R2) holding the gate low; the solenoid (L1) is energised from the 12 V rail and a flyback diode (D1) clamps the inductive spike at switch-off. Access is authorised by an RC522 RFID reader (U2) on the SPI bus, with its own decoupling. All devices share a single onboard network: a combined Wi-Fi access point and Gigabit switch (U3) links the Jetson, the Kinova arm and the LiDAR over wired Ethernet, and serves the head and chest tablets and the .NET MAUI teleoperation client over Wi-Fi, so that all inference stays on the local network. The emergency-stop button (SW1, normally closed) energises a safety relay (K1, with flyback diode D2) whose contact breaks the +48V_MOT rail of the power schematic, giving a hardware-level motion cut independent of software. The architecture is presented in figure 3.26:

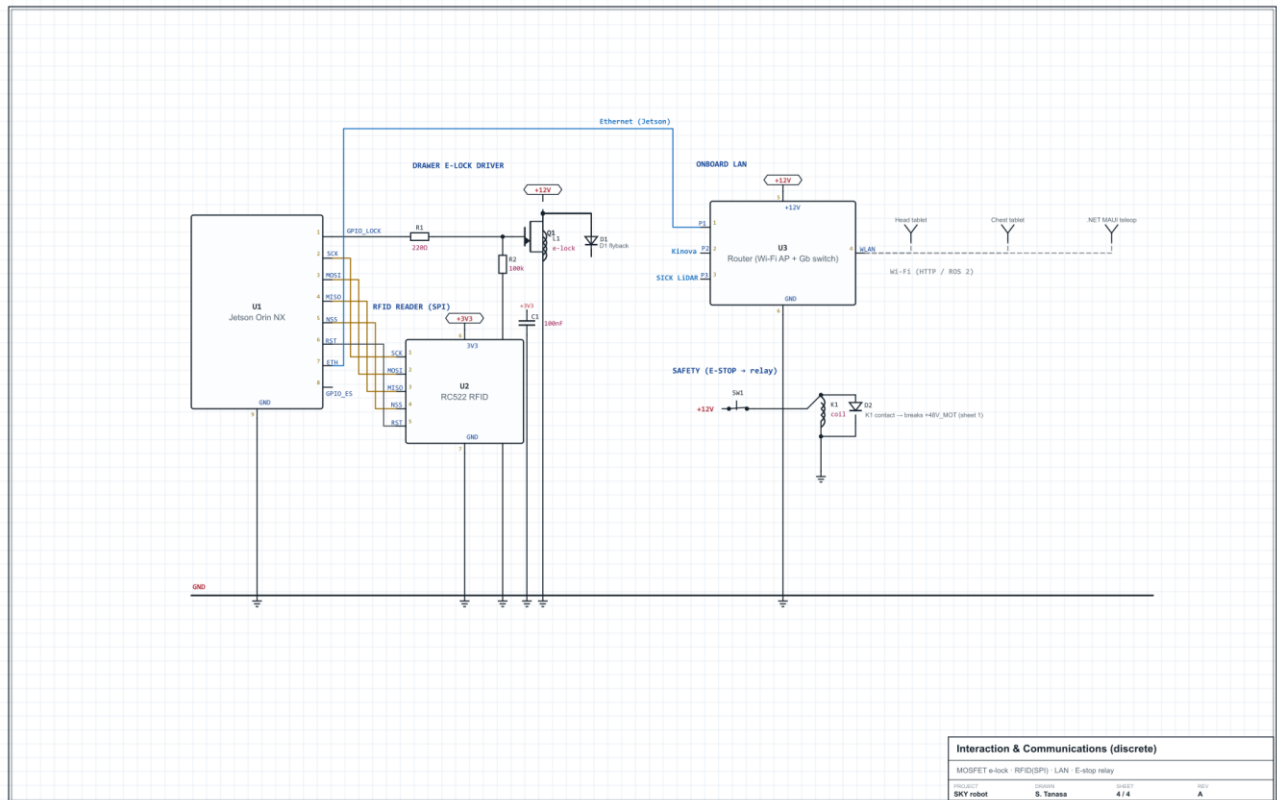


Figure 3.26 - Interaction and communications architecture

3.12. Structural verification of the arm-mounting flange by finite element analysis

The analytical dimensioning of the arm-mounting flange in sub-chapters 3.6 and 3.7 showed that the bending stress produced by the fully extended arm stays far below the yield limit of the selected alloy. To confirm this result on the real geometry - including the central bore, the twelve M6 torso holes and the four locating-pin bores, which introduce local stress concentrations that a hand calculation cannot capture - a linear-static finite-element analysis (FEA) was carried out in CATIA V5. The purpose is to verify that the flange stays well within the elastic domain under the worst-case load and to confirm that its 22 mm thickness is governed by stiffness (angular deflection amplified over the 0.902 m arm reach) rather than by strength.

The flange is a $\varnothing 135 \times 22$ mm disc of wrought aluminum alloy EN AW-7075 T651 ($E = 71.7$ GPa, $\nu = 0.33$, $\sigma_{\text{yield}} \approx 460$ MPa). The part was discretized with parabolic tetrahedral elements, with local refinement around the holes and the central bore, as shown in figure 3.27. For more details, please check Appendix B.

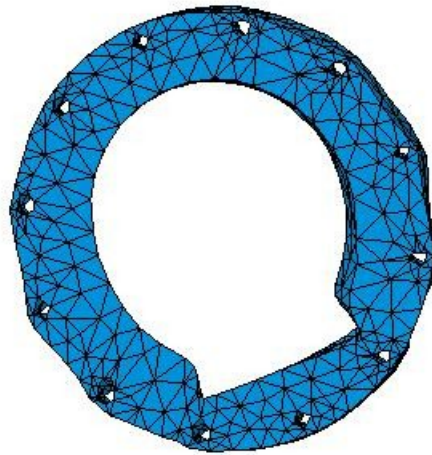


Figure 3.27 - Finite-element mesh of the arm-mounting flange

The twelve M6 torso-fixing holes on the Ø125 mm bolt circle were fully restrained, reproducing the clamped connection to the torso upper plate. The worst-case dynamic bending moment $M_{dyn} = 145 \text{ N}\cdot\text{m}$ — derived in 3.7.1 from the static moment of $72.4 \text{ N}\cdot\text{m}$ and a dynamic factor $k_d = 2$ — was applied at the arm-side interface, representing the arm fully extended in the horizontal plane. Figure 3.28 shows the restraints (blue) and the applied load (yellow).

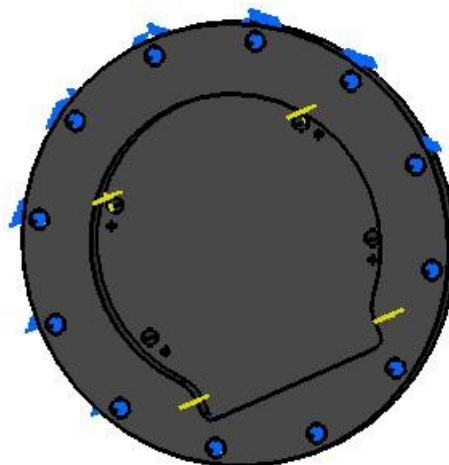


Figure 3.28 - Boundary conditions applied to the arm-mounting flange

The resulting equivalent von Mises stress field is shown in figure 3.29. The peak equivalent stress reaches 6.39 MPa, localized at the loaded interface and around the torso-hole restraints, while the

bulk of the part carries less than 2 MPa. The stress concentrations remain small and highly localized, with no sign of net-section overload at any hole.

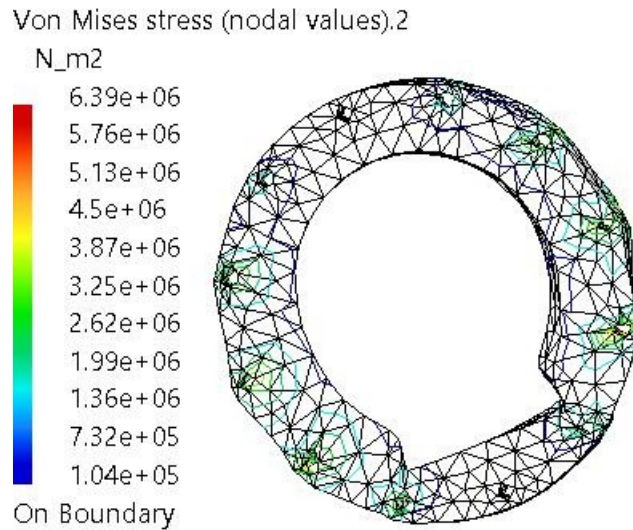


Figure 3.29 - Von Mises stress distribution (nodal values)

The maximum shear-stress field, presented in figure 3.30, follows the same pattern with a peak value of 2.45 MPa, consistent with the von Mises result and again confined to the immediate vicinity of the restrained holes. Both fields confirm a uniform, low-stress load path through the disc.

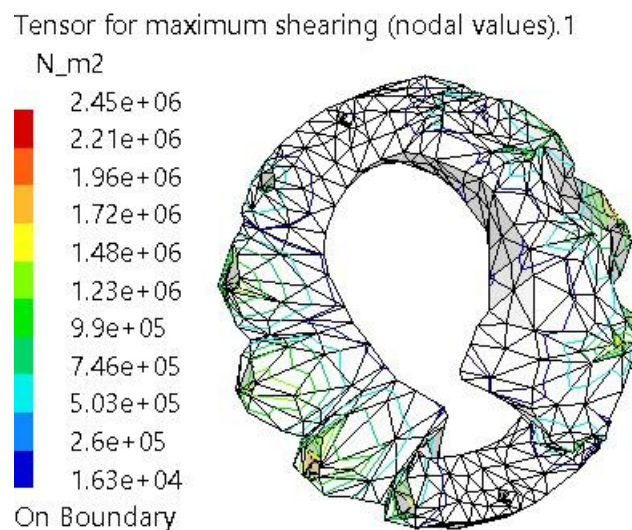


Figure 3.30 - Maximum shear-stress distribution (nodal values)

Taking the peak von Mises stress of 6.39 MPa against the 460 MPa yield strength of EN AW-7075 T651 gives a structural safety factor $SF = \sigma_{\text{yield}} / \sigma_{\text{max}} \approx 72$. The complete set of inputs and results is summarized in table 3.15.

Table 3.15 – Finite-element analysis results for the arm-mounting flange

Quantity	Value	Remark
Applied dynamic bending moment, M_{dyn}	145 N·m	Worst case, arm horizontal (figure 3.7.1)
Restraint	12 × M6 torso holes	Fixed (clamped)
Material	EN AW-7075 T651	$\sigma_{\text{yield}} \approx 460$ MPa
Von Mises stress (nodal), max	6.39 MPa	Figure 3.29
Maximum shear stress (nodal), max	2.45 MPa	Figure 3.30
Structural safety factor, SF	≈ 72	$\sigma_{\text{yield}} / \sigma_{\text{von Mises}}$

The analysis confirms the conclusion anticipated in 3.7.1: even under the worst-case dynamic load, the flange operates roughly two orders of magnitude below yield, so it is not a strength-critical part. Its 22 mm thickness and the choice of the high-stiffness EN AW-7075 T651 alloy are justified not by stress but by the need to minimize angular deflection at the base, since any small tilt of the mounting plane is amplified into a tool-point positioning error over the 0.902 m reach of the manipulator. The FEA therefore validates the analytical design and demonstrates that the flange satisfies both the strength and the stiffness requirements of the arm-to-torso interface.

3.13. Overall drawings

The complete geometric definition of the robot is provided as a set of technical drawings, grouped in the appendices. Three drawings are included. The overall assembly drawing of the SKY robot, in A0 format with the associated bill of materials, is given in Appendix C; it shows the spatial arrangement of the head, neck, torso, arm and mobile base, together with the interfaces between the main sub-assemblies. The detail drawing of the arm-mounting flange, fully dimensioned and having the tolerances according to the specifications derived in sub-chapter 3.7 (ISO 286-1 fits, ISO 1101 geometric tolerances and ISO 1302 surface texture), is given also in Appendix C. Together, these drawings constitute the manufacturing and integration reference for the mechanical parts treated analytically in this chapter.

4. Prototype

4.1. Purpose and scope of the scale demonstrator

In order to validate the concepts that define SKY as a system, a functional scale demonstrator was designed, built and programmed. The demonstrator is not a miniature copy of the mechanical design; it is a functional analogue, in which every major subsystem of the full-scale robot is represented by an accessible component that fulfils the same role in the overall architecture.

The demonstrator validates, at reduced scale: (i) the layered electrical architecture with a protected battery pack, star-point power distribution and independently regulated voltage rails, mirroring the philosophy of sub-chapter 3.11; (ii) the multi-sensor perception concept, with ultrasonic ranging, physiological sensing and RFID identification operating concurrently on shared buses; (iii) the dual operating mode of the mobile base – autonomous obstacle avoidance and manual tele-operation from a mobile application – corresponding to the behavior arbitration described in sub-chapter 3.8; and (iv) the access-controlled manipulation scenario, in which a validated RFID scan unlocks a servo-driven mechanism and triggers a sequenced arm motion, reproducing at model scale the medication-handling use case of the full robot.

Table 4.1 summarizes the correspondence between the full-scale architecture and the demonstrator hardware. This mapping is deliberate: each substitution preserves the engineering problem (bus arbitration, power budgeting, real-time constraints) while replacing only the cost and scale of the component.

Table 4.1 – Correspondence between the full-scale architecture and the scale demonstrator

Full-scale SKY (Chapter 3)	Scale demonstrator
NVIDIA Jetson Orin NX + ROS 2 stack	Arduino Mega 2560 (single non-blocking firmware)
SICK TiM571 2-D LiDAR + RealSense D435i	3 × HC-SR04 ultrasonic ranging sensors
Kinova Gen3 7-DOF manipulator + Robotiq 2F-140	3 × MG996R servo arm segments + 1 × MG996R lock actuator
Maxon EC-i 40 mecanum drive (4 wheels)	4 × DC gearmotors driven by an L298N dual H-bridge
OB1203 pulse / SpO2 sensor	MAX30100 pulse-oximetry module
LiFePO4 48 V 20 Ah battery + BMS	3S Li-Ion 18650 pack (11.1 V) + 3S 20 A BMS
Dedicated safety layer (ISO 13482 analysis)	Fuse hierarchy, master disconnect, firmware fail-safes
ESP32 facial expression displays (ESP-NOW)	Identical – reused directly from the full-scale concept
.NET MAUI Android application (Bluetooth)	Identical – same application, HC-05 SPP link

4.2. Electrical architecture and power distribution

The demonstrator is powered by a 3S lithium-ion pack built from three 18650 cells housed in a series holder, giving a nominal 11.1 V (12.6 V fully charged). The pack is supervised by a dedicated 3S 20 A battery management system (BMS) which provides over-charge, over-discharge, over-current and short-circuit protection, together with passive cell balancing. Because the holder exposes only the end terminals, the two intermediate cell junctions required by the BMS balance inputs (B1 = 4.2 V tap, B2 = 8.4 V tap) were soldered directly onto the holder contact rivets, after identifying the correct junctions by voltage measurement against the pack negative. All charging and discharging take place exclusively through the protected P+/P– outputs of the BMS; the pack is charged with a 12.6 V CC/CV charger.

Power distribution follows the same star-point philosophy adopted for the full-scale robot in sub-chapter 3.11.1, scaled to the demonstrator currents. The protected battery output feeds, in series: a 10 A main fuse placed immediately after P+ (also acting as the master disconnect of the whole robot), and the positive star node from which every consumer branch originates. The ground return of every board in the system is brought to a single star ground node, guaranteeing a common reference for all logic signals and keeping the high motor return currents out of the logic ground paths. Two additional protection elements complete the hierarchy: a 5 A fuse on the H-bridge branch – the only branch fed directly from the battery without an intermediate converter – and a 1.5 A switch on the 5 V logic rail, which allows the controller to be power-cycled independently of the rest of the system.

Three LM2596 step-down converters generate the regulated rails: converter #1 supplies 5.0 V to the Arduino Mega and its logic peripherals; converter #2 supplies a dedicated 6.0 V rail for the four MG996R servomotors, buffered by a 1000 μ F electrolytic capacitor that absorbs the current transients at motion onset; converter #3 supplies 5.0 V to the two ESP32 modules and the TFT display. Separating the servo rail from the logic rail is essential: a single MG996R can draw over 2 A at stall, and sharing its supply with the microcontroller would cause voltage sags and brown-out resets. Each converter was adjusted to its target voltage with a multimeter before any load was connected, since the factory potentiometer position of an LM2596 is undefined and can exceed 20 V.

Two logic-level constraints are handled explicitly in the wiring. The RC522 RFID reader is supplied strictly from the 3.3 V pin of the Arduino Mega, as its transceiver is not 5 V tolerant. The HC-05 Bluetooth module is supplied at 5 V, but its RX input is a 3.3 V pin; the Mega TX1 line therefore reaches it through a 1 k Ω / 2 k Ω resistive divider, while the HC-05 TX output drives the Mega RX1 input directly, a 3.3 V high level being safely read as logic HIGH by the 5 V microcontroller. In figures 4.1, 4.2, and 4.3, will be presented the power distribution diagram, the battery holder with the BMS board connected, and the three LM2596 modules with the fuses.

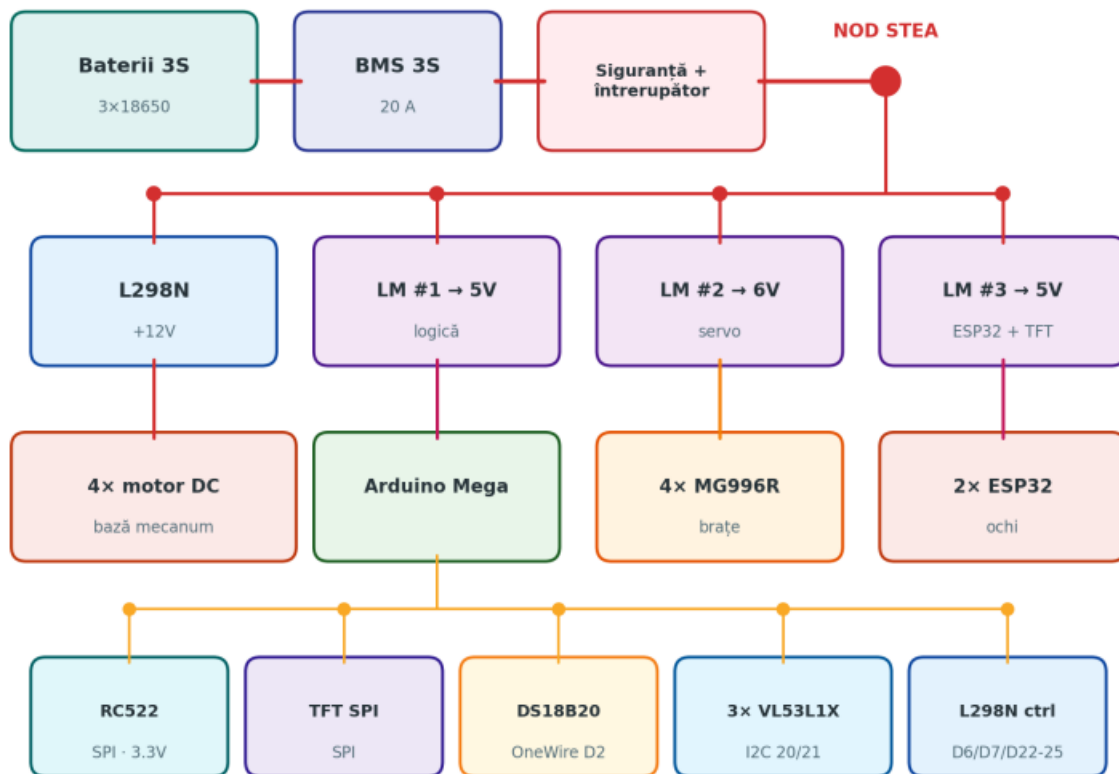


Figure 4.1 - Power distribution architecture of the demonstrator

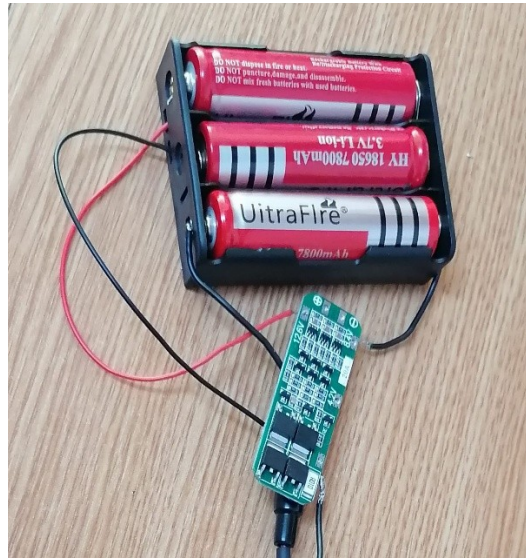


Figure 4.2 – Battery pack assembly: holder with soldered balance taps and the 3S 20 A BMS

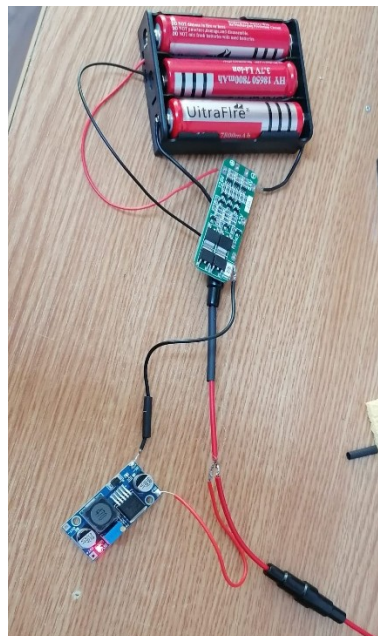


Figure 4.3 – Voltage regulation and protection stage: the three LM2596 converters, main fuse, branch fuse and master disconnect

4.3. Actuation

4.3.1. Mobile base

The mobile base uses four DC gearmotors driven by a single L298N dual H-bridge. The two motors on each side are connected in parallel on one bridge channel, so the base is controlled as a differential (skid-steer) platform: forward, reverse, and pivot turns are obtained by the sign of each channel, while speed is set by PWM on the enable inputs (ENA – left pair, ENB – right pair). The H-bridge power stage is fed at battery voltage through the 5 A branch fuse; its logic ground is tied to the star ground node, without which the direction inputs would have no valid reference.

Two firmware measures protect the power system during base operation. First, PWM is applied through an acceleration ramp (+8 PWM counts every 10 ms) rather than as a step: the inrush current of four simultaneously starting motors, combined with the relatively high internal resistance of the

cells, would otherwise sag the pack voltage and reset the controller. Second, the stop command bypasses the ramp entirely and zeroes both channels immediately – deceleration is never traded for safety. In figure 4.4 it is shown how the H-bridge is connected to the Arduino Mega 256 and to the four DC motors.

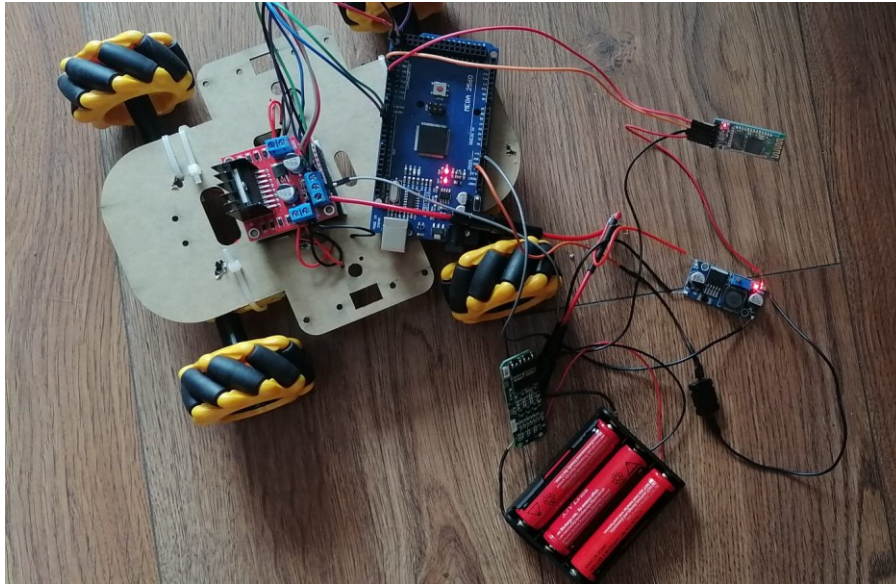


Figure 4.4 - L298N H-bridge wiring: battery branch with 5 A fuse, paired motor outputs and control lines to the Arduino Mega

4.3.2. Lock and arm servomotors

Four MG996R servomotors implement the access-control scenario. One servo acts as the lock: on a validated RFID scan it rotates from 90° to 0° , emulating the retraction of a solenoid latch, and returns to 90° at the end of the sequence – shown in figure 4.5. The remaining three servos form the model arm – shown in figure 4.5; they are driven strictly one at a time, in sequence, both to reproduce a deliberate, legible motion and to keep the instantaneous current on the 6 V rail within the capability of a single LM2596. All servo motion is generated incrementally (1° every 15 ms) rather than by direct position jumps, which limits both the mechanical jerk and the current transient of each step. When the sequence completes, the firmware detaches all servo outputs: the motors stop drawing holding current, and the characteristic idle jitter of analogue servos is eliminated.



Figure 4.5 – Servo group of the demonstrator: lock actuator and the three arm segments

4.4. Sensing

4.4.1. Obstacle detection – HC-SR04 ultrasonic sensors

Three HC-SR04 ultrasonic ranging sensors – facing left, front and right – provide the spatial awareness for the autonomous mode, standing in for the 2-D LiDAR of the full-scale design. Each sensor emits a 40 kHz ultrasonic burst and measures the time until the echo returns; the distance is derived from this echo time and the speed of sound, giving a usable range of roughly 0.02–4 m within a beam of about 15°. Because ranging is acoustic rather than optical, the readings are unaffected by target colour or ambient lighting, although acoustically soft or sharply angled surfaces reflect the pulse weakly and are the main source of missed returns. Each sensor uses a trigger output and an echo input on dedicated Mega pins (left Trig/Echo = D26/D27, front = D28/D29, right = D34/D35). To respect the non-blocking design rule of the firmware, the sensors are driven through the NewPing library in timer-interrupt mode, with a single sensor pinged at a time in round-robin; this both avoids stalling the loop on the echo pulse and prevents acoustic cross-talk between the three transducers. Each new distance is stored as soon as its echo is timed and consumed by the autonomous-drive state machine on the next pass. Figure 4.6 presents the placement of the three HC-SR04 sensors on the mobile base, and figure 4.7 shows the HC-SR04 initialization.



Figure 4.6 - Placement of the three HC-SR04 sensors on the mobile base

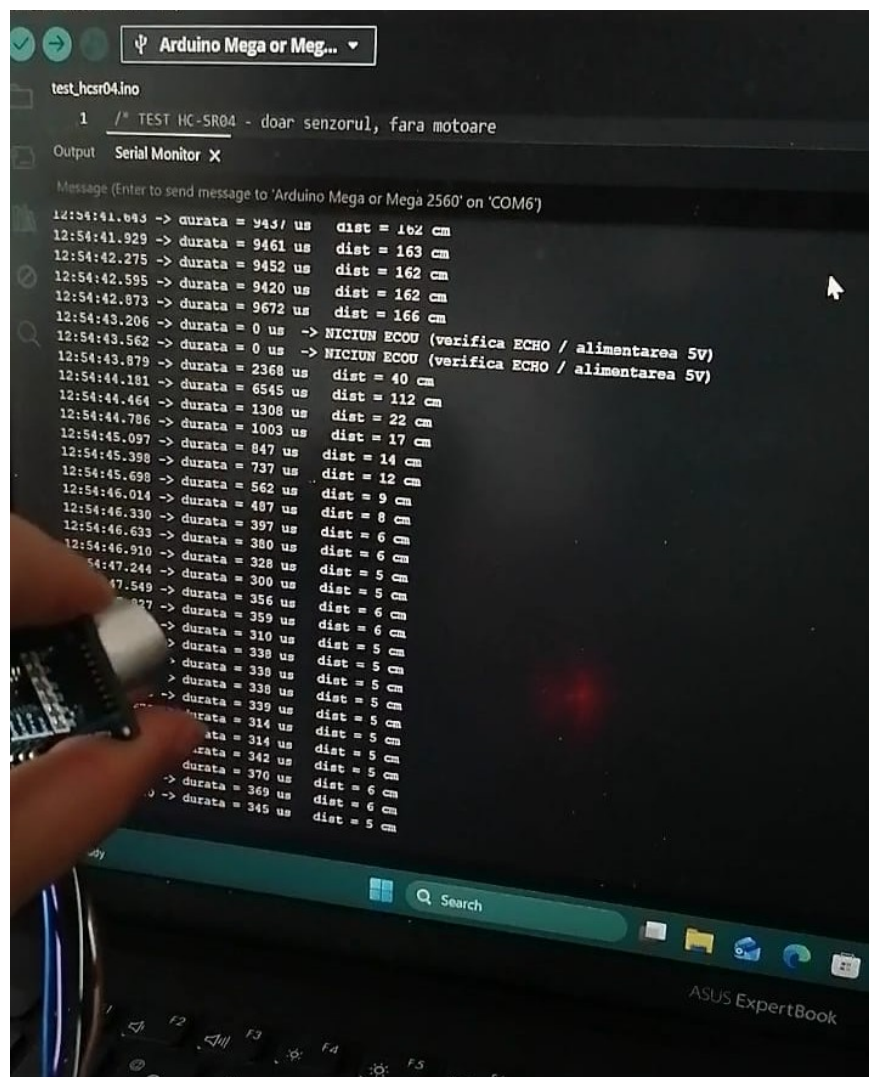


Figure 4.7 - Results of HC-SR04

4.4.2. Physiological sensing – MAX30100 pulse oximeter

The MAX30100 module measures heart rate and blood-oxygen saturation by photoplethysmography: a red and an infrared LED illuminate the fingertip and a photodiode measures the pulsatile absorption of blood. It is the demonstrator counterpart of the OB1203 sensor selected in sub-chapter 3.9.3 and shares the I2C bus with the ranging sensors (fixed address 0x57). The sensor imposes the strictest real-time constraint in the whole firmware: its driver must be serviced continuously – any blocking delay in the main loop corrupts the beat detection. This single requirement shaped the architecture of the entire program, as described in sub-chapter 4.6. The infrared LED current is set to 7.6 mA, a value determined experimentally to give stable readings at model scale. Measured values are displayed live on the TFT panel and included in the telemetry stream sent to the mobile application. In figure 4.8 it is shown the sensor and in figure 4.9, the results after scanning a fingerprint.

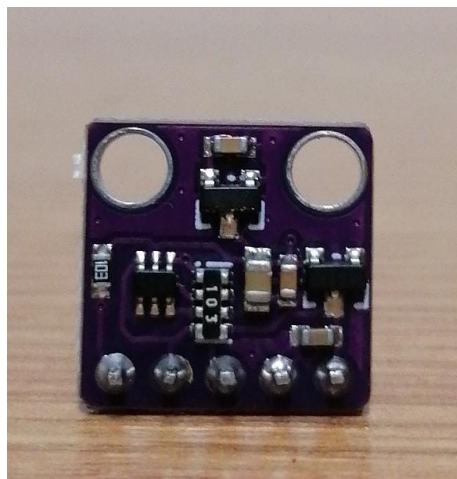


Figure 4.8 – MAX30100 pulse-oximetry

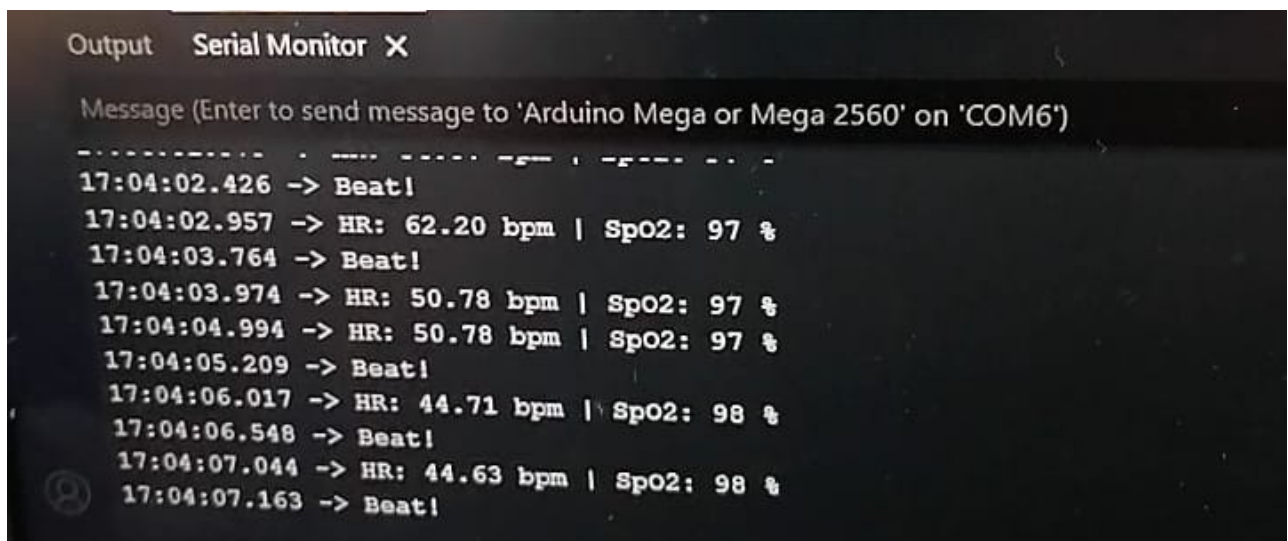


Figure 4.9 - Result of MAX30100 sensor

4.4.3. Temperature sensing – DS18B20

A DS18B20 digital thermometer provides temperature measurement over the single-wire (OneWire) protocol, on pin D4 with the mandatory 4.7 kΩ pull-up resistor between the data line and

5 V. The default library behavior – blocking for up to 750 ms during a conversion – is incompatible with the pulse-oximeter constraint, so the sensor is operated asynchronously: the firmware issues a conversion request, continues executing, and collects the result on a later loop pass once the conversion time (188 ms at the selected 10-bit resolution) has elapsed. One reading every two seconds is ample for the thermal time constant of the environment. The sensor is shown in figure 4.10.

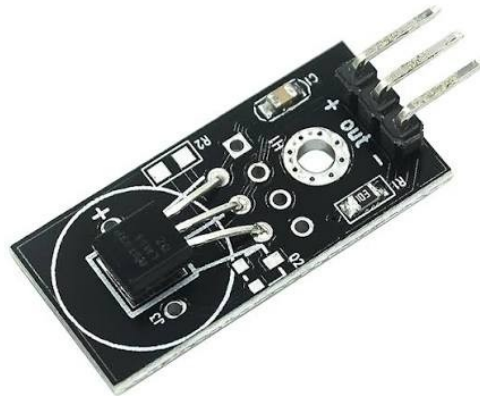


Figure 4.10 – DS18B20 temperature sensor with its 4.7 kΩ pull-up resistor

4.4.4. Identification — RC522 RFID reader

An RC522 reader operating at 13.56 MHz implements the access-control function. It communicates over the hardware SPI bus (SCK = 52, MOSI = 51, MISO = 50) with chip-select on D53 and reset on D49, and is powered from the 3.3 V rail of the Mega. The firmware polls for card presence on every loop pass, reads the 4-byte UID, and compares it against a whitelist of authorized cards stored in the sketch. A validated UID triggers the lock. A rejected UID is signaled on the display. A 2 s cool-down after each scan prevents a card left on the antenna from re-triggering the sequence continuously. The RC522 and the two cards are shown in figure 4.11; and in figure 4.12 it is presented the results of the simulated operation.



Figure 4.11 – RC522 RFID reader and the authorized tags

```

-----
16:51:58.826 -> Versiune firmware RC522: 0x82
16:51:58.826 -> Apropie un card...
16:53:44.626 -> Versiune firmware RC522: 0x82
16:53:44.626 -> Apropie un card...
16:53:46.912 -> UID: E4 C8 F0 06
16:53:46.912 -> >> Acces PERMIS - Card 1
16:54:04.896 -> Versiune firmware RC522: 0x82
16:54:04.896 -> Apropie un card...
16:54:06.249 -> UID: 87 75 63 06
16:54:06.249 -> >> Acces PERMIS - Card 2

```

Figure 4.12 – RFID UID acquisition and validation

4.5. Human–machine interface and communications

The demonstrator communicates with its operator through three channels. A TFT display connected to the shared SPI bus (with its own chip-select, D48) shows the live system state: heart rate and SpO2 from the pulse oximeter, ambient temperature, the active driving mode, the three measured distances and the result of the last RFID scan. The display is refreshed by partial redraws only – each value is rewritten solely when it changes – because full-screen redraws would monopolizes the SPI bus and starve the pulse-oximeter driver.

The facial expression subsystem is carried over unchanged from the full-scale concept: two ESP32 modules, each driving an eye display, are powered from the third 5 V rail through their VIN regulator inputs and synchronize their animations over ESP-NOW in a master–slave arrangement. No data wiring exists between the eyes and the Mega; the eye subsystem is electrically integrated (shared power and ground) but logically autonomous.

Manual tele-operation uses the existing .NET MAUI Android application over a Bluetooth SPP link provided by an HC-05 module on the hardware Serial1 port (pins 18/19) at 9600 baud. Serial1 is used deliberately instead of the default Serial port: pins 0/1 are shared with the USB bootloader, and a module connected their blocks sketch uploads. The command protocol is line-oriented and human-readable: MOVE:F/B/L/R/S selects the base motion, SPD:l,r sets the per-side PWM values, and MODE:A / MODE:M switches between autonomous and manual operation. The firmware acknowledges every command and additionally streams a one-line telemetry frame (mode, front distance, heart rate, SpO2, temperature) once per second, allowing the application to display the robot state in real time. In figure 4.13 it is shown the TFT screen during a live measurement. Figure 4.14 shows how the ESP32 display the eye.

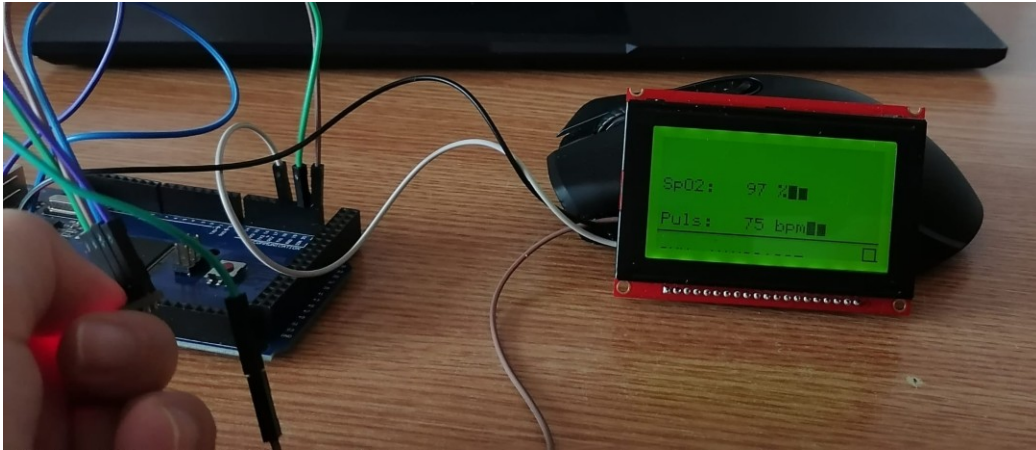


Figure 4.13 – TFT status display during operation

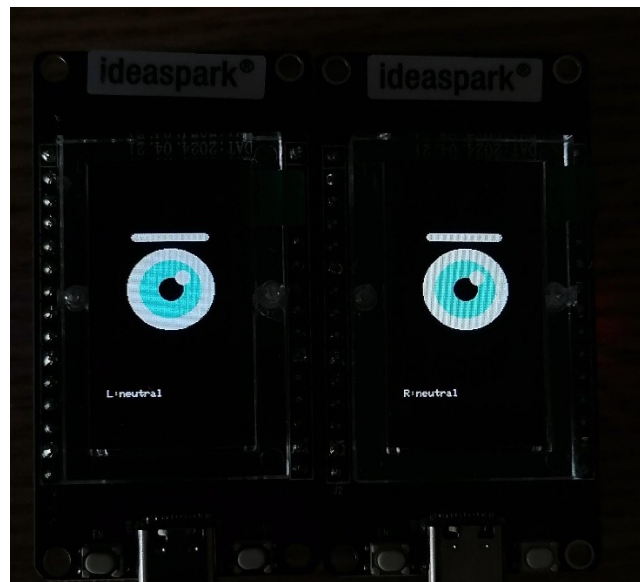


Figure 4.14 – ESP32 eye modules with synchronized expressions (ESP-NOW master–slave)

4.6. Firmware architecture

All subsystems run in a single Arduino sketch structured as a non-blocking cooperative loop. The design rule is absolute: no function called from `loop()` may block. Every time-dependent behavior - servo trajectories, display refresh, telemetry cadence, sensor conversion delays, autonomous maneuvers – is implemented with `millis()`-based scheduling, and the pulse-oximeter driver update is the first call of every pass, giving it the highest effective service rate. This rule was not an aesthetic choice: during incremental integration it was established experimentally that a single blocking `delay()` anywhere in the loop degrades the MAX30100 beat detection beyond usability.

Sequential behaviors are expressed as finite state machines. The access-control sequence is a ten-state machine (lock opening, three arm extension states, hold, three retraction states, lock closing, servo detach) in which each state advances the active servo by one degree per 15 ms tick and hands over to the next state on target arrival; the mobile base is halted for the duration of the sequence. The autonomous driving mode is a second, smaller state machine: the base drives forward while the front distance exceeds 300 mm; on detecting an obstacle it stops and either turns for a fixed 500 ms towards the side with more free space (if that side exceeds 250 mm) or reverses for 600 ms before turning. The timed maneuvers eliminate the left-right oscillation that a purely reactive rule produces in corners. Two safeguards protect the platform: the base advances only while a valid front reading

confirms clearance beyond 300 mm, and any mode change or arm-sequence start forces an immediate motor stop. A sequence from the entire code for the main loop of the firmware is presented in listing 4.1. Listing 4.2 and 4.3 are presented the finite state machine of the lock-and-arm sequence; respectively the autonomous driving state machine with timed avoidance maneuvers.

```

1 void loop() {
2   // Regula absoluta: pulsoximetrul este servit primul, la fiecare pas,
3   // pentru a-i garanta cea mai mare rata efectiva de actualizare.
4   pox.update(); // MAX30100 - detectia batailor
5
6   unsigned long now = millis();
7
8   // --- Planificator cooperativ, totul pe baza de millis() ---
9   if (now - tSensors >= SENSOR_PERIOD) { // citire senzori (100 ms)
10    tSensors = now;
11    readTemperature(); // DS18B20, non-blocant
12    frontDistance = readFrontDistance(); // HC-SR04 fata (ultrasonic)
13  }
14
15  if (now - tTelemetry >= 1000) { // telemetrie 1 Hz
16    tTelemetry = now;
17    sendTelemetry(); // STAT:mode,dist,HR,SpO2,temp
18  }
19
20  // --- Comportamente secventiale ca masini de stari ---
21  if (armSequenceActive) runArmSequence(now);
22  if (driveMode == AUTONOMOUS && !armSequenceActive) autonomousDrive(now);
23
24  // --- Interfete de comunicatie ---
25  handleBluetooth(); // comenzi HC-05 (Serial1)
26  handleRFID(now); // RC522, cu cooldown
27  updateDisplay(now); // TFT, redesenari partiale
28  serviceServos(now); // detach-when-idle
29 }

```

Listings 4.1 - Main loop of the firmware: cooperative non-blocking scheduling with the pulse-oximeter serviced first

```

1 // Cele zece stari ale secventei de acces: deblocare -> extindere brat
2 // (3 servo-uri) -> mentinere -> retragere -> blocare -> eliberare servo-uri.
3 enum ArmState {
4   IDLE, UNLOCK,
5   EXTEND_1, EXTEND_2, EXTEND_3,
6   HOLD,
7   RETRACT_3, RETRACT_2, RETRACT_1,
8   LOCK, DETACH
9 };
10
11 ArmState armState = IDLE;
12 unsigned long tArm = 0, tHold = 0;
13 const unsigned long ARM_TICK = 15; // 1 grad la fiecare 15 ms
14
15 void runArmSequence(unsigned long now) {
16   if (now - tArm < ARM_TICK) return; // planificare non-blocanta
17   tArm = now;
18
19   switch (armState) {
20     case UNLOCK: if (stepTowards(servoLock, LOCK_OPEN)) armState = EXTEND_1; break;
21     case EXTEND_1: if (stepTowards(servoArm1, ARM1_OUT)) armState = EXTEND_2; break;
22     case EXTEND_2: if (stepTowards(servoArm2, ARM2_OUT)) armState = EXTEND_3; break;
23     case EXTEND_3: if (stepTowards(servoArm3, ARM3_OUT)) { armState = HOLD; tHold = now; } break;
24     case HOLD: if (now - tHold >= HOLD_MS) armState = RETRACT_3; break;
25     case RETRACT_3: if (stepTowards(servoArm3, ARM3_IN)) armState = RETRACT_2; break;
26     case RETRACT_2: if (stepTowards(servoArm2, ARM2_IN)) armState = RETRACT_1; break;
27     case RETRACT_1: if (stepTowards(servoArm1, ARM1_IN)) armState = LOCK; break;
28     case LOCK: if (stepTowards(servoLock, LOCK_CLOSED)) armState = DETACH; break;
29     case DETACH:
30       detachAllServos(); // eliberare pentru a reduce jitter si consum
31       armSequenceActive = false;
32       armState = IDLE;
33       break;
34     default: break;
35   }
36 }
37
38 // Avanseaza servo-ul cu un grad spre tinta; intoarce true la sosire.
39 bool stepTowards(Servo &s, int target) {
40   int pos = s.read();
41   if (pos == target) return true;
42   s.write(pos < target ? pos + 1 : pos - 1);
43   return false;
44 }

```

Listings 4.2 – Finite state machine of the lock-and-arm sequence

```

1 // Praguri: liber > 300 mm; viraj 500 ms daca latura aleasa > 250 mm; altfel
2 // mers inapoi 600 ms inainte de viraj. Manevrele temporizate elimina oscilatia
3 // stanga-dreapta pe care o produce o regula pur reactiva in colturi.
4 // Citirile intorc distanta in mm, sau MAX_RANGE (4000) daca nu exista ecou:
5 // la HC-SR04 lipsa ecoului inseamna cale libera, nu obstacol.
6 enum DriveState { FORWARD, MANEUVER };
7 DriveState dState = FORWARD;
8 unsigned long tManeuver = 0, maneuverDuration = 0;
9
10 void autonomousDrive(unsigned long now) {
11     switch (dState) {
12         case FORWARD:
13             if (frontDistance <= 300) { // obstacol in raza de 30 cm
14                 stopMotors();
15                 long freeLeft = readLeftDistance();
16                 long freeRight = readRightDistance();
17                 if (max(freeLeft, freeRight) > 250) {
18                     (freeLeft > freeRight) ? turnLeft() : turnRight();
19                     maneuverDuration = 500; // viraj 500 ms
20                 } else {
21                     reverse();
22                     maneuverDuration = 600; // mers inapoi 600 ms
23                 }
24                 tManeuver = now;
25                 dState = MANEUVER;
26             }
27             else driveForward(); // > 300 mm: cale libera
28             break;
29
30         case MANEUVER:
31             if (now - tManeuver >= maneuverDuration) {
32                 stopMotors();
33                 dState = FORWARD;
34             }
35             break;
36     }
37 }

```

Listings 4.3 – Autonomous driving state machine with timed avoidance maneuvers

4.7. Companion software layer: AI services and mobile application

The intelligence of the demonstrator is deliberately split into two layers. The Arduino Mega handles everything with hard real-time constraints — motor control, the servo state machine, sensor scheduling and the safety behaviors described in sub-chapter 4.6. The perceptual and conversational functions designed for the full-scale robot in sub-chapter 3.8 are reproduced at reduced scale by a companion software layer distributed over commodity hardware: a Python inference server hosting the AI models, an acquisition client running under Termux on an Android tablet mounted on the model, the ESP32 facial-expression firmware, and the C# mobile application used for tele-operation. The five devices communicate over three transports – HTTP over Wi-Fi, ESP-NOW and Bluetooth SPP – each chosen for the latency and reliability profile of its link. In figure 4.15 is presented the communication between the tablet and the server.

```

20:16  speak("Goodbye!")
        break
    print("Taking photo...")
    take_photo()
    if record_audio(5):
        response, emotion, objects = send_audio_with_vision()
        speak(response)
    else:
        print("Recording failed.")
except KeyboardInterrupt:
    speak("Goodbye")
    break

if __name__ == "__main__":
    main()
"""
code.close()
print("Done!")
ENDOFSCRIPT
Done!
- $ python3 ~/sky_tablet.py
Sky Medical Assistant
Server connected!
Sky: Hello! I am Sky. How can I help you?
>
Taking photo...
Listening for 5 seconds... SPEAK NOW!
Recording started: /data/data/com.termux/files/usr/tmp/sky_rec.m4a
Max Duration: 00:05
No recording to stop
M4A size: 11322 bytes
WAV exists: True
You: showered mom
Emotion: neutral
Sky: No, let's review the lab results again, I want to rule out any underlying conditions that might be causing a neutral demeanor.
> ^C
Sky: Goodbye
- $

```

Figure 4.15 - Communication between the tablet and server

4.7.1. AI inference server (Python / Flask)

All artificial-intelligence functions are concentrated in a single Python process exposing a REST API built with Flask. Four models are served behind this interface: Vosk performs offline speech-to-text transcription; a YOLOv8 network provides object detection on the camera frames; DeepFace classifies the facial expression of the interlocutor into seven emotion categories; and a locally hosted Llama 3.2 model, served through Ollama, generates the conversational responses. Serving four heterogeneous models in real time through one API was identified early as one of the three major integration challenges of the project: the solution adopted is a single long-lived process in which every model is loaded once at start-up and each function is mapped to its own endpoint returning JSON, so that no request pays the model-loading cost. The emotion label produced by DeepFace is additionally forwarded to the facial-expression subsystem, closing the loop between what the robot perceives and what it displays. In listing 4.4 is presented the REST endpoints of the AI inference server (Python / Flask).

```

1 # Server unic: fiecare model este incarcat o singura data la pornire, iar
2 # fiecare functie este mapata pe propriul endpoint care intoarce JSON, astfel
3 # incat nicio cerere nu plateste costul de incarcare a modelului.
4 from flask import Flask, request, jsonify
5
6 app = Flask(__name__)
7
8 vosk_model = Model("model") # speech-to-text offline
9 yolo_model = YOLO("yolov8n.pt") # detectie de obiecte
10 # DeepFace (emotie) si Ollama/Llama 3.2 (dialog) sunt apelate la cerere
11
12 @app.route("/talk", methods=["POST"])
13 def talk():
14     text = transcribe_vosk(request.files["audio"]) # Vosk
15     reply = ask_ollama(text) # Llama 3.2 prin Ollama
16     return jsonify({"text": text, "reply": reply})
17
18 @app.route("/analyze", methods=["POST"])
19 def analyze():
20     frame = decode_image(request.files["image"])
21     objects = detect_objects(frame) # YOLOv8
22     emotion = DeepFace.analyze(frame, actions=["emotion"])[0]["dominant_emotion"]
23     set_emotion(emotion) # transmite starea la ESP32
24     return jsonify({"objects": objects, "emotion": emotion})
25
26 @app.route("/talk_with_vision", methods=["POST"])
27 def talk_with_vision():
28     text = transcribe_vosk(request.files["audio"])
29     frame = decode_image(request.files["image"])
30     objects = detect_objects(frame)
31     emotion = DeepFace.analyze(frame, actions=["emotion"])[0]["dominant_emotion"]
32     reply = ask_ollama(build_prompt(text, objects, emotion))
33     set_emotion(emotion)
34     return jsonify({"reply": reply, "objects": objects, "emotion": emotion})
35
36 @app.route("/emotion", methods=["GET"])
37 def emotion():
38     return jsonify({"emotion": current_emotion})
39
40 @app.route("/set_emotion", methods=["POST"])
41 def set_emotion_ep():
42     global current_emotion
43     current_emotion = request.json["emotion"]
44     return jsonify({"status": "ok", "emotion": current_emotion})
45
46 if __name__ == "__main__":
47     app.run(host="0.0.0.0", port=5000)

```

Listings 4.4 - REST endpoints of the AI inference server (Python / Flask)

4.7.2. Acquisition client on the Android tablet (Termux)

The tablet mounted on the demonstrator acts as the sensory head for the interaction functions. Running under Termux, a lightweight client captures microphone audio and camera frames and posts them to the inference server over the local network; the structured reply – transcription, detected objects, recognized emotion and the generated answer – drives the spoken and displayed interaction. This split deliberately decouples the acquisition hardware from the compute: the tablet contributes exactly the sensors and the screen, while all heavy inference runs on the server, mirroring at model scale the edge-offload pattern argued for the full-scale robot. The Audio and vision acquisition client running under Termux is presented in listing 4.5.

```

1 #!/data/data/com.termux/files/usr/bin/bash
2 # Ruleaza pe tableta montata pe demonstrator: capteaza audio si un cadru,
3 # le trimite serverului de inferenta, apoi reda raspunsul structurat.
4 SERVER="http://192.168.1.100:5000"
5
6 while true; do
7   # 1. Inregistrare audio de la microfon (M4A)
8   termux-microphone-record -f /tmp/rec.m4a -l 5
9   sleep 5
10  termux-microphone-record -q
11
12  # 2. Conversie M4A -> WAV 16 kHz mono (format cerut de Vosk)
13  ffmpeg -y -i /tmp/rec.m4a -ar 16000 -ac 1 /tmp/rec.wav 2>/dev/null
14
15  # 3. Captura cadru de la camera
16  termux-camera-photo -c 0 /tmp/frame.jpg
17
18  # 4. Trimitere catre server si redare vocala a raspunsului
19  RESP=$(curl -s -X POST "$SERVER/talk_with_vision" \
20    -F "audio=@/tmp/rec.wav" \
21    -F "image=@/tmp/frame.jpg")
22
23  REPLY=$(echo "$RESP" | jq -r '.reply')
24  echo "SKY: $REPLY"
25  termux-tts-speak "$REPLY"
26 done

```

Listings 4.5 – Audio and vision acquisition client running under Termux

4.7.3. Facial-expression firmware (ESP32)

Two ESP32 modules, each driving one eye display, implement the affective interface. The master module receives the current emotional state and mirrors it to the slave over ESP-NOW – a connectionless protocol chosen because it requires no router, no pairing and delivers frames with millisecond-scale latency, which keeps the two eyes visually synchronised. Each expression is implemented as an animation state machine in the firmware, so a single short state message is sufficient to switch both displays consistently. The Master–slave expression synchronization over ESP-NOW is presented in listing 4.6.

```

1 // Modulul master primește starea emoțională și o oglindește pe slave printr-un
2 // protocol fără conexiune, fără router, fără împerechere, cu latență de ordinul
3 // ms - astfel cei doi ochi rămân sincronizați vizual.
4 #include <esp_now.h>
5 #include <WiFi.h>
6
7 uint8_t slaveAddress[] = { 0x24, 0x6F, 0x28, 0xAA, 0xBB, 0xCC };
8
9 typedef struct {
10     uint8_t emotion; // 0=neutru, 1=fericit, 2=trist, 3=surprins, ...
11 } ExpressionMsg;
12
13 ExpressionMsg msg;
14
15 void setup() {
16     WiFi.mode(WIFI_STA);
17     esp_now_init();
18
19     esp_now_peer_info_t peer = {};
20     memcpy(peer.peer_addr, slaveAddress, 6);
21     peer.channel = 0;
22     peer.encrypt = false;
23     esp_now_add_peer(&peer);
24 }
25
26 // --- MASTER: apelat când sosește o nouă stare emoțională de la server ---
27 void setExpression(uint8_t emotion) {
28     msg.emotion = emotion;
29     playAnimation(emotion); // display local
30     esp_now_send(slaveAddress, (uint8_t*)&msg, sizeof(msg)); // oglindire pe slave
31 }
32
33 // --- SLAVE: callback la recepția stării de la master ---
34 void onReceive(const uint8_t* mac, const uint8_t* data, int len) {
35     ExpressionMsg incoming;
36     memcpy(&incoming, data, sizeof(incoming));
37     playAnimation(incoming.emotion); // aceeași animație pe al doilea ochi
38 }

```

Listings 4.6 – Master–slave expression synchronization over ESP-NOW

4.7.4. Mobile application (C#, .NET MAUI)

Manual tele-operation and telemetry display are provided by an Android application written in C# on .NET MAUI, structured on the MVVM pattern with the CommunityToolkit libraries. The application maintains a Bluetooth SPP connection to the HC-05 module and speaks the line-oriented protocol of the firmware: MOVE: commands for direction, SPD: for per-side speed, MODE: for switching between manual and autonomous operation. In the opposite direction it parses the STAT: telemetry frames emitted at 1 Hz – driving mode, front distance, heart rate, oxygen saturation and temperature – and presents them to the operator. The same application is used unchanged for the demonstrator and in the full-scale concept, which is itself a validation of the communication architecture. The Bluetooth communication service of the .NET MAUI application is presented in listing 4.7.

```

1 // Mentine o conexiune SPP catre HC-05 si vorbeste protocolul orientat pe linii
2 // al firmware-ului: MOVE: (directie), SPD: (viteza/parte), MODE: (manual/autonom).
3 // Parseaza cadrele STAT: emise la 1 Hz si le expune interfetei (MVVM).
4 public class BluetoothService
5 {
6     private IStreamConnection _connection; // SPP peste HC-05
7
8     // --- Comenzi trimise catre robot ---
9     public Task MoveAsync(char dir) // F / B / L / R / S
10     => SendAsync($"MOVE:{dir}");
11
12     public Task SetSpeedAsync(int left, int right)
13     => SendAsync($"SPD:{left},{right}");
14
15     public Task SetModeAsync(char mode) // A = autonom, M = manual
16     => SendAsync($"MODE:{mode}");
17
18     public Task EmergencyStopAsync()
19     => SendAsync("MOVE:S"); // E-STOP
20
21     private async Task SendAsync(string cmd)
22     {
23         var bytes = Encoding.ASCII.GetBytes(cmd + "\n");
24         await _connection.WriteAsync(bytes, 0, bytes.Length);
25     }
26
27     // --- Receptia telemetriei STAT: emisa la 1 Hz ---
28     public event EventHandler<Telemetry> TelemetryReceived;
29
30     private void OnLineReceived(string line)
31     {
32         if (!line.StartsWith("STAT:")) return; // STAT:mode,dist,HR,SpO2,temp
33         var p = line.Substring(5).Split(',');
34         TelemetryReceived?.Invoke(this, new Telemetry {
35             Mode = p[0],
36             FrontDist = int.Parse(p[1]),
37             HeartRate = int.Parse(p[2]),
38             SpO2 = int.Parse(p[3]),
39             Temperature = float.Parse(p[4])
40         });
41     }
42 }

```

Listings 4.7 – Bluetooth communication service of the .NET MAUI application

4.8. Integration, testing and lessons learned

The demonstrator was integrated incrementally: each subsystem was first brought up and validated in isolation with a dedicated test sketch (base motion over Bluetooth, RFID with servo response, pulse oximetry), and only then merged into the integrated firmware. Every hardware stage was accepted against an explicit checklist before the next stage was connected – rail voltages verified in no-load conditions, ground continuity confirmed between all boards, and current draw measured where anomalies were suspected. This discipline localised every fault to the most recently added element and kept debugging tractable.

Several lessons of general engineering value emerged from the integration. First, contact quality dominates reliability at these power levels: the majority of electrical faults encountered were high-resistance joints – screw terminals clamped on insulation, cold solder joints on nickel-plated contacts – which manifest confusingly as voltage present upstream and absent downstream, or as BMS protection events under load. The systematic remedy was measurement-driven fault isolation: measuring voltage segment by segment along a branch until the drop is located, and distinguishing a damaged module from a bad contact by measuring the input resistance of the disconnected module. Second, the protection hierarchy proved its worth in practice: when an intermittent fault produced a sustained over-current, the BMS disconnected the pack before any component was damaged, and the fault was diagnosed from the residual voltages on its balance inputs. Third, real-time constraints must be designed in, not patched in: the non-blocking architecture, the display partial-redraw strategy and

the asynchronous temperature acquisition all exist because a single slow operation anywhere in the loop measurably degrades the pulse sensor. Finally, power-rail separation (logic / servo / peripheral) turned what would have been system-wide brown-out failures into localised, diagnosable events.

With all subsystems integrated, the demonstrator performs the complete target scenario: it navigates autonomously among obstacles or under manual control from the mobile application, continuously displays physiological and environmental measurements, and executes the access-controlled arm sequence on presentation of an authorized RFID tag – a functional, scaled validation of the architectural decisions taken for the full-scale SKY robot in Chapter 3.

SUMMARY OF THE FIRST THREE MAJOR CHALLENGES OF THE PROJECT

The first major challenge

The most mechanically complex problem encountered during the design phase concerned the wrist roll joint – the kinematic interface between the servo and the gripper at the end of Link 3 of the arm. The functional requirement is straightforward: the MX-28 servo must rotate the gripper around the longitudinal axis of the arm. However, the physical implementation of this requirement revealed a fundamental constrain that no standard component could resolve directly.

The core problem is that the servo body must remain rigidly fixed to Link 3, while the gripper mounting flange must rotate independently around it. An assembly such as rigidly coupling the motor output shaft to the flange without a bearing interface would cause the entire motor body to co-rotate with the gripper, which is both mechanically incorrect and structurally unsafe in a clinical environment where unpredictable contact forces are expected.

The first step was a precise measurement of the MX-28 output shaft, which yielded a diameter of $\varnothing 5.7\text{mm}$ – a non-standard dimension that does not correspond to any ISO or DIN shaft series. This single measurement eliminated all commercially available standard bushing options. The only viable path was a custom transition bushing, with following dimensions: inner diameter $\varnothing 5.7\text{mm}$, outer diameter $\varnothing 15\text{mm}$, length 15mm, and a $\varnothing 3\text{mm}$ transverse through-hole for a locking pin to prevent rotational slip between the bushing and shaft.

The bearing selection followed from the bushing outer diameter. An SKF 61802 deep groove ball bearing ($d = 15\text{mm}$, $D=24\text{mm}$, $B=5\text{mm}$) was selected and a Seeger retaining ring $\varnothing 24\text{mm}$ secures the assembly axially. The mounting flange – which is a detachable piece from the gripper body – is the element that accommodates the bearing outer race directly, eliminating the need for a separate intermediate bracket and simplifying the assembly sequence: the flange is removed from the gripper, the bearing and bushing subassembly is inserted, the retaining ring is installed, and the flange is then reattached to the gripper body. Four M5 stainless screw fix the flange to the rotating side, while four M2.5 screws fix the motor body bracket to Link 3 on the fixed side.

The complete assembly was modelled, verifying clearances, screw accessibility, and the correct separation between fixed and rotating elements.

The second major challenge

The second major challenge was the simultaneous integration of four independent AI models into a single operational system capable of serving multiple hardware clients in real time, without dedicated AI accelerator hardware and without cloud connectivity.

The four models – Vosk for offline speech recognition, Llama 3.2 via Ollama for natural language generation, YOLOv8 for object and person detection, and DeepFace for facial emotion recognition – each carry distinct computational profiles and latency characteristics. Running them fully sequentially would make conversational interaction unacceptably slow; running them all in parallel without resource management would cause memory contention on a standard Windows PC. The solution was a priority-based execution architecture in which the speech recognition and LLM pipeline are kept on the critical conversational path, while YOLOv8 and DeepFace operate on a lower-frequency polling cycle – sufficient for emotion-adaptive response and fall detection without blocking verbal interaction.

A concrete implementation challenge arose from the audio pipeline between the Android tablet and the Flask server. The Termux voice client on the tablet records audio using the Android native

microphone API, which produces M4A files encoded in AAC at 8000 Hz – a format that Vosk cannot process directly, as it requires 16-bit PCM at 16000 Hz. The solution was a ffmpeg conversation step executed on the tablet before transmission: the M4A file is resampled to 16000 Hz mono PCM and converted to WAV format, then sent to the Flask server via HTTP POST to the /talk endpoint. This conversation step added negligible latency (measured at under 20ms for a 3-second audio clip at 176x speed factor in ffmpeg) while resolving the format incompatibility entirely.

The Flask server [69] exposes dedicated endpoints for each subsystem: /talk for full speech-to-LLM-to-TTS conversational loop, /emotion for DeepFace polling, and /detect for YOLOv8 inference. The two ESP32 modules poll /emotion independently over Wi-Fi at a fixed interval, rendering synchronized animated eye expressions on their OLED display in real time – each eye reacting to the detected patient emotional state without requiring a dedicated microcontroller. The .NET MAUI application communicates with the Arduino Mega via Bluetooth SPP for motor commands, entirely bypassing the Flask server for latency-critical drive control. This separation of the motor control path from the AI inference path was a deliberate architectural decision that ensures navigation responsiveness is never degraded by AI processing load.

All AI inference runs exclusively in the local Windows PC, with no data transmitted to external servers. This design choice was not only a practical constraint – offline operation in network-restricted environments – but also a privacy requirement, ensuring that patient audio, video, and biometrical data never leave the local network.

In the figure S.1, the conversational pipeline for the practical part is described:

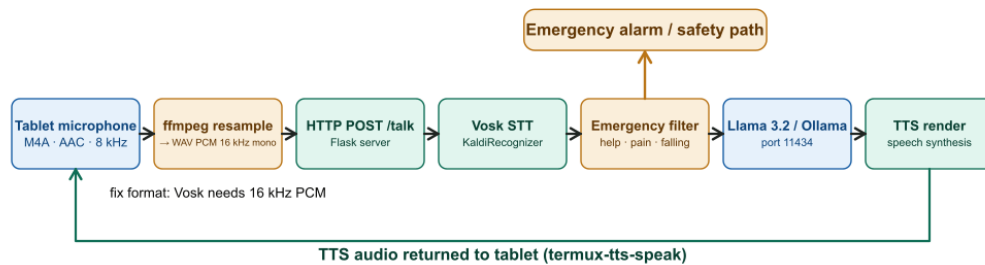


Figure S.1 – Conversational pipeline for the practical part

The third major challenge

The third challenge was the design and debugging of a coherent communication architecture across five heterogeneous hardware components operating simultaneously: a Windows PC running the Flask inference server, an Android tablet running a Termux Python clients for voice and camera streaming, an Android phone running a .NET MAUI application for manual motor control, and an Arduino Mega [71] handling low-level motor and sensor interfacing, and two ESP32 Wi-Fi modules driving the animated face displays.

The central architectural decision – to route all intelligence through a single Flask API on the PC rather than distributing logic across devices – was not obvious at the outset. An alternative approach would have placed separate inference modules on each device, which conflicts, duplicated state management, and made debugging across five simultaneous devices effectively intractable. The single-API design created one authoritative source of system state, simplified the communication graph, and allowed each device to be developed and tested independently against well-defined HTTP endpoints.

Three distinct communication protocols operate in parallel within the architecture. HTTP REST over Wi-Fi connects the Flask server to the tablet, the ESP32 modules. Bluetooth SPP connects the .NET MAUI smartphone application to the Arduino Mega, providing a low-latency channel for real-

time directional motor commands – D-pad input, speed sliders, arm joint control, and gripper actuation – without routing through the PC. The ESP32 modules communicate exclusively via Wi-Fi, polling the Flask server's /emotion endpoint and rendering synchronized animated eye expression independently on their OLED displays in real time. Once firmware is deployed, all three hardware components – Arduino Mega and both ESP32 modules – operate autonomously on battery power, with no wired connection to the PC required during normal operation.

One of the most significant debugging challenges arose from the Bluetooth pairing process between the HC-05 module connected to the Arduino Mega and the smartphone application. Initial pairing attempts consistently failed, requiring multiple configuration iterations on both the module and Android Bluetooth stack before a stable connection was established. Once pairing was resolved, the communication channel proved reliable for real-time motor control.

A secondary debugging challenge arose during the development of the .NET MAUI motor control application. After adding new UI elements — speed sliders and a battery level progress bar — the application crashed silently on startup immediately after installation, with no error message displayed on the device. The root cause was a XAML data binding incompatibility: the ProgressBar.Progress property expects a double value between 0.0 and 1.0, while the BatteryLevel variable was declared as an int in the range 0–100. This type mismatch caused the application to fail at runtime without raising an explicit exception, making diagnosis non-trivial. A separate issue arose from Android 12+ Bluetooth permission requirements: the BLUETOOTH_CONNECT and BLUETOOTH_SCAN permissions must be explicitly requested at runtime in MainActivity.cs, rather than being declared solely in AndroidManifest.xml. Once both issues were resolved, the application operated stably, providing reliable real-time motor control over Bluetooth SPP.

The communication architecture for the practical part is described in figure S.2:

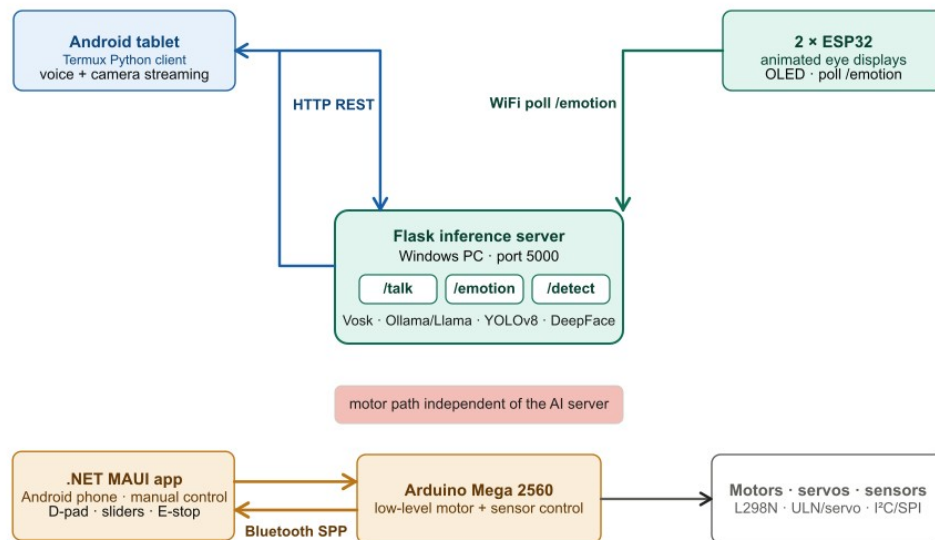


Figure S.2 – Communication architecture for the practical part

Communication between the inference server and the animated displays following a polling, decoupling the two subsystems. As shown in listing S.1 the server exposes the most recently detected emotion through a lightweight read-only endpoint, which the two EPS32 modules query independently over Wi-Fi. This stateless approach allows additional display client to be added without modifying the server logic.

```
# Shared state: holds the most recently detected patient emotion.
# Read by the /emotion endpoint so the ESP32 eye displays can poll it.

current_emotion_global = {"emotion": "neutral"}

@app.route("/emotion", methods=["GET"])
def get_emotion():
    """
    Returns the most recently detected patient emotion.
    Polled by the ESP32 eye displays (via WiFi) to update their animations
    in real time based on what the camera sees.
    """
    return jsonify(current_emotion_global)
```

Listing S.1 – Shared-state endpoint polled over WIFI by the ESP32

CONCLUSIONS

This thesis set out to design and develop a hybrid robotic healthcare assistant capable of simultaneously handling hospital logistics, monitoring patients and sustaining meaningful social interaction – a combination that, as the benchmark analysis demonstrated, no currently deployed clinical platform offers on a single system. That objective has been met at two complementary levels, and the convergence of the two is what gives the work its sense of completeness.

At the conceptual level, the full-scale platform was specified and validated analytically. Every load-bearing assembly was dimensioned and verified against the yield strength of the selected aluminium alloy with safety factors above the adopted thresholds, the platform's tipping stability was confirmed across all defined load cases, and the complete drive, power and autonomy budget was established. The manipulator was resolved through an explicit make-vs-buy evaluation, retaining commercial arm for the realised configuration while fully dimensioning an in-house alternative as a documented upgrade path. The result is a coherent, internally consistent mechatronic design backed by traceable calculations rather than by assertion.

At the functional level, a working prototype was actually built and tested. A single Python inference pipeline integrating offline speech recognition, object and person detection, facial-emotion analysis and a locally hosted language model was demonstrated in real time. Low-level motion control was implemented and validated on embedded hardware, and an animated emotional-expression interface was driven by dedicated microcontrollers. Most importantly, the agentic behaviour layer realises the project's central originality claim: the robot weighs competing objectives and autonomously lets a clinical or social need pre-empt a logistical one, while a deterministic safety path always overrides the probabilistic AI. The entire pipeline runs on-board with no cloud dependency, satisfying the privacy requirement intrinsic to medical data, and its hardware-decoupled design allows it to be reused beyond this specific body.

The significance of the work lies in showing that the long-standing trade-off between physically and socially capable medical robots is not fundamental but architectural, and that an integrated system covering all four functional dimensions is achievable with offline, privacy-preserving, commodity-grade components. The limitations are bounded – the principal one being the absence of a fully assembled full-scale prototype – and the further-development path, from full hardware integration through on-target navigation to regulatory evaluation under ISO 13482 and IEC 60601-1, is defined as a direct continuation of the validated subsystem. The project therefore delivers both a substantiated

design and a functioning demonstration of its core intelligence, establishing a credible foundation for a clinically deployable hybrid healthcare robot.

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Planning of robotic/automated production	
Client-server applications in robotics	
Quality assurance in automated processes	
Autonomous robotic systems	
Vision systems in robotics	

Action of robotic systems	
Managing the supply chain for robotic production	
Robotic systems command and control	
Information management in robotic systems	
Manufacturing intelligent systems	
Robotics optimisation	
High precision robotic systems	
Mini and micro-robotics	
Flexible manufacturing systems	
Robotics mechanical design	
Human-robot and multi-robot cooperation applications	
Other (specify):	

Related domains

Manufacturing automation	
Automation	
Intelligent industrial products	
Industrial automation	
All type of mechatronics modules	
Internet of Things (IoT)	
Smart consumer products	
Modules in the structure of manufacturing equipment	
Numerically control manufacturing equipment	
Drive, servo drives and electric drives	
Hydraulics and pneumatics	
Microcontrollers	

Sensorics	
Medical equipment and devices	
Intelligent buildings	
Automated manufacturing technologies	
Monitoring and control of processes and technologies	
Production planning systems	
Virtual reality and related technologies	
Applied electronics	
Technologies in physics and chemistry (lasers, holography, etc.)	
Non-destructive inspection, measurement and control systems	
Any technical or technological innovation	
Other (specify):	

Assessment of the bachelor thesis by the scientific coordinator

This space is allocated to the scientific coordinator

Criterium ↓	Level →	Unsatisfactory	Satisfactory	Good	High	Excellent
		1	2	3	4	5
Quality of expression in Romanian/English		☐	☐	☐	☐	☐
Aesthetic aspect of the bachelor thesis		☐	☐	☐	☐	☐
Subject actuality		☐	☐	☐	☐	☐
Clarity of the written documentation		☐	☐	☐	☐	☐
Originality		☐	☐	☐	☐	☐
Technical quality (fairness solutions, calculations etc.)		☐	☐	☐	☐	☐
Quality of the bibliographic documentation		☐	☐	☐	☐	☐
Quality of the drawings		☐	☐	☐	☐	☐
Relevance for the social or economic environment		☐	☐	☐	☐	☐

Signature of scientific coordinator: _____

Checklist

Compulsory component	Approached in the bachelor thesis	
	YES	NO
Calculation of the main mechanical components		
Calculation of the drive system		
Technological calculation for a mechanical reference		
Definition of geometric parameters and tolerances for a mechanical reference		
Overall drawings		
A .mp4 video clip of 60 sec with the presentation of your project (for marketing purposes)		

Optional components / considering the specific of the subject	Approached in the bachelor thesis	
	YES	NO
Automation, electronic and sensorics		
Programming		

Signature of scientific coordinator: _____